

筑波大学集中講義2025

原子核理論特講II 「ハンズオンで学ぶ核力の基礎」

ー 第六回 「カイラル有効場理論と最近の話題」 ー

福井徳朗

九州大学 基幹教育院

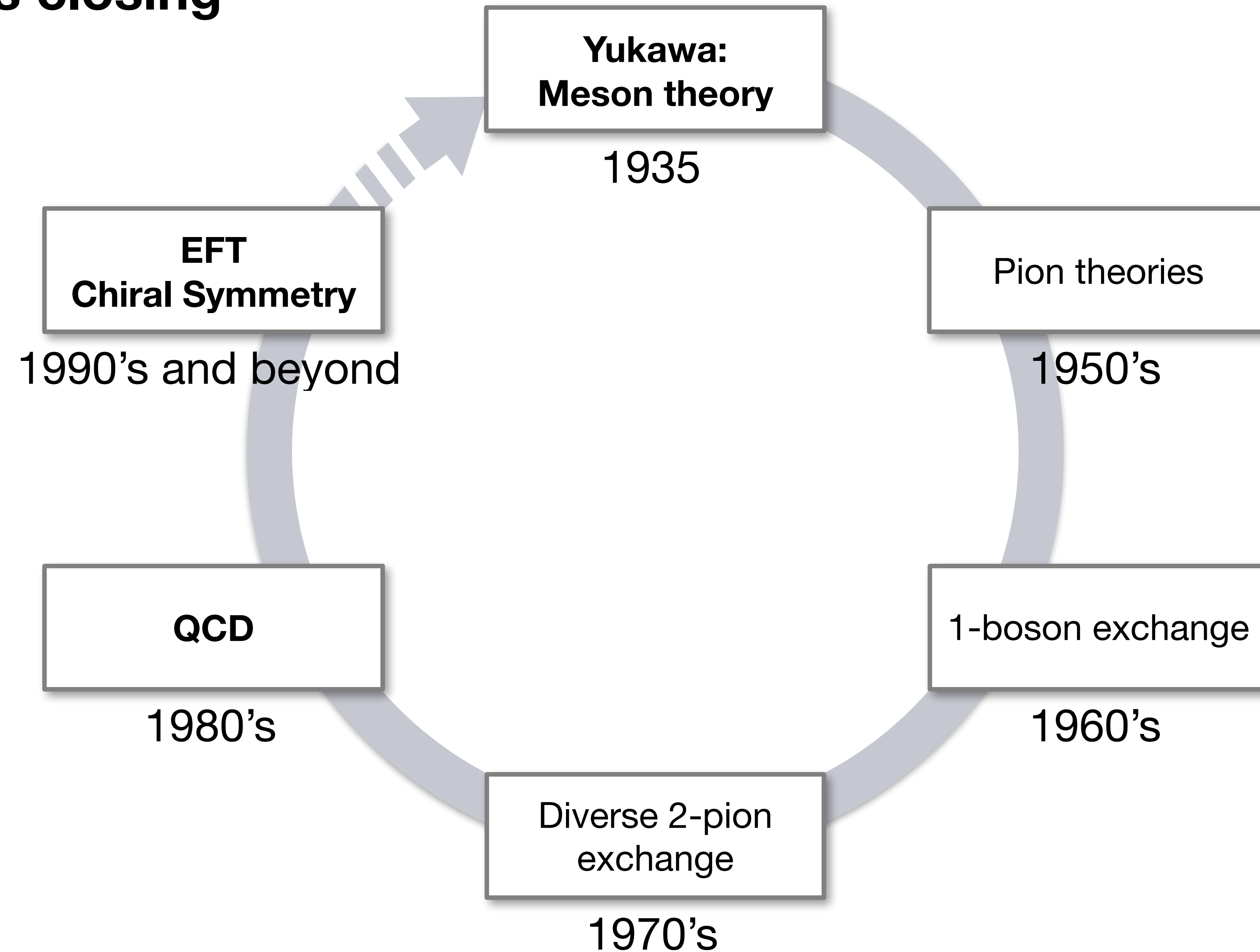
tokuro.fukui@artsci.kyushu-u.ac.jp

27/February/2025

Brief introduction of chiral EFT

“the circle of history is closing”

Machleidt & Entem, PR **503**, 1 (2011)



Concept of chiral EFT

Chiral effective field theory (EFT)

Weinberg, PA **96**, 327 (1979)
Epelbaum +, RMP **81**, 1773 (2009)
Machleidt & Entem, PR **503**, 1 (2011)

Degrees of freedom and symmetry

Nucleons and pions
Chiral symmetry



Soft scale: $Q \sim m_\pi$
Hard scale: $\Lambda_\chi \sim m_\rho$

Perturbative expansion of Lagrangian

$\left(\frac{Q}{\Lambda_\chi}\right)^\nu$ Power counting
Theoretical uncertainty

Many-body forces on an equal footing

Leading 3NF at N²LO ($n = 3$)

$$\mathcal{L}_{\text{eff}} = \mathcal{L}_\pi + \mathcal{L}_{\pi N} + \mathcal{L}_{NN} + \dots$$

$$\mathcal{L}_\pi = \frac{F^2}{4} \text{Tr}(\nabla^\mu U \nabla_\mu U^\dagger + \chi_+) + \dots$$

$$\mathcal{L}_{\pi N} = \bar{N}(i v \cdot D + g_A \cdot S)N + \dots$$

$$\mathcal{L}_{NN} = -\frac{1}{2}C_S(\bar{N}N)^2 + 2C_T(\bar{N}S N)^2 + \dots$$

More details

Lagrangian

$$\mathcal{L}_{\text{eff}} = \mathcal{L}_{\pi} + \mathcal{L}_{\pi N} + \mathcal{L}_{NN} + \cdots$$

$$\mathcal{L}_{\pi}^{(0)} = \frac{F^2}{4} \langle \nabla^\mu U \nabla_\mu U^\dagger + \chi_+ \rangle,$$

$$\mathcal{L}_{\pi N}^{(0)} = \bar{N} (i v \cdot D + \mathring{g}_A u \cdot S) N,$$

$$\mathcal{L}_{NN}^{(0)} = -\frac{1}{2} C_S (\bar{N} N) (\bar{N} N) + 2 C_T (\bar{N} S N) \cdot (\bar{N} S N),$$

$$\begin{aligned} \mathcal{L}_{\pi}^{(2)} = & \frac{l_3}{16} \langle \chi_+ \rangle^2 + \frac{l_4}{16} [2 \langle \nabla_\mu U \nabla^\mu U^\dagger \rangle \langle \chi_+ \rangle \\ & + 2 \langle \chi^\dagger U \chi^\dagger U + \chi U^\dagger \chi U^\dagger \rangle - 4 \langle \chi^\dagger \chi \rangle - \langle \chi_- \rangle^2] \\ & + \cdots, \end{aligned}$$

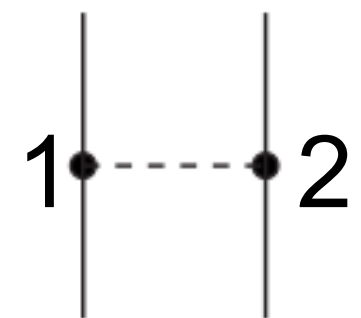
$$\begin{aligned} \mathcal{L}_{\pi N}^{(2)} = & \bar{N} \left(\frac{1}{2\mathring{m}} (v \cdot D)^2 - \frac{1}{2\mathring{m}} D \cdot D + d_{16} S \cdot u \langle \chi_+ \rangle \right. \\ & \left. + i d_{18} S^\mu [D_\mu, \chi_-] + \cdots \right) N, \end{aligned}$$

$$\begin{aligned} \mathcal{L}_{NN}^{(2)} = & -\tilde{C}_1 \{ (\bar{N} D N) \cdot (\bar{N} D N) + [(D \bar{N}) N] \cdot [(D \bar{N}) N] \} \\ & - 2(\tilde{C}_1 + \tilde{C}_2) (\bar{N} D N) \cdot [(D \bar{N}) N] \\ & - \tilde{C}_2 (\bar{N} N) \cdot [(D^2 \bar{N}) N + \bar{N} D^2 N] + \cdots \end{aligned}$$

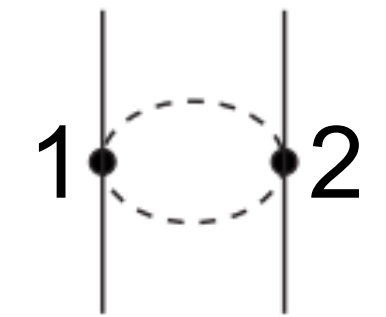
Weinberg, PA **96**, 327 (1979)
Epelbaum +, RMP **81**, 1773 (2009)
Machleidt & Entem, PR **503**, 1 (2011)

Nucleon field	N
Pion field	$\boldsymbol{\pi}$
Nucleon four-velocity	v_μ
Covariant spin vector	$S_\mu \equiv (1/2) i \gamma_5 \sigma_{\mu\nu} v^\nu$
Pion decay constant	F
Axial vecor coupling const.	$\mathring{g}_A$
Low-energy constant	C_S and C_T
Unitary 2×2 matrix	$U(\boldsymbol{\pi}) = u^2(\boldsymbol{\pi})$ $U(\boldsymbol{\pi}) = 1 + \frac{i}{F} \boldsymbol{\tau} \cdot \boldsymbol{\pi} - \frac{1}{2F^2} \boldsymbol{\pi}^2 + \mathcal{O}(\pi^3)$
Covariant derivative (nucleon)	$D_\mu = \partial_\mu + [u^\dagger, \partial_\mu u]/2$
Covariant derivative (pion)	$u_\mu = i(u^\dagger \partial_\mu u - u \partial_\mu u^\dagger)$
Chiral symmetry breaking	$\chi_+ = u^\dagger \chi u^\dagger + u \chi^\dagger u$ $\chi = 2B\mathcal{M}$
Finite light quark masses	$\mathcal{M} = \text{diag}(m_u, m_d)$
Scalar quark condensate	$\langle 0 \bar{u} u 0 \rangle = -F^2 B$
Pion-quark-mass relation	$M_\pi^2 = 2Bm_q + \mathcal{O}(m_q^2)$

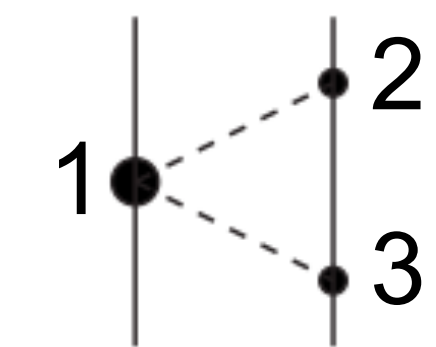
■ Examples: $\nu = -4 + 2N + 2L + \sum_i \Delta_i, \Delta_i = d_i + \frac{n_i}{2} - 2$

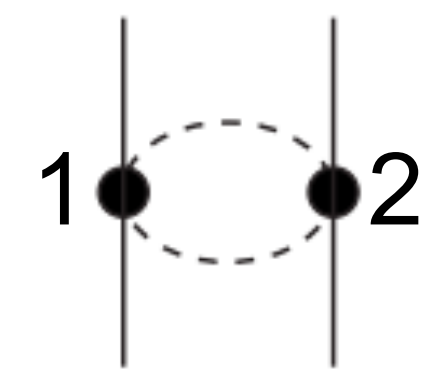
 $N = 2, L = 0, \Delta_1 = \Delta_2 = 0$
 $\nu = 0$

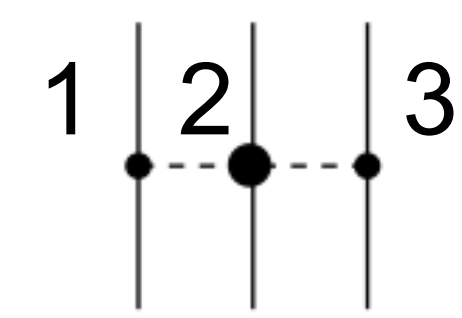
 $N = 2, L = 0, \Delta = 0$
 $\nu = 0$

 $N = 2, L = 1, \Delta_1 = \Delta_2 = 0$
 $\nu = 2$

 $N = 2, L = 0, \Delta = 2$
 $\nu = 2$

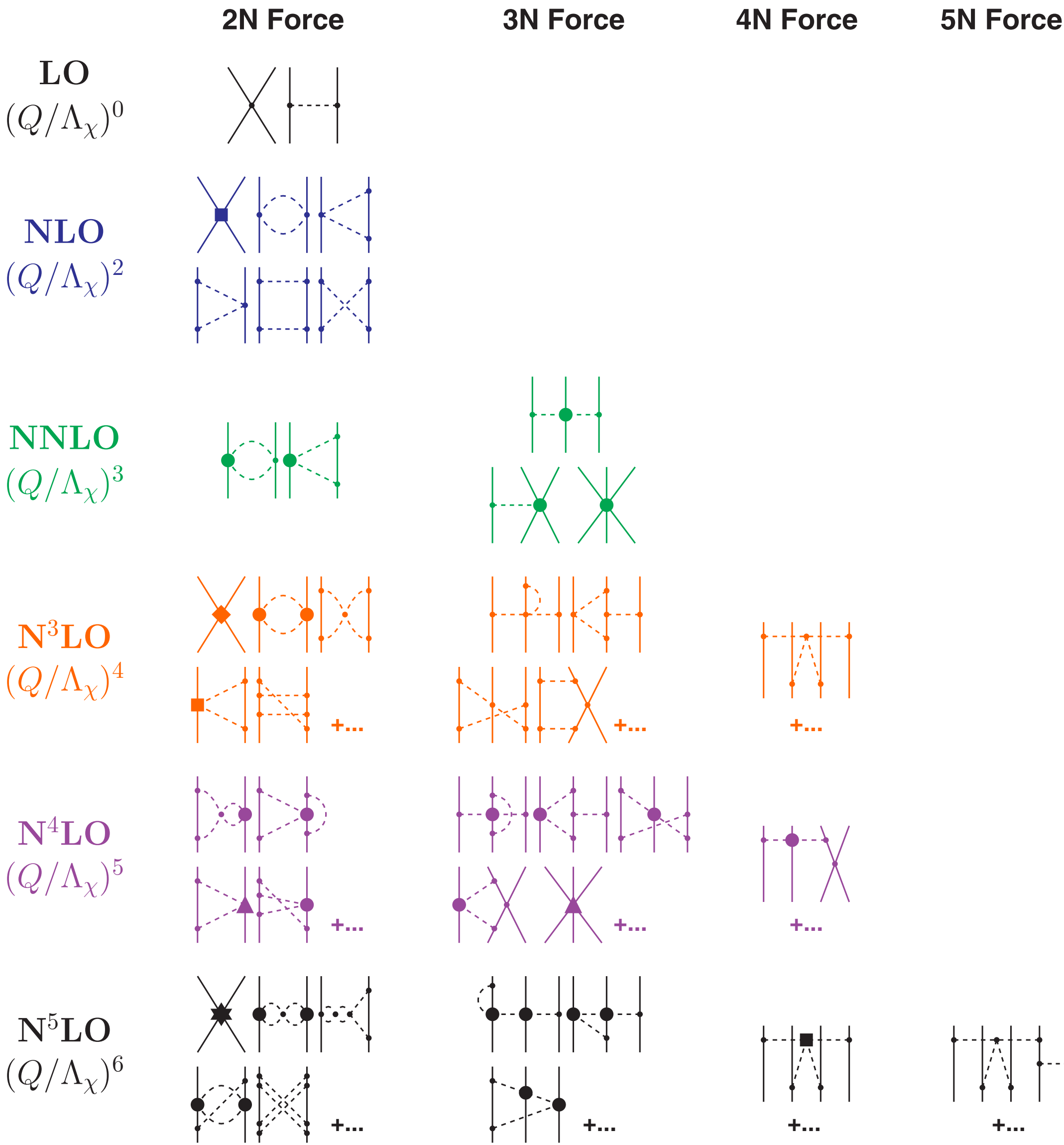
 $N = 2, L = 1, \Delta_1 = 1, \Delta_2 = \Delta_3 = 0$
 $\nu = 3$

 $N = 2, L = 1, \Delta_1 = \Delta_2 = 1$
 $\nu = 4$

 $N = 3, L = 0, \Delta_1 = \Delta_3 = 0, \Delta_2 = 1$
 $\nu = 3$

dot : $\Delta = 0$
circle : $\Delta = 1$
square : $\Delta = 2$

Hierarchy of chiral EFT



Entem +, PRC **96**, 024004 (2017)

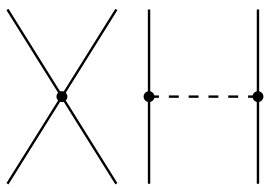
Hierarchy of Δ -full chiral EFT

Machleidt & Entem, PR 503, 1 (2011)

2NF

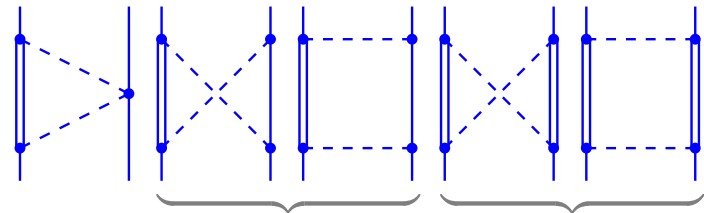
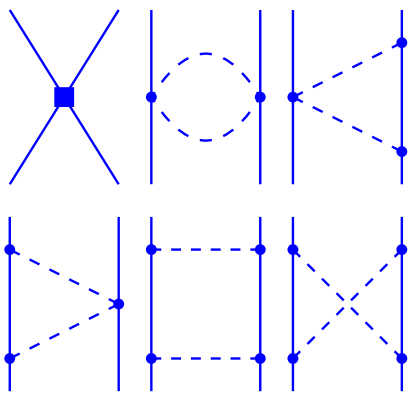
LO
 $(Q/\Lambda_\chi)^0$

Δ -less

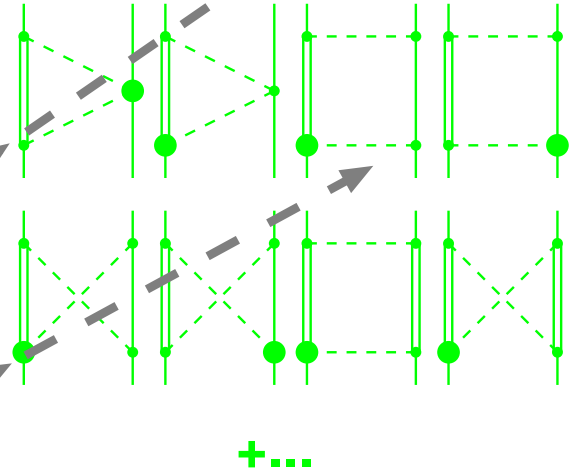
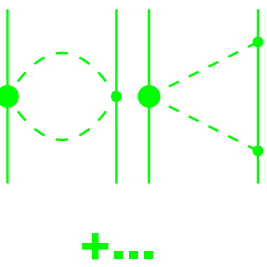


Additional in Δ -full

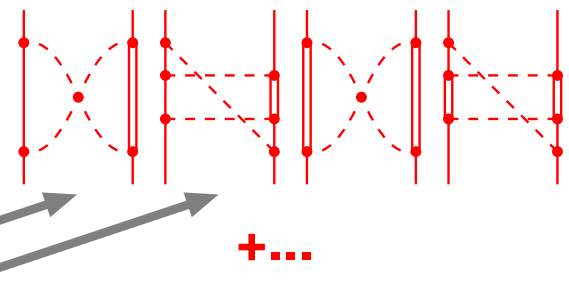
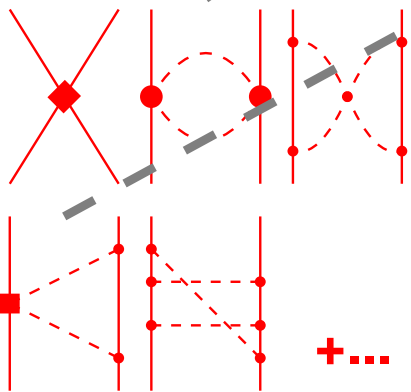
NLO
 $(Q/\Lambda_\chi)^2$



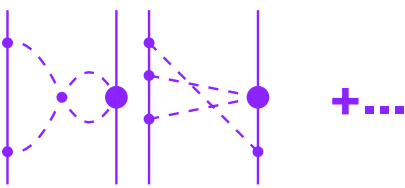
NNLO
 $(Q/\Lambda_\chi)^3$



N³LO
 $(Q/\Lambda_\chi)^4$



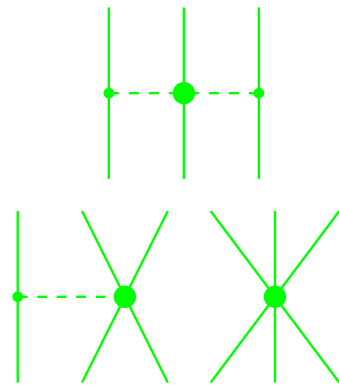
N⁴LO
 $(Q/\Lambda_\chi)^5$



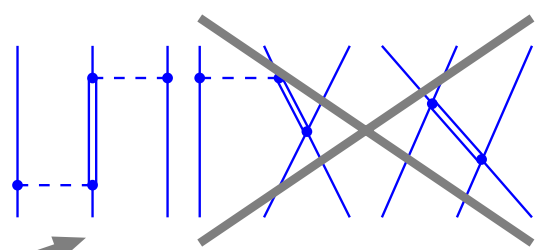
3NF

LO
 $(Q/\Lambda_\chi)^0$

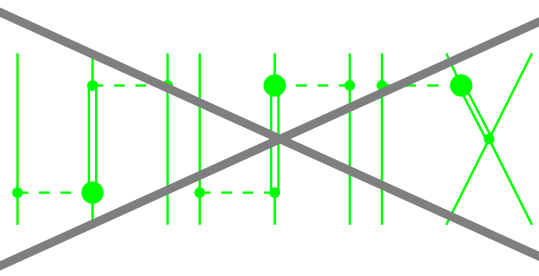
Δ -less



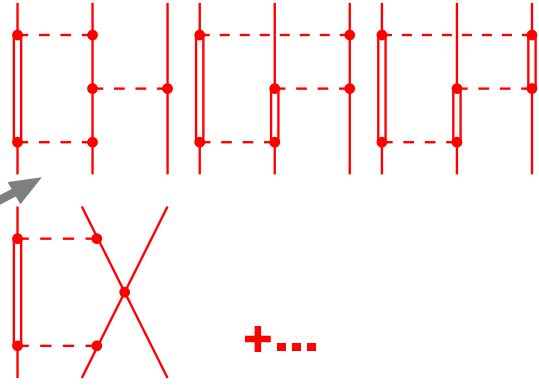
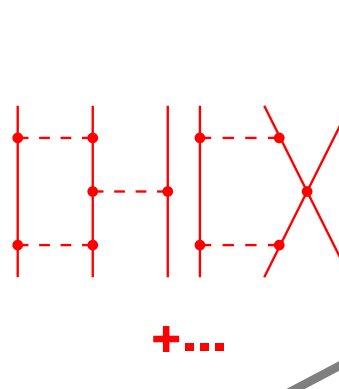
Additional in Δ -full



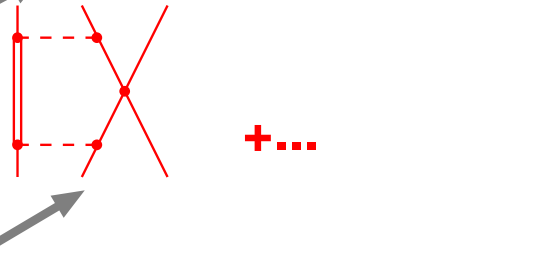
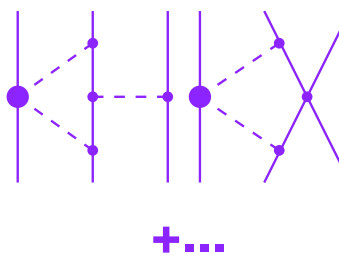
NLO
 $(Q/\Lambda_\chi)^2$



NNLO
 $(Q/\Lambda_\chi)^3$



N³LO
 $(Q/\Lambda_\chi)^4$



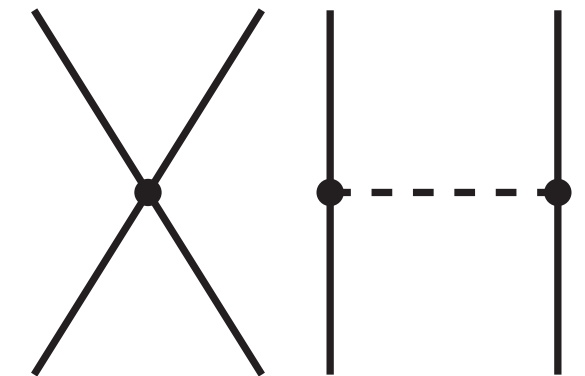
N⁴LO
 $(Q/\Lambda_\chi)^5$

LO potential

Machleidt & Entem, PR **503**, 1 (2011)

$$\hat{V}_{2N}^{(\text{LO})} = \hat{V}_{1\pi} + \hat{V}_{\text{ct}}^{(0)}$$

$$\text{LO} \\ (Q/\Lambda_\chi)^0$$



$$v_{1\pi}(\boldsymbol{p}', \boldsymbol{p}) = - \left(\frac{g_A}{2f_\pi} \right)^2 \frac{(\boldsymbol{\sigma}_1 \cdot \boldsymbol{q})(\boldsymbol{\sigma}_2 \cdot \boldsymbol{q})}{q^2 + m_\pi^2} \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2$$

Central + Tensor

$$v_{\text{ct}}^{(0)}(\boldsymbol{p}', \boldsymbol{p}) = C_S + C_T \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2$$

Central

Momentum transfer

$$\boldsymbol{q} = \boldsymbol{p}' - \boldsymbol{p}$$

NLO potential

Machleidt & Entem, PR **503**, 1 (2011)

$$\hat{V}_{2N}^{(\text{NLO})} = \hat{V}_{2\pi}^{(2)} + \hat{V}_{\text{ct}}^{(2)}$$

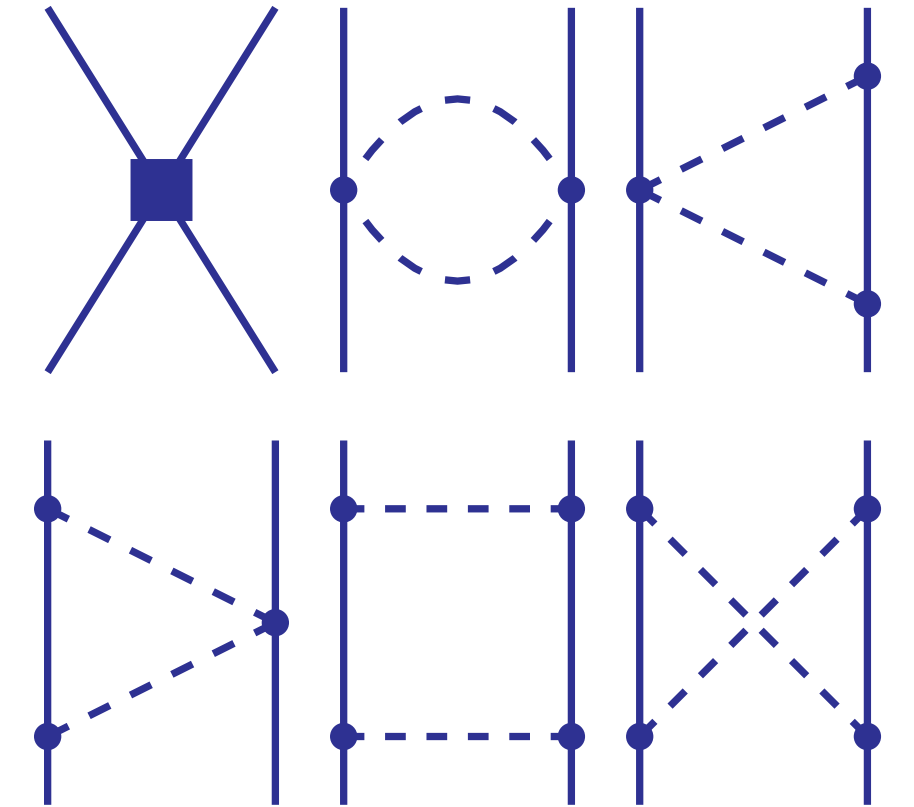
$$v_{2\pi}^{(2)}(\mathbf{p}', \mathbf{p}) = W_C^{(2)}(q, Q) \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 + V_S^{(2)}(q, Q) \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 \\ + V_T^{(2)}(q, Q) (\boldsymbol{\sigma}_1 \cdot \mathbf{q}) (\boldsymbol{\sigma}_2 \cdot \mathbf{q})$$

Central + Tensor

$$v_{\text{ct}}^{(2)}(\mathbf{p}', \mathbf{p}) = C_1 q^2 + C_2 Q^2 + (C_3 q^2 + C_4 Q^2) \boldsymbol{\sigma}_a \cdot \boldsymbol{\sigma}_b + C_5 [-i \mathbf{S} \cdot (\mathbf{q} \times \mathbf{Q})] \\ + C_6 (\boldsymbol{\sigma}_1 \cdot \mathbf{q}) (\boldsymbol{\sigma}_2 \cdot \mathbf{q}) + C_7 (\boldsymbol{\sigma}_1 \cdot \mathbf{Q}) (\boldsymbol{\sigma}_2 \cdot \mathbf{Q})$$

Central + SO + Tensor

NLO
 $(Q/\Lambda_\chi)^2$



Average momentum

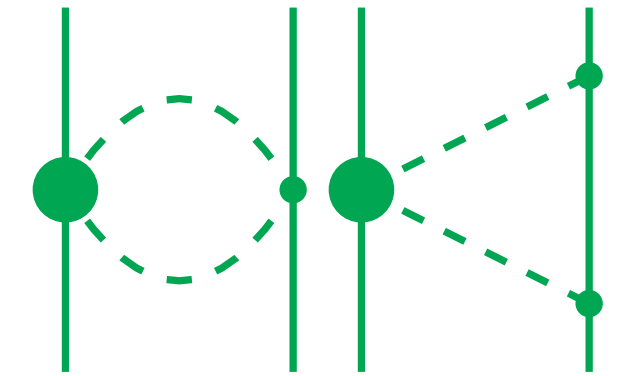
$$Q = (\mathbf{p}' + \mathbf{p})/2$$

N²LO potential

Machleidt & Entem, PR **503**, 1 (2011)

$$\hat{V}_{2N}^{(\text{N}^2\text{LO})} = \hat{V}_{2\pi}^{(3)}$$

NNLO
 $(Q/\Lambda_\chi)^3$



$$\begin{aligned} v_{2\pi}^{(3)}(\boldsymbol{p}', \boldsymbol{p}) = & V_C^{(3)}(q, Q) + W_C^{(3)}(q, Q) \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 \\ & + \left[V_S^{(3)}(q, Q) + W_S^{(3)}(q, Q) \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 \right] \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 \\ & + \left[V_{LS}^{(3)}(q, Q) + W_{LS}^{(3)}(q, Q) \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 \right] [-i\boldsymbol{S} \cdot (\boldsymbol{q} \times \boldsymbol{Q})] \\ & + \left[V_T^{(3)}(q, Q) + W_T^{(3)}(q, Q) \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 \right] (\boldsymbol{\sigma}_1 \cdot \boldsymbol{q}) (\boldsymbol{\sigma}_2 \cdot \boldsymbol{q}) . \end{aligned}$$

Central + SO + Tensor

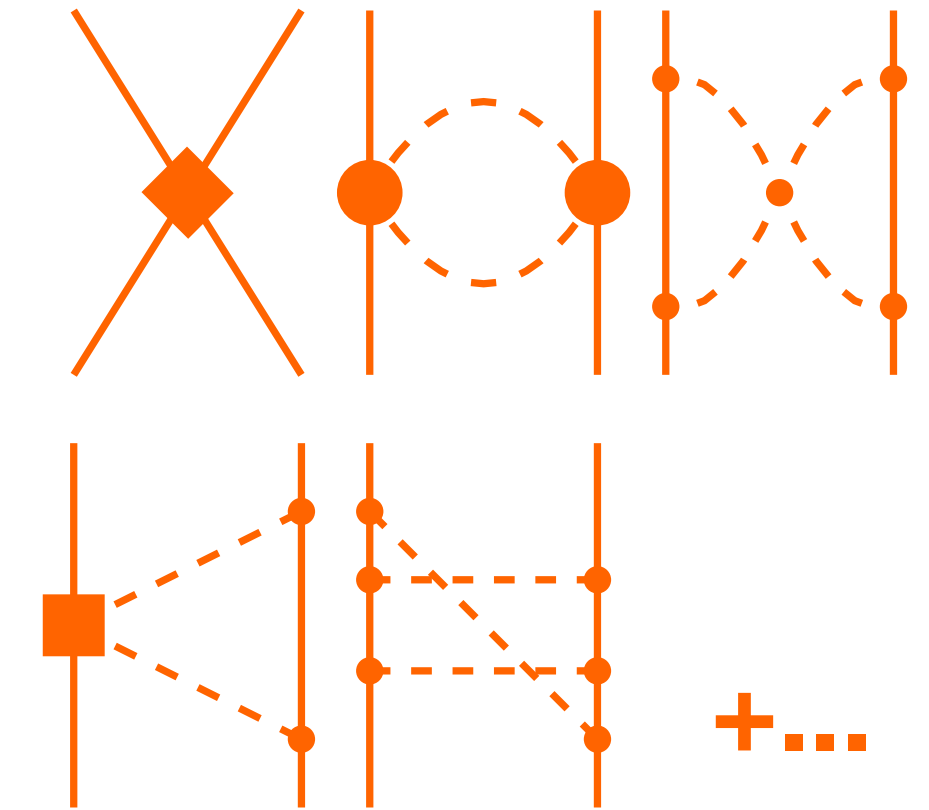
N³LO potential

Machleidt & Entem, PR **503**, 1 (2011)

$$\hat{V}_{2N}^{(\text{N}^3\text{LO})} = \hat{V}_{2\pi}^{(4)} + \hat{V}_{\text{ct}}^{(4)}$$

$$\begin{aligned} v_{\text{ct}}^{(4)}(\mathbf{p}', \mathbf{p}) = & D_1 q^4 + D_2 Q^4 + D_3 q^2 Q^2 + D_4 (\mathbf{q} \times \mathbf{Q})^2 \\ & + \left[D_5 q^4 + D_6 Q^4 + D_7 q^2 Q^2 + D_8 (\mathbf{q} \times \mathbf{Q})^2 \right] \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 \\ & + [D_9 q^2 + D_{10} Q^2] [-i\mathbf{S} \cdot (\mathbf{q} \times \mathbf{Q})] \\ & + [D_{11} q^2 + D_{12} Q^2] (\boldsymbol{\sigma}_1 \cdot \mathbf{q}) (\boldsymbol{\sigma}_2 \cdot \mathbf{q}) \\ & + [D_{13} q^2 + D_{14} Q^2] (\boldsymbol{\sigma}_1 \cdot \mathbf{Q}) (\boldsymbol{\sigma}_2 \cdot \mathbf{Q}) \\ & + D_{15} [\boldsymbol{\sigma}_1 \cdot (\mathbf{q} \times \mathbf{Q})] [\boldsymbol{\sigma}_2 \cdot (\mathbf{q} \times \mathbf{Q})] \end{aligned}$$

N³LO
(Q/Λ_χ)⁴



Central + SO + Tensor

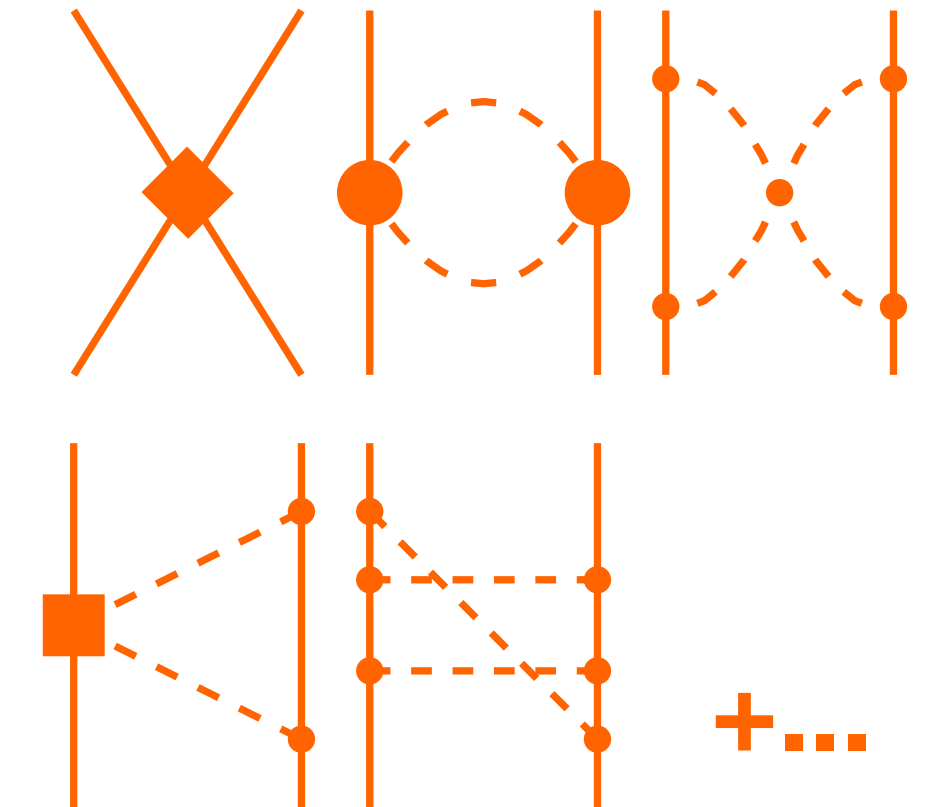
N³LO potential

Machleidt & Entem, PR **503**, 1 (2011)

$$\hat{V}_{2N}^{(\text{N}^3\text{LO})} = \hat{V}_{2\pi}^{(4)} + \hat{V}_{\text{ct}}^{(4)}$$

$$\begin{aligned} v_{2\pi}^{(4)}(\mathbf{p}', \mathbf{p}) = & V_C^{(c_i^2)}(q, Q) + W_S^{(c_i^2)}(q, Q) (\boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2) (\boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2) + W_T^{(c_i^2)}(q, Q) (\boldsymbol{\sigma}_1 \cdot \mathbf{q}) (\boldsymbol{\sigma}_2 \cdot \mathbf{q}) (\boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2) \\ & + V_C^{(c_i/m_N)}(q, Q) + W_C^{(c_i/m_N)}(q, Q) \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 + W_S^{(c_i/m_N)}(q, Q) (\boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2) (\boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2) \\ & + \left[V_{LS}^{(c_i/m_N)}(q, Q) + W_{LS}^{(c_i/m_N)}(q, Q) \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 \right] [-i\mathbf{S} \cdot (\mathbf{q} \times \mathbf{Q})] \\ & + W_T^{(c_i/m_N)}(q, Q) (\boldsymbol{\sigma}_1 \cdot \mathbf{q}) (\boldsymbol{\sigma}_2 \cdot \mathbf{q}) (\boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2) \\ & + V_C^{(m_N^{-2})}(q, Q) + W_C^{(m_N^{-2})}(q, Q) \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 \\ & + \left[V_S^{(m_N^{-2})}(q, Q) + W_S^{(m_N^{-2})}(q, Q) \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 \right] \boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2 \\ & + \left[V_{LS}^{(m_N^{-2})}(q, Q) + W_{LS}^{(m_N^{-2})}(q, Q) \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 \right] [-i\mathbf{S} \cdot (\mathbf{q} \times \mathbf{Q})] \\ & + \left[V_T^{(m_N^{-2})}(q, Q) + W_T^{(m_N^{-2})}(q, Q) \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 \right] (\boldsymbol{\sigma}_1 \cdot \mathbf{q}) (\boldsymbol{\sigma}_2 \cdot \mathbf{q}) \\ & + V_{\sigma L}^{(m_N^{-2})}(q, Q) [\boldsymbol{\sigma}_1 \cdot (\mathbf{q} \times \mathbf{Q})] [\boldsymbol{\sigma}_2 \cdot (\mathbf{q} \times \mathbf{Q})] \\ & + V_C^{(2L)}(q, Q) + W_C^{(2L)}(q, Q) \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 \\ & + \left[V_S^{(2L)}(q, Q) + W_S^{(2L)}(q, Q) \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 \right] (\boldsymbol{\sigma}_1 \cdot \boldsymbol{\sigma}_2) \\ & + \left[V_T^{(2L)}(q, Q) + W_T^{(2L)}(q, Q) \boldsymbol{\tau}_1 \cdot \boldsymbol{\tau}_2 \right] (\boldsymbol{\sigma}_1 \cdot \mathbf{q}) (\boldsymbol{\sigma}_2 \cdot \mathbf{q}) \end{aligned}$$

$$\text{N}^3\text{LO} \\ (Q/\Lambda_\chi)^4$$

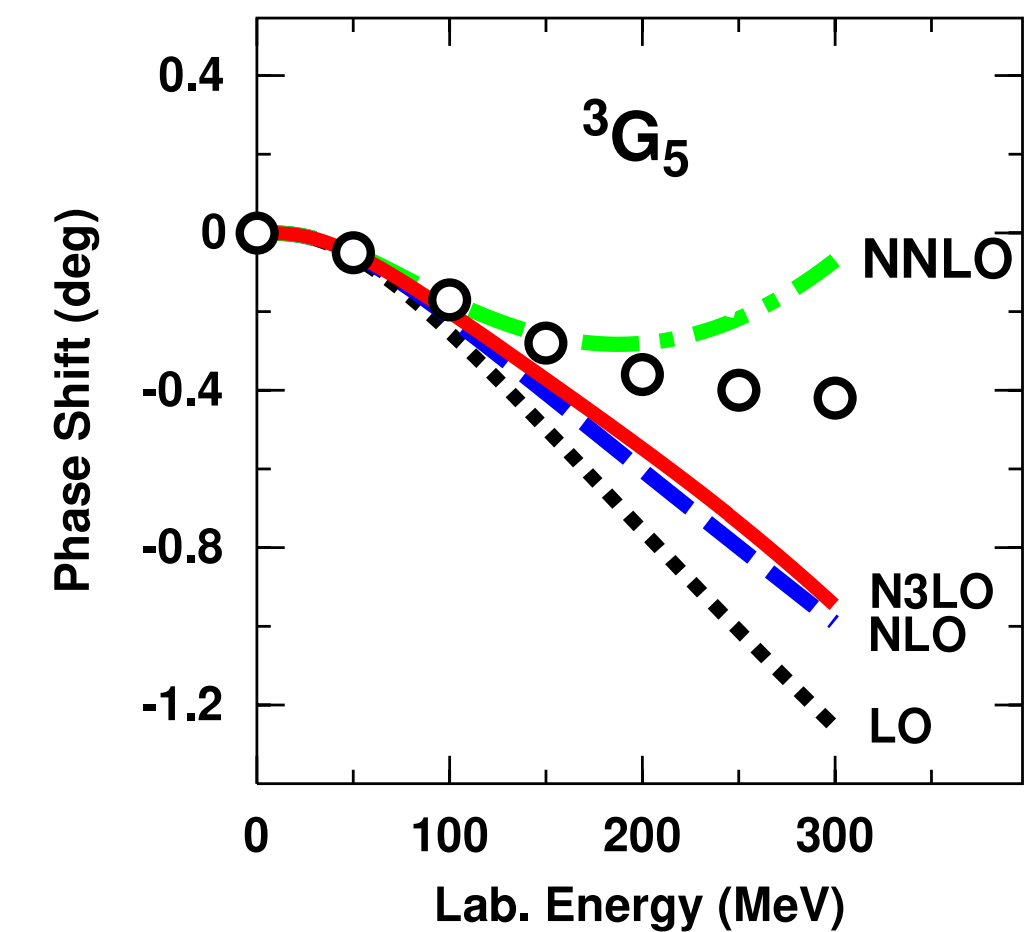
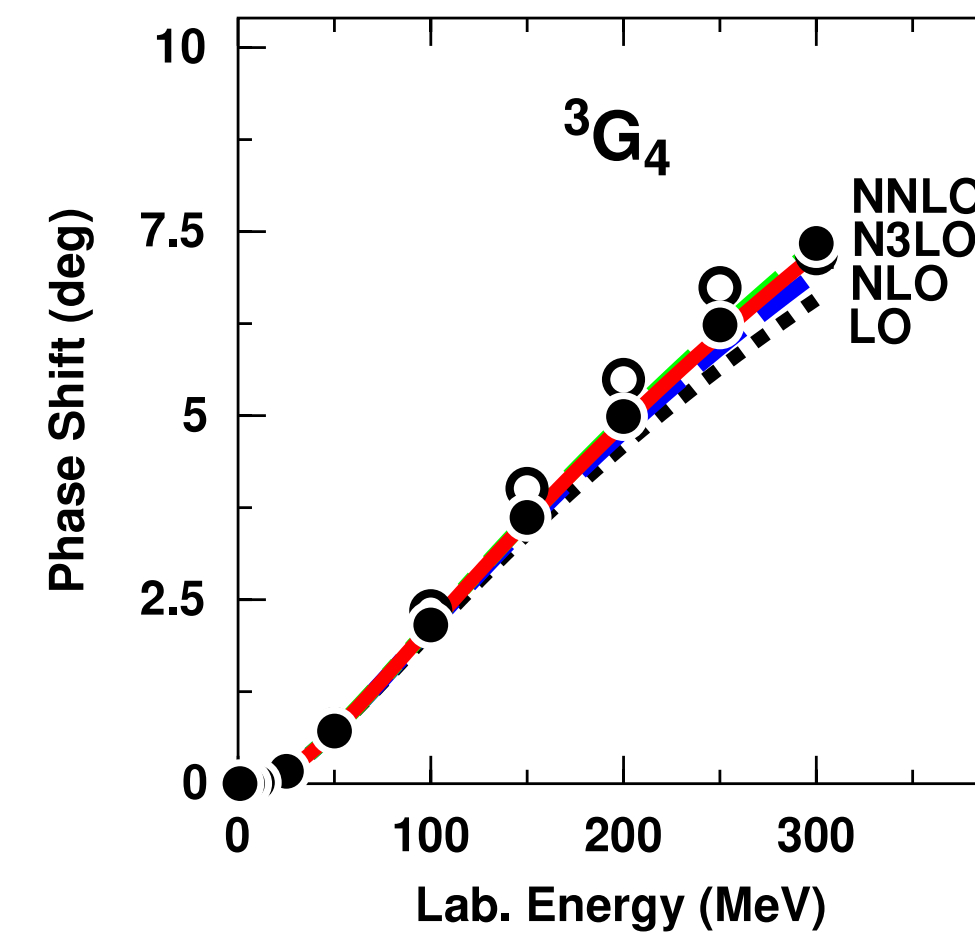
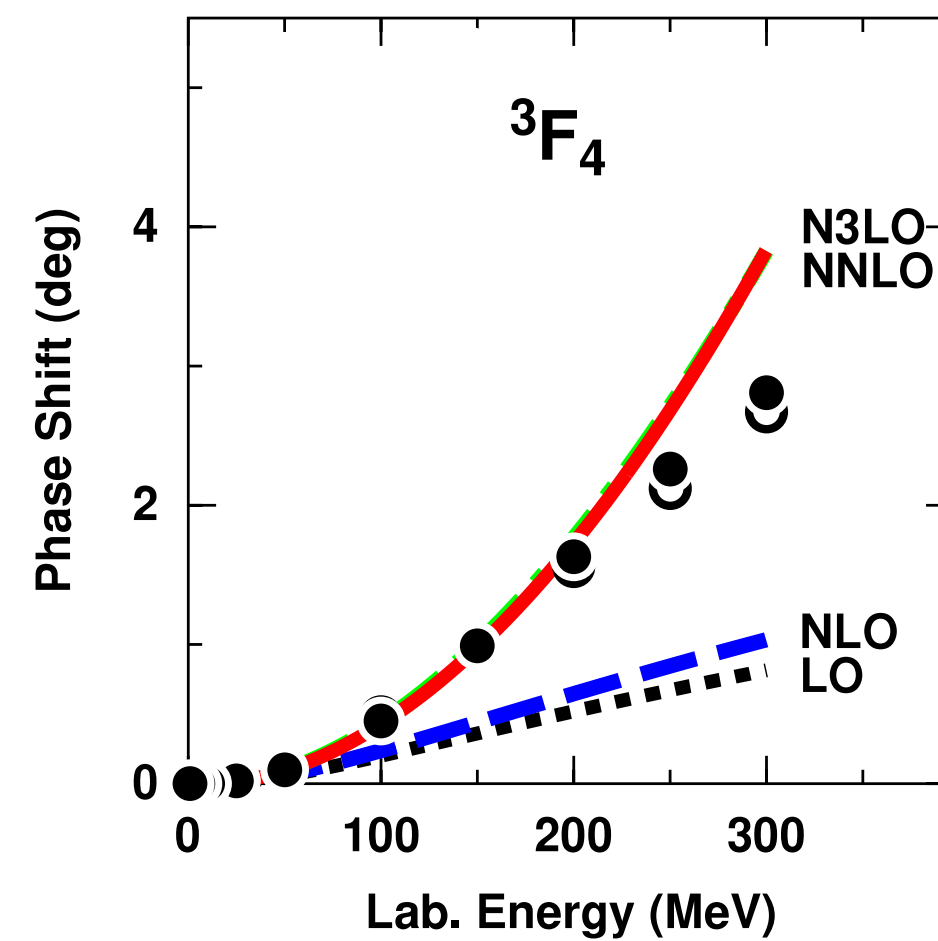
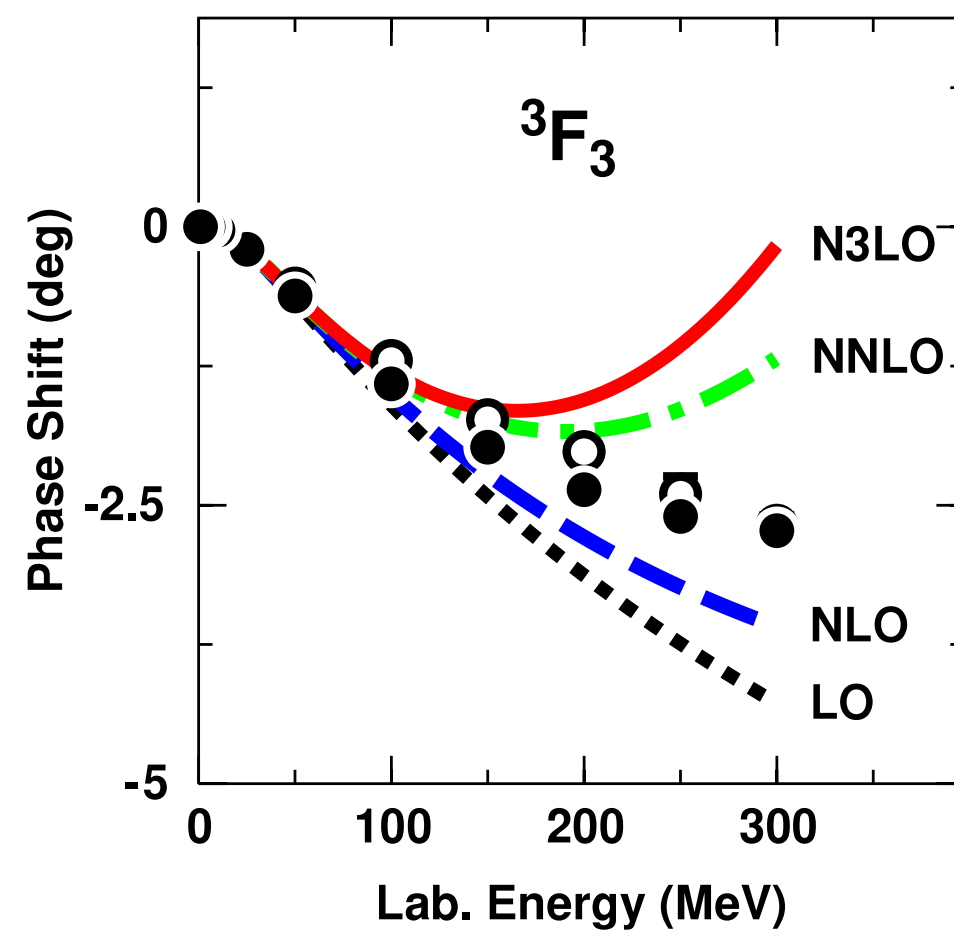
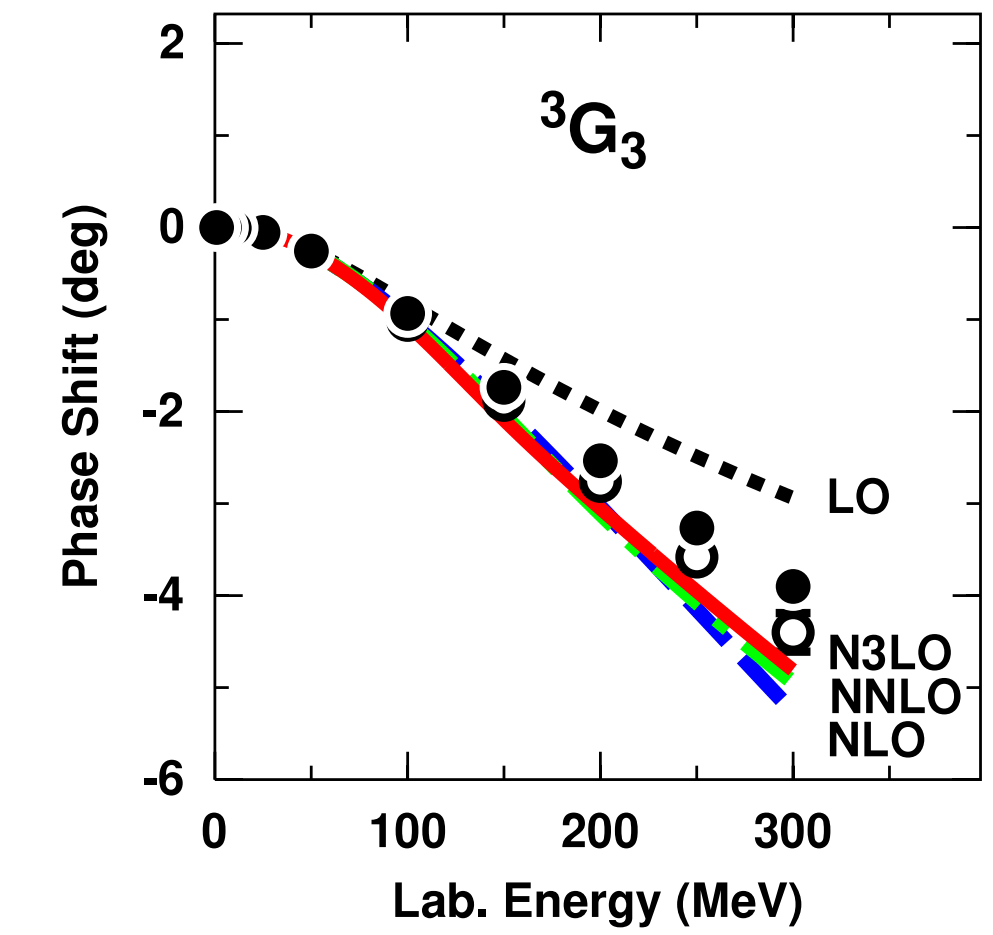
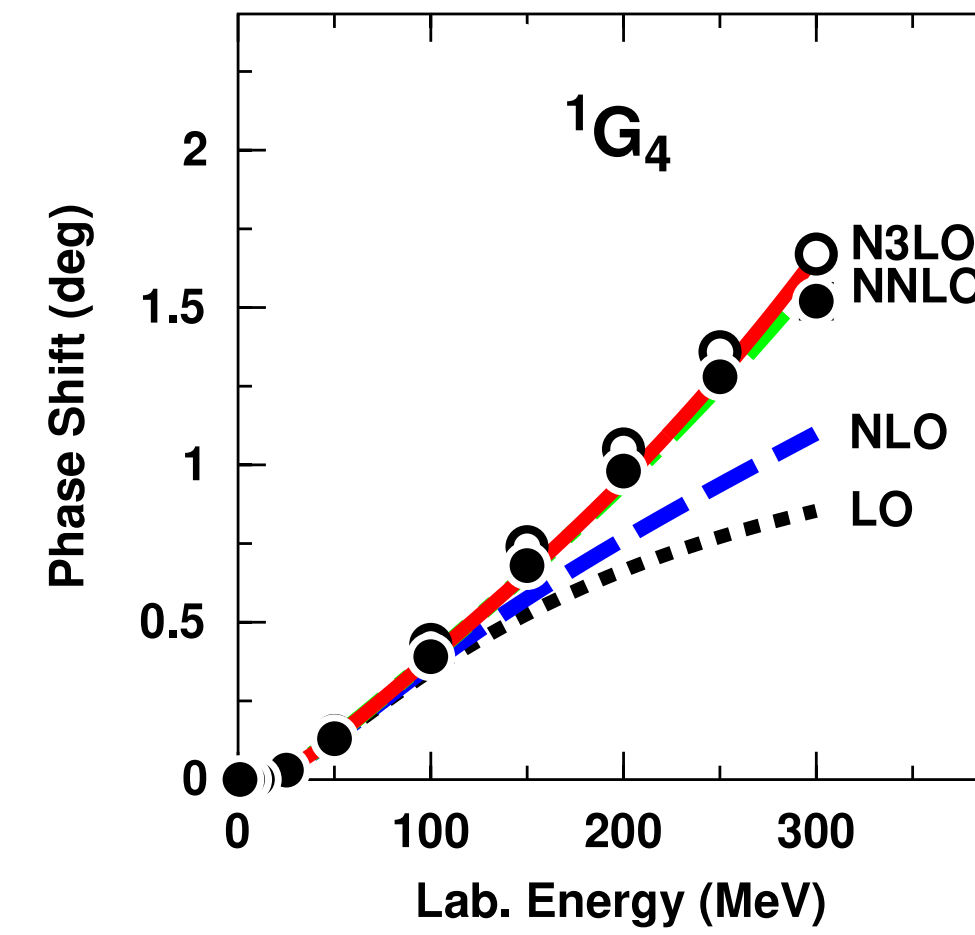
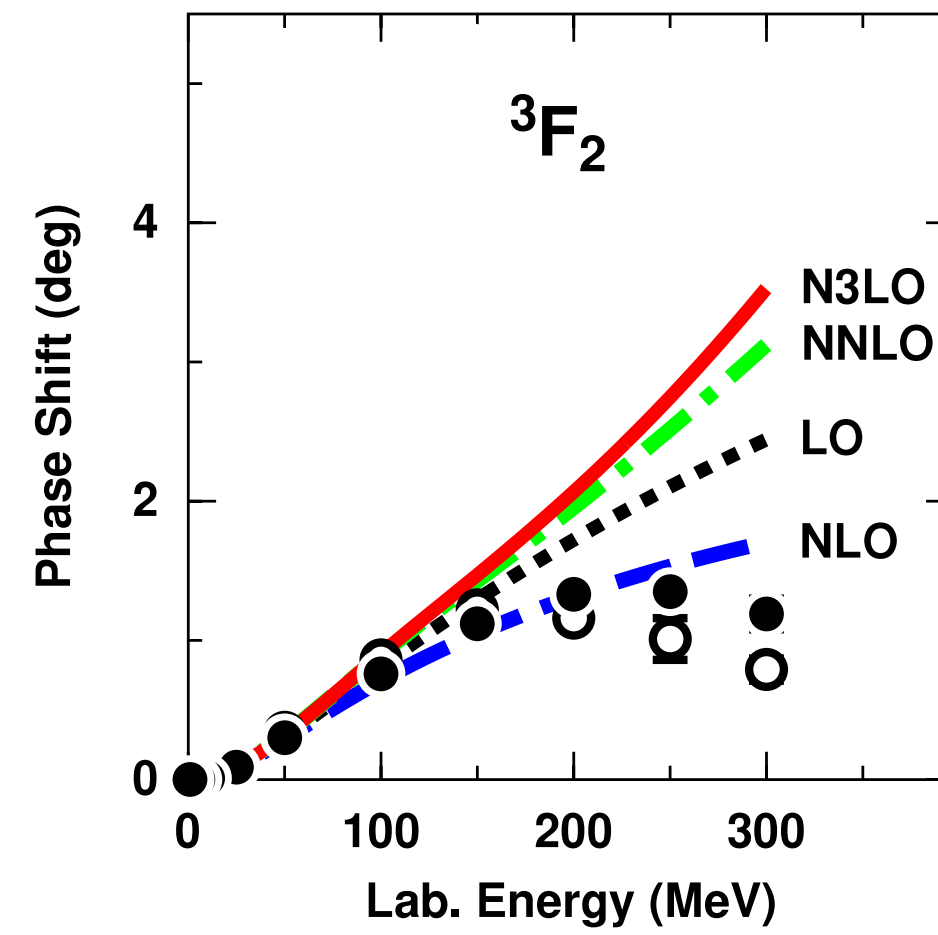
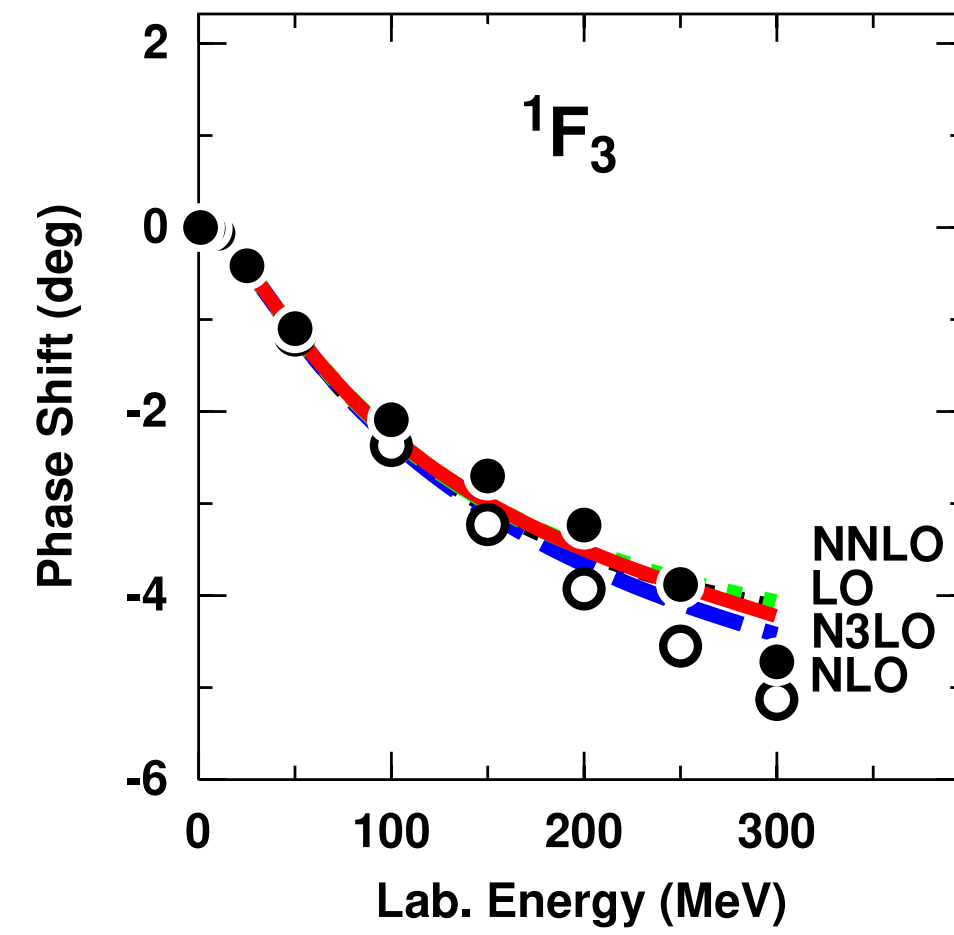


Central + SO + Tensor

NN-phase shift by chiral EFT at N³LO

Peripheral interaction ($1\pi + 2\pi$)

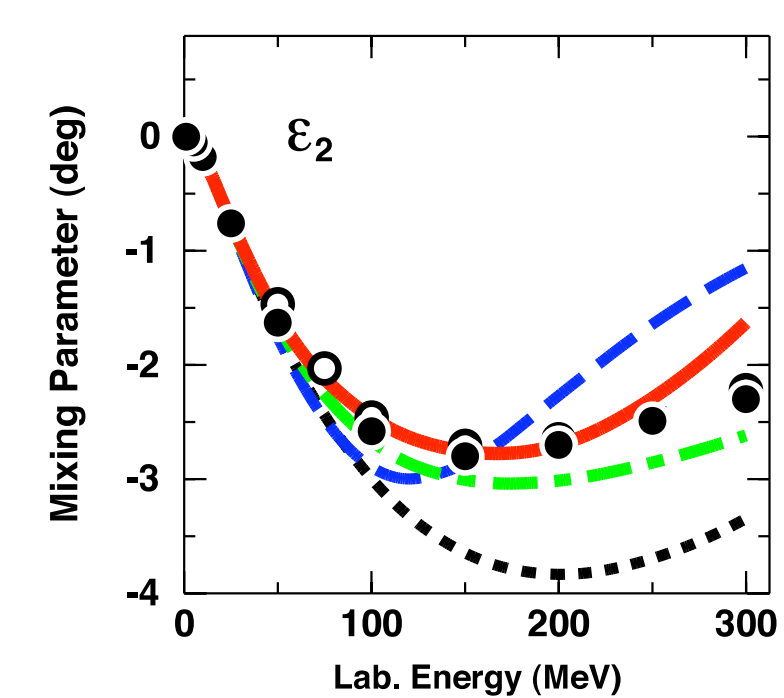
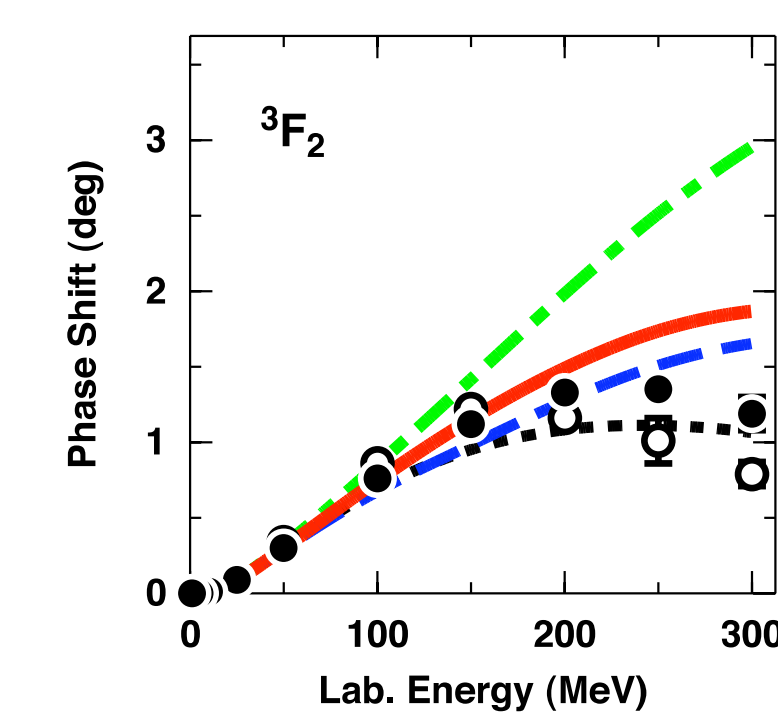
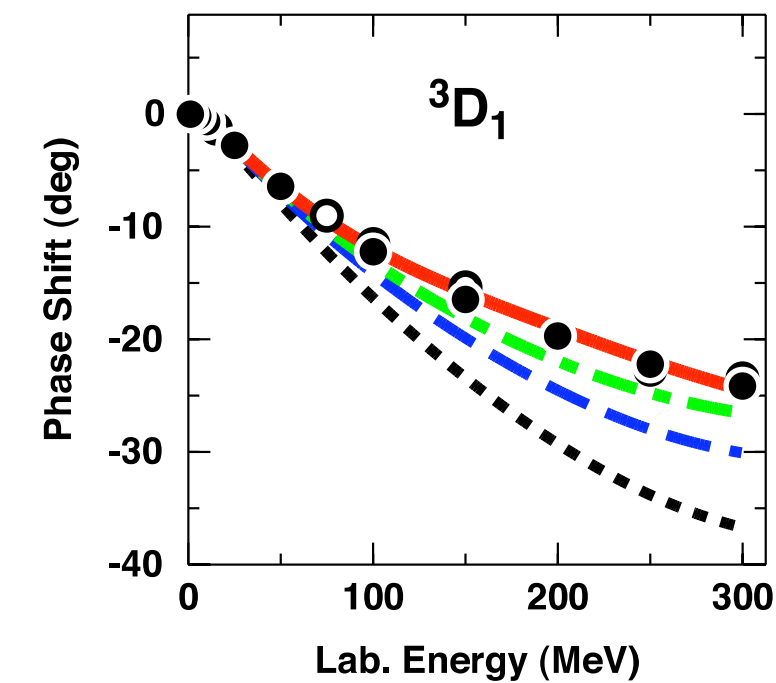
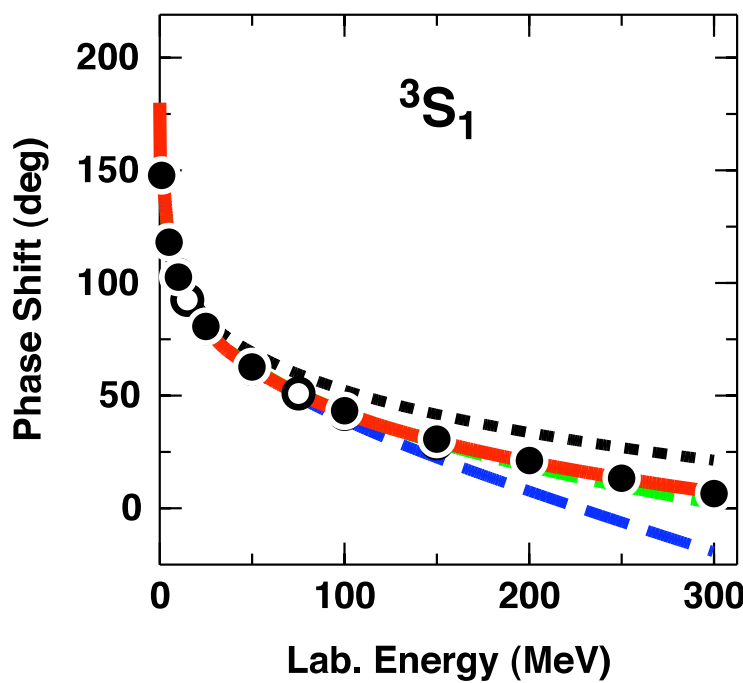
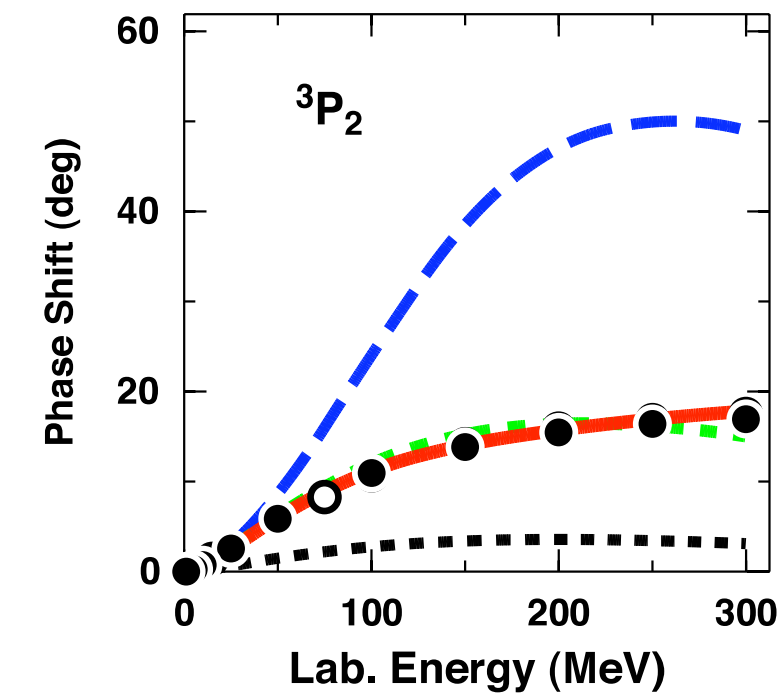
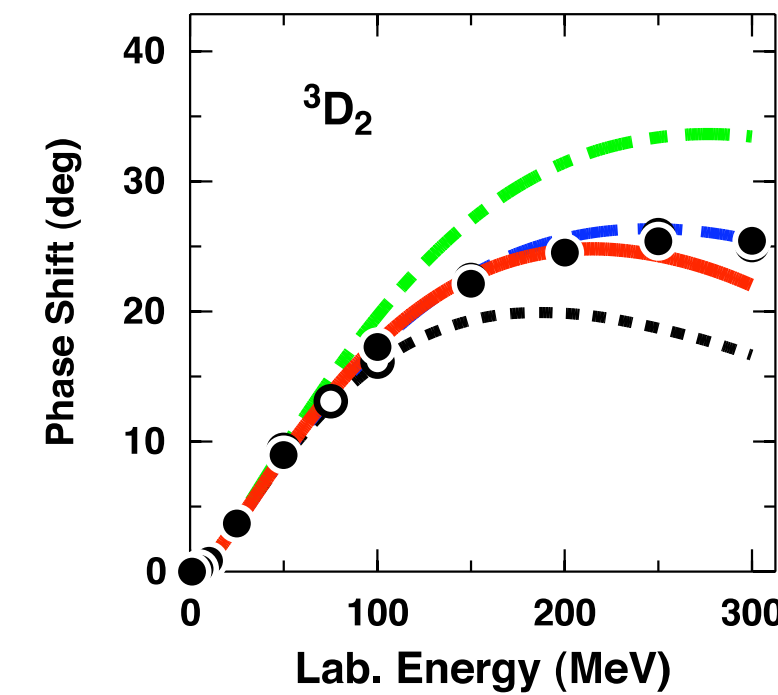
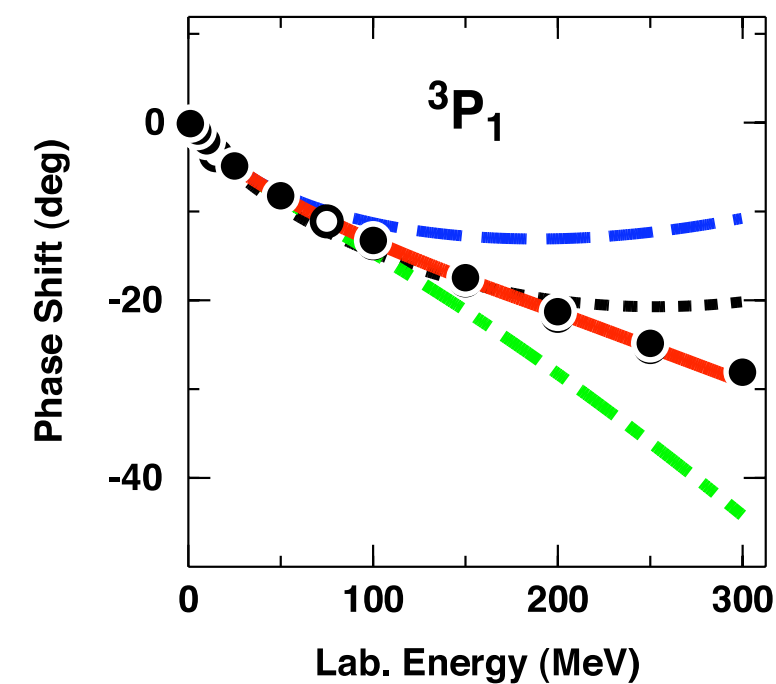
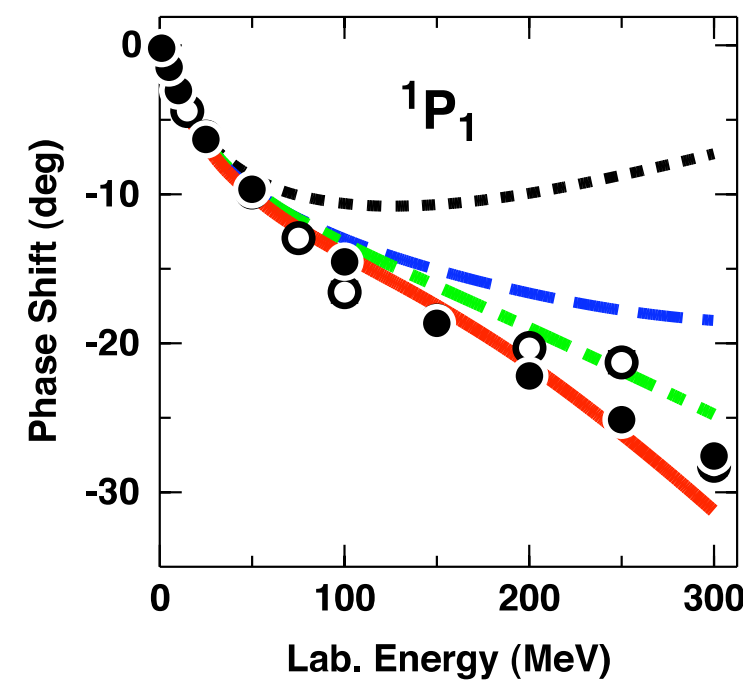
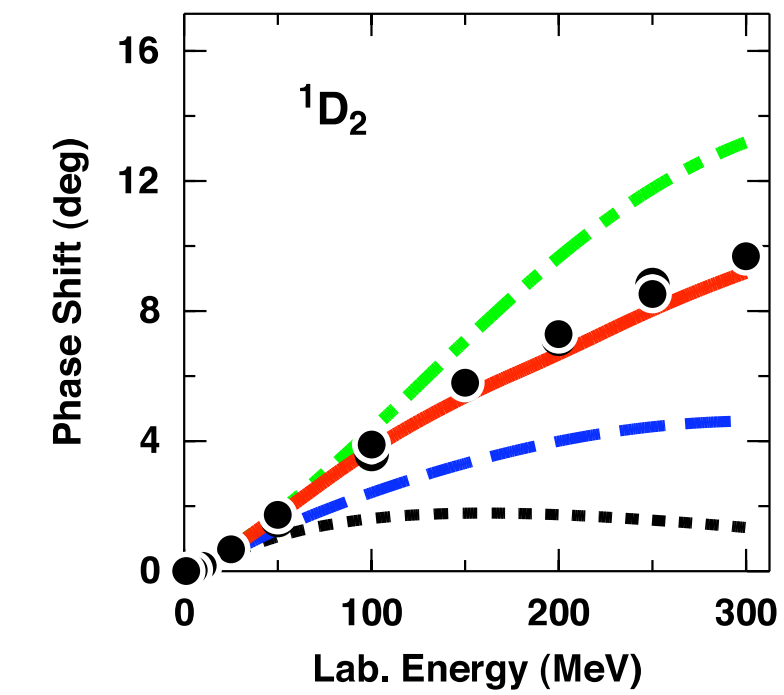
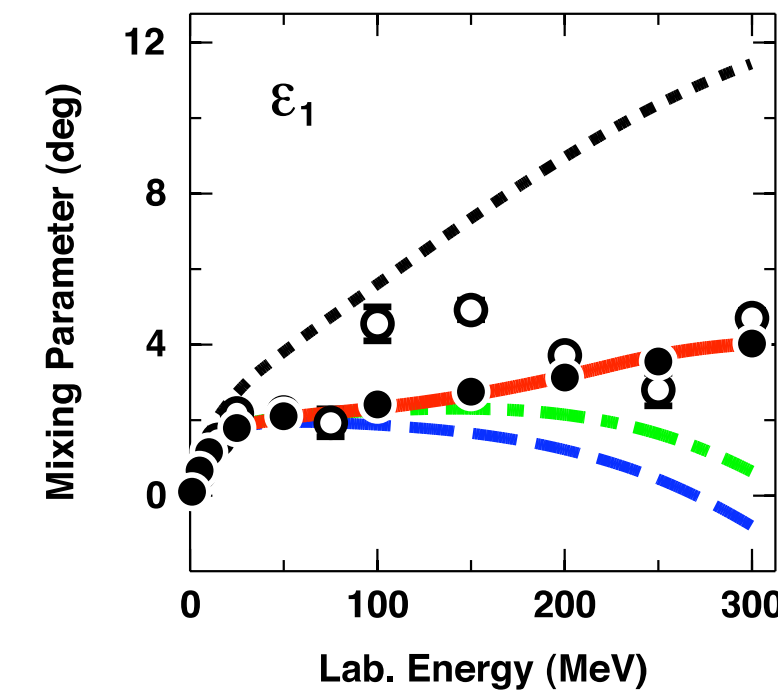
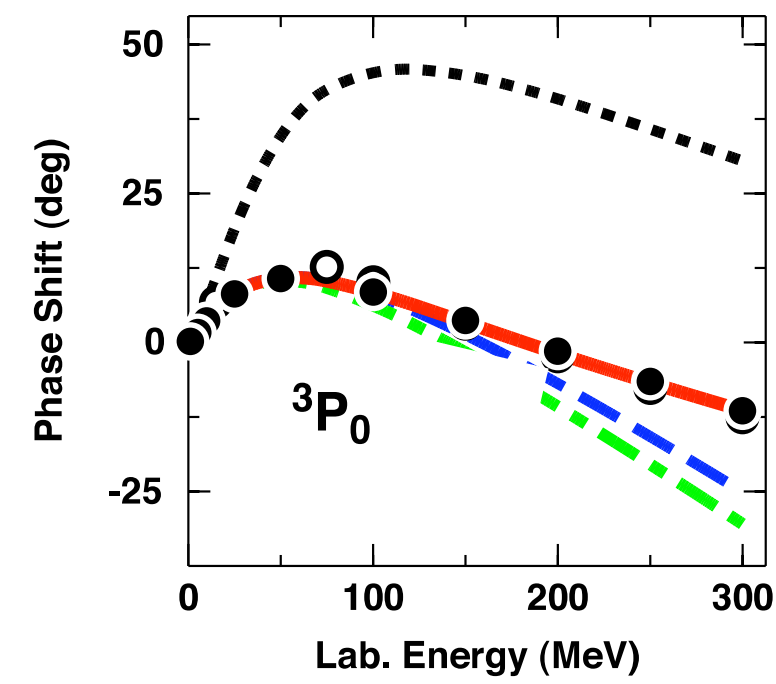
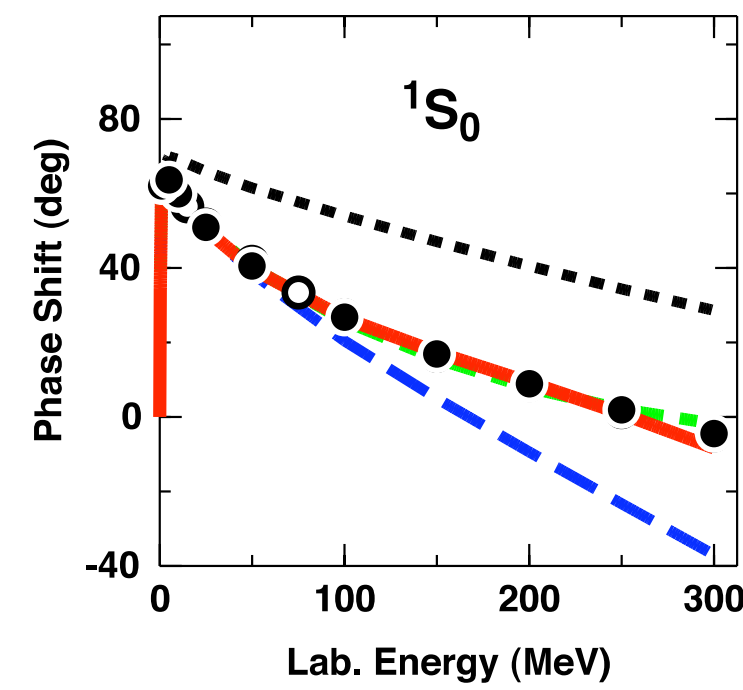
Machleidt & Entem, PR 503, 1 (2011)



NN -phase shift by chiral EFT at $N^3\text{LO}$

Full interaction (peripheral + contact)

Machleidt & Entem, PR 503, 1 (2011)



NN -phase shift by chiral EFT at N^4 LO

Order-by-order convergence

Entem +, PRC **96**, 024004 (2017)

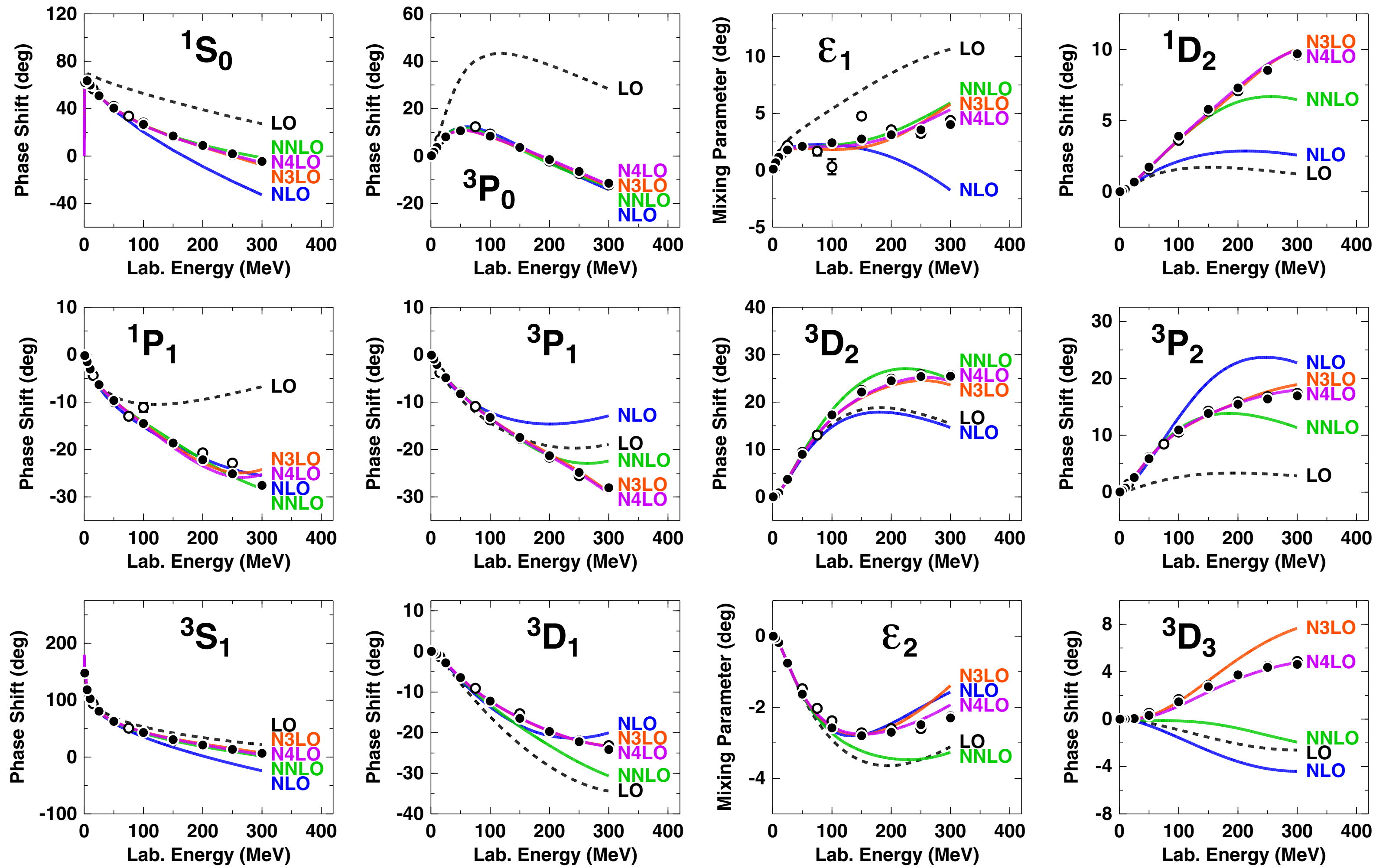
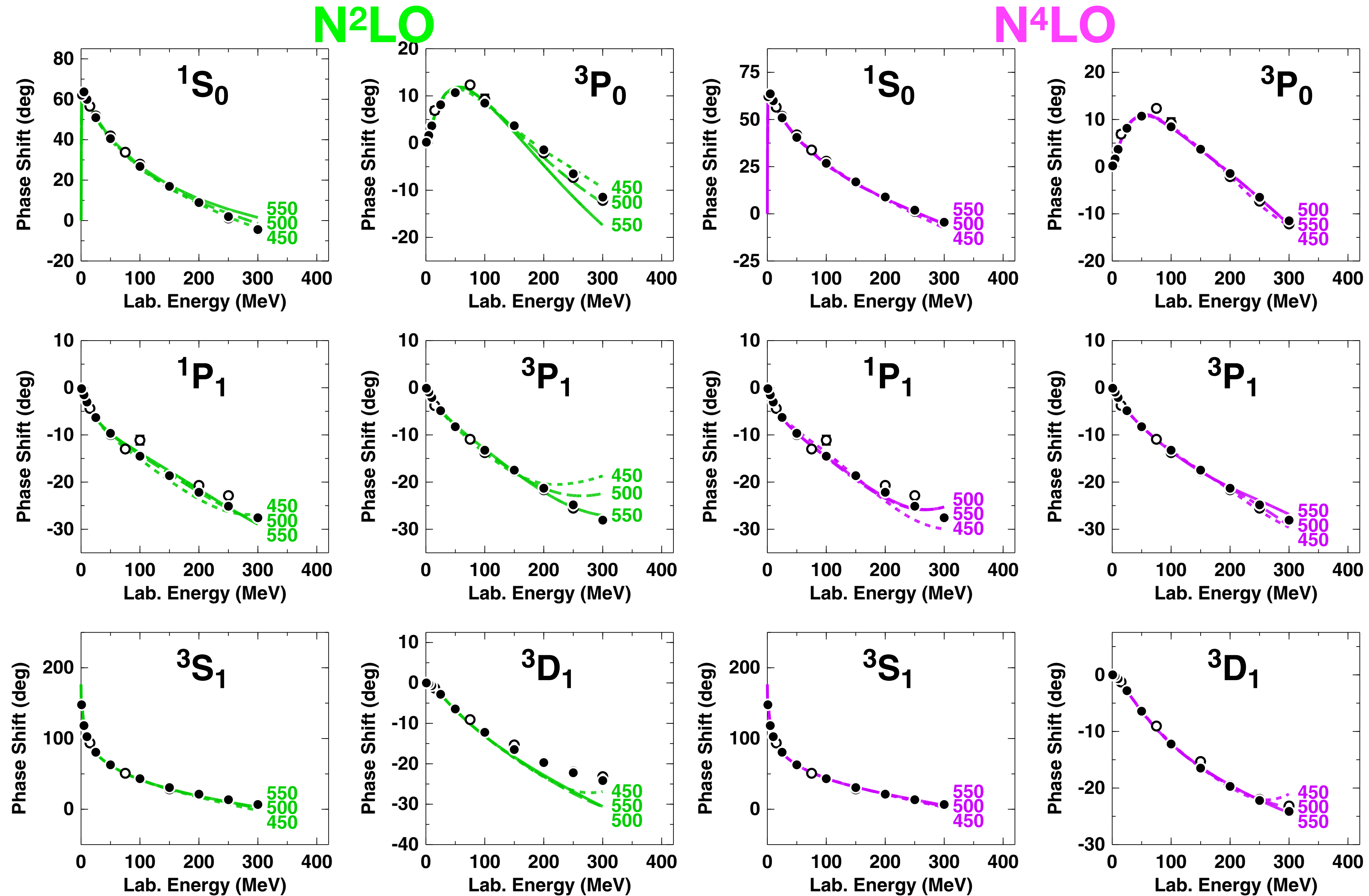


FIG. 2. Chiral expansion of neutron-proton scattering as represented by the phase shifts in S , P , and D waves and mixing parameters ϵ_1 and ϵ_2 . Five orders ranging from LO to N^4 LO are shown as denoted. A cutoff $\Lambda = 500$ MeV is applied in all cases. The filled and open circles represent the results from the Nijmegen multienergy np phase-shift analysis [80] and the GWU single-energy np analysis SP07 [102], respectively.

NN -phase shift by chiral EFT at N^4LO

Cutoff dependence

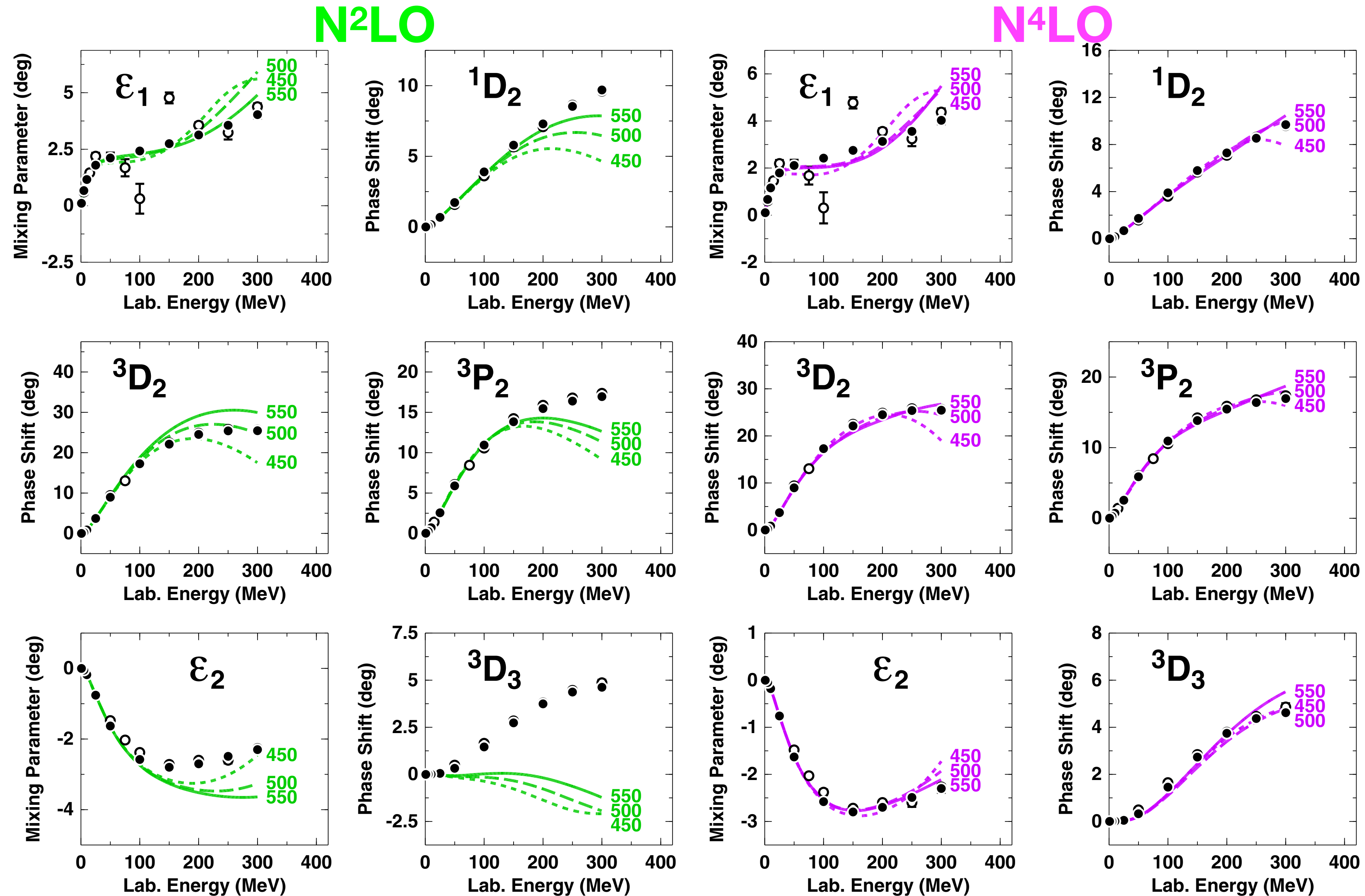
Entem +, PRC **96**, 024004 (2017)



NN -phase shift by chiral EFT at N^4LO

Cutoff dependence

Entem +, PRC **96**, 024004 (2017)



NN-phase shift by chiral EFT at N⁴LO

χ^2 and summary

Machleidt & Entem, PR **503**, 1 (2011)

Entem +, PRC **96**, 024004 (2017)

Columns three to five display the χ^2 /datum for the reproduction of the 1999 ***np* database** [173] (subdivided into energy intervals) by various *np* potentials. For the chiral potentials, the χ^2 /datum is stated in terms of ranges which result from a variation of the cutoff parameters used in the regulator functions. The values of these cutoff parameters in units of MeV are given in parentheses. T_{lab} denotes the kinetic energy of the incident nucleon in the laboratory system.

T_{lab} bin (MeV)	# of np data	Idaho N ³ LO [68] (500–600)	Juelich N ³ LO [171] (600/700–450/500)	Argonne V_{18} [174]
0–100	1058	1.0–1.1	1.0–1.1	0.95
100–190	501	1.1–1.2	1.3–1.8	1.10
190–290	843	1.2–1.4	2.8–20.0	1.11
0–290	2402	1.1–1.3	1.7–7.9	1.04

TABLE VIII. χ^2 /datum for the fit of the *pp* plus *np* data up to 190 MeV and two- and three-nucleon bound-state properties as produced by *NN* potentials at NNLO and N⁴LO applying different values for the cutoff parameter Λ of the regulator function Eq. (2.43). For some of the notation, see Table VII, where also empirical information on the deuteron and triton can be found.

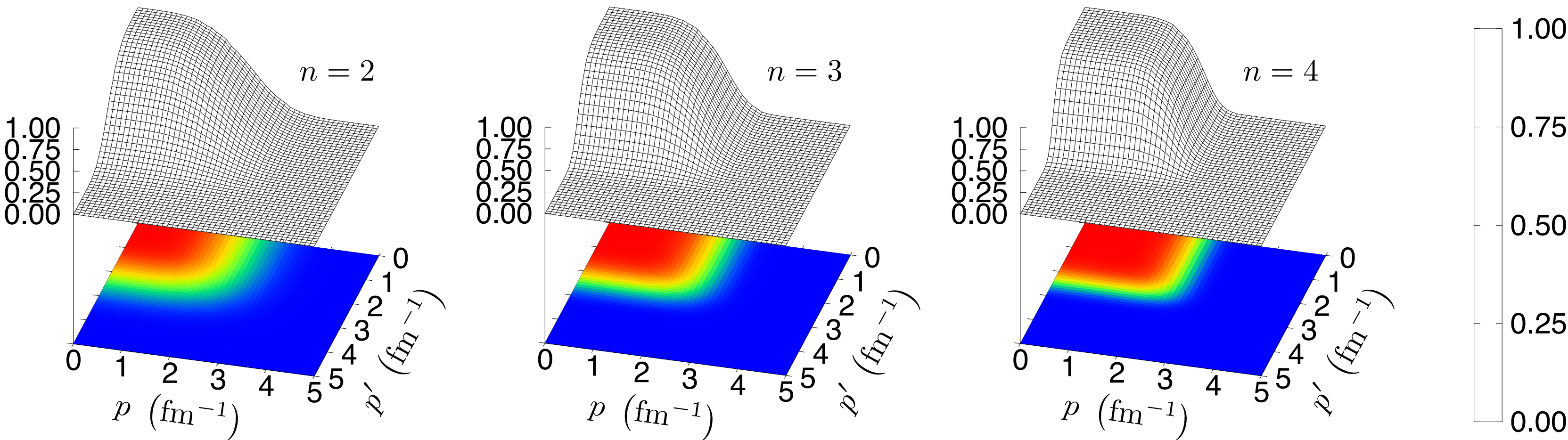
Λ (MeV)	NNLO			N ⁴ LO		
	450	500	550	450	500	550
χ^2 /datum <i>pp</i> and <i>np</i>						
0–190 MeV (2903 data)	4.12	3.27	3.32	1.17	1.08	1.25
Deuteron						
B_d (MeV)	2.224575	2.224575	2.224575	2.224575	2.224575	2.224575
A_S (fm ^{−1/2})	0.8847	0.8844	0.8843	0.8852	0.8852	0.8851
η	0.0255	0.0257	0.0258	0.0254	0.0258	0.0257
r_{str} (fm)	1.967	1.968	1.968	1.966	1.973	1.971
Q (fm ²)	0.269	0.273	0.275	0.269	0.273	0.271
P_D (%)	3.95	4.49	4.87	4.38	4.10	4.13
Triton						
B_t (MeV)	8.35	8.21	8.10	8.04	8.08	8.12

Nonlocal regularization

$$v_{2N}(\mathbf{p}', \mathbf{p}) \rightarrow v_{2N}(\mathbf{p}', \mathbf{p}) u_n(p, p')$$
$$u_n(p, p') = \exp \left[- \left(\frac{p}{\Lambda_0} \right)^{2n} - \left(\frac{p'}{\Lambda_0} \right)^{2n} \right]$$

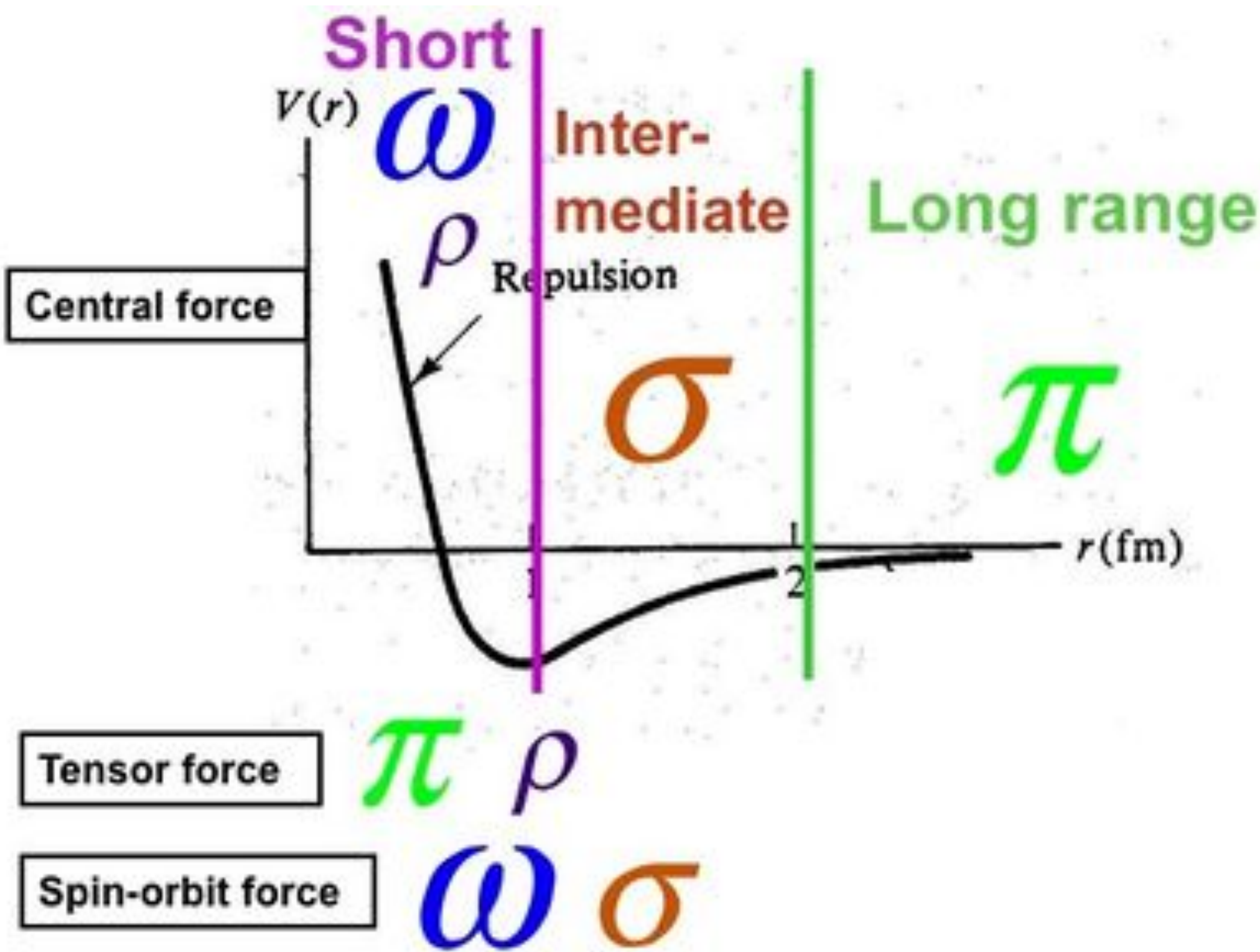
n	4 (One-pion exchange term at LO)
	3 (Contact terms at LO)
	2 (All other terms at NLO, N ² LO, and N ³ LO)
Λ_0	500 MeV

Machleidt & Entem, PR **503** 1 (2011)



Chiral EFT and meson theory

Machleidt, PRC 63, 024001 (2001)
scholarpedia.org/article/Nuclear_Forces



	Chiral EFT	Conventional meson theory
1PE		
2PE		
Short range		

$\frac{1}{m_\omega^2 + Q^2} \approx \frac{1}{m_\omega^2} \left[1 - \frac{Q^2}{m_\omega^2} + \frac{Q^4}{m_\omega^4} - \dots \right]$

$\approx$

Renormalization, power counting, and regularization

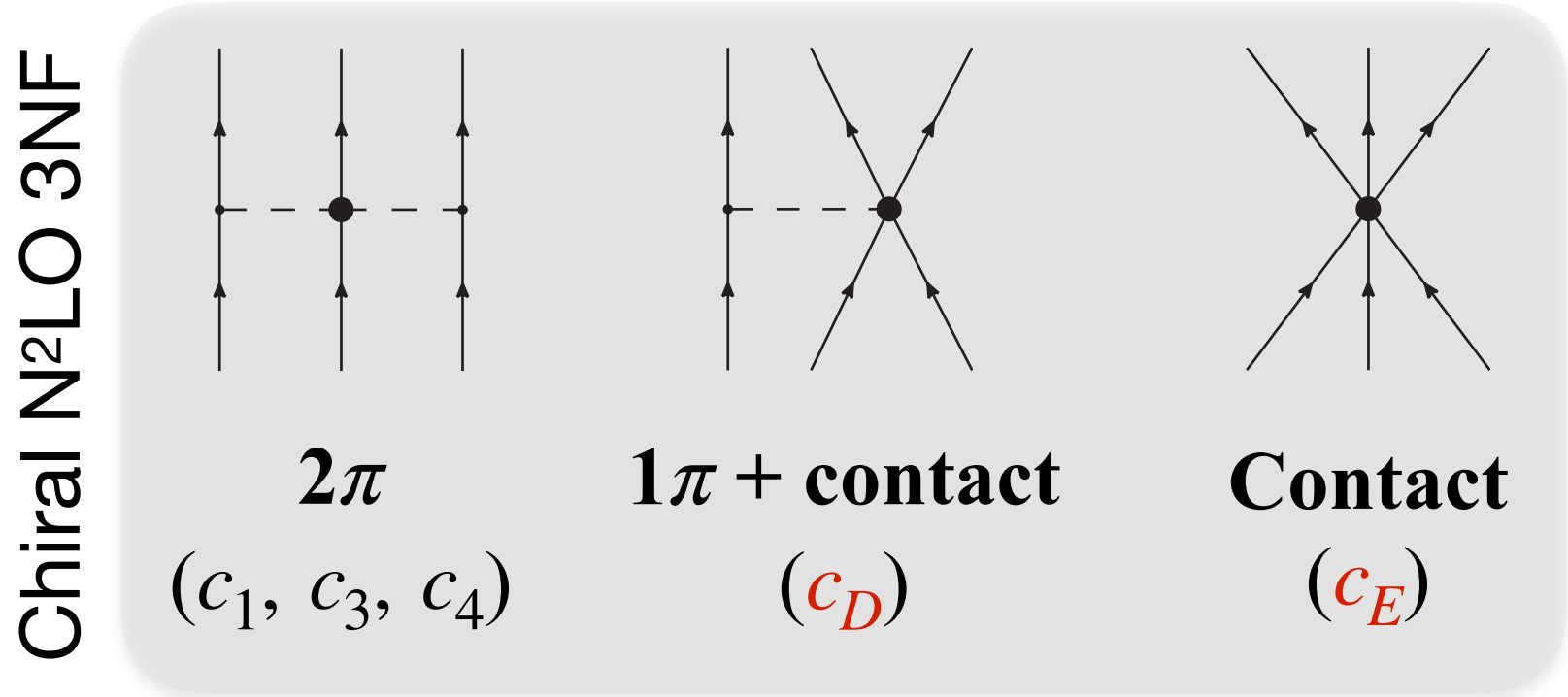
Weinberg power counting
with naive dimension analysis
→ Issues even at LO:

- Amplitudes and observables are not obtained
- Renormalization of 1π (nonperturbative)
- Lacks of contacts

→ Alternative power counting and/or soft cutoff?

Intensive works ongoing!

3NF LECs



The LECs, c_D and c_E , cannot be fixed from 2N observables.

→ Their fitting procedures are not well settled.

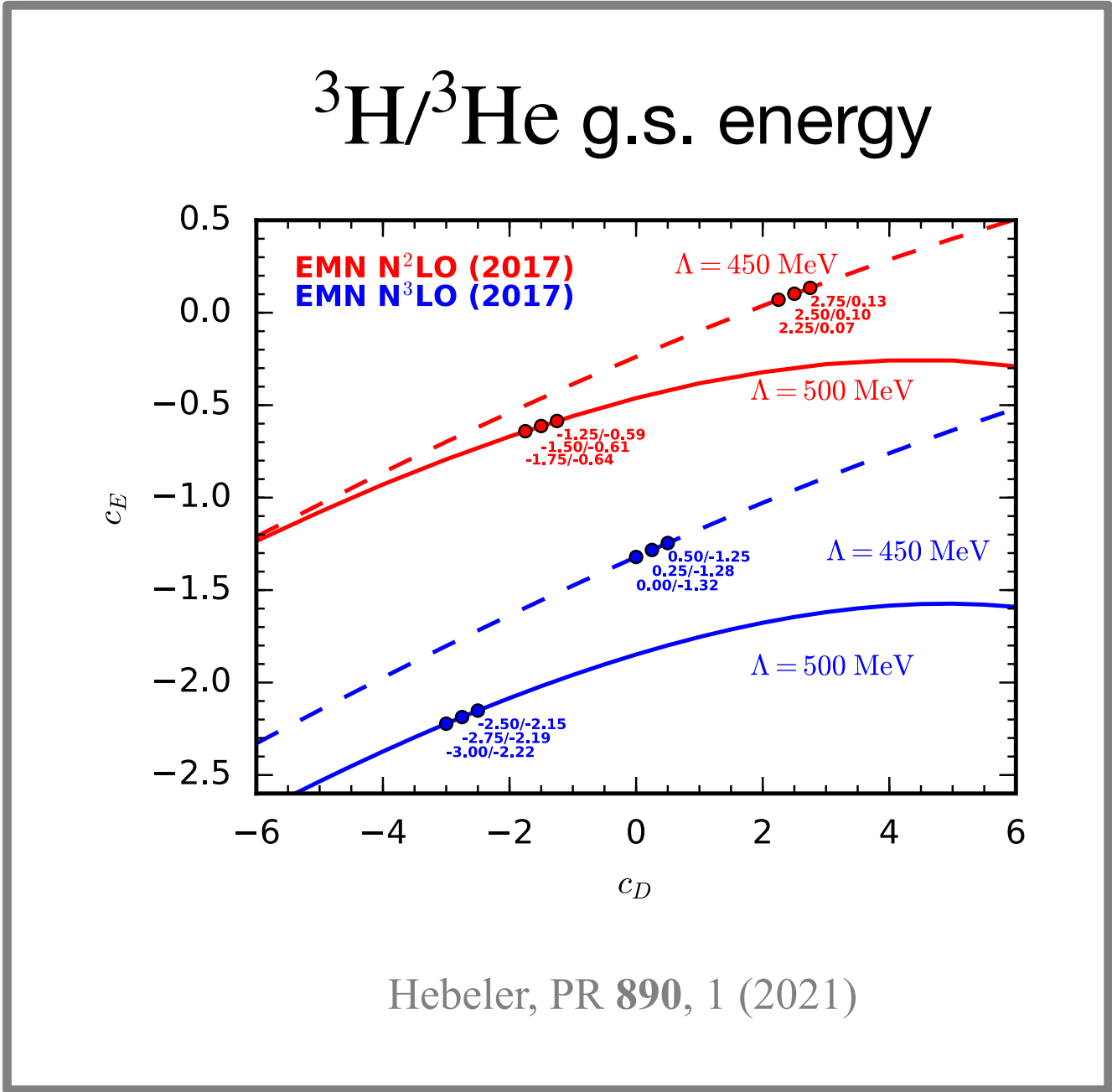
Phenomenologically succeeded!

Comments on fitting procedures of 3N-LECs

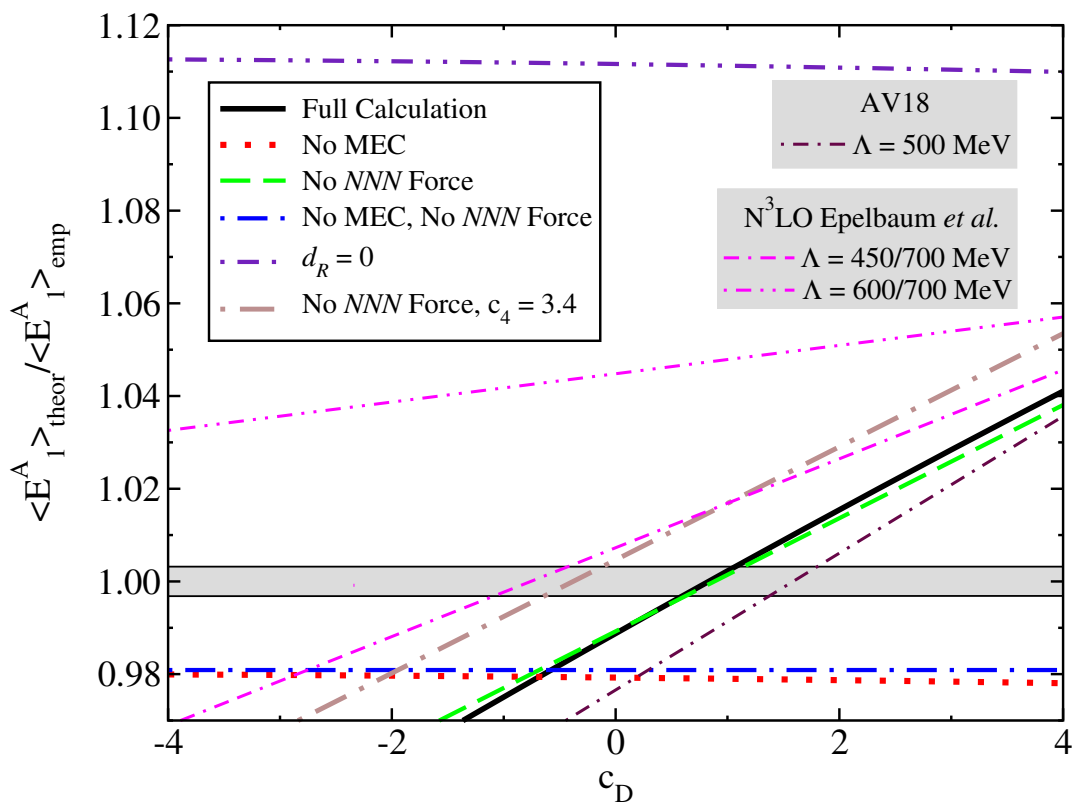
Approach 1: Fixed from purely 3-body systems

I personally believe this approach is appropriate.

Phillips, NPA 107, 209 (1968)



^3H -Gamow-Teller trans.

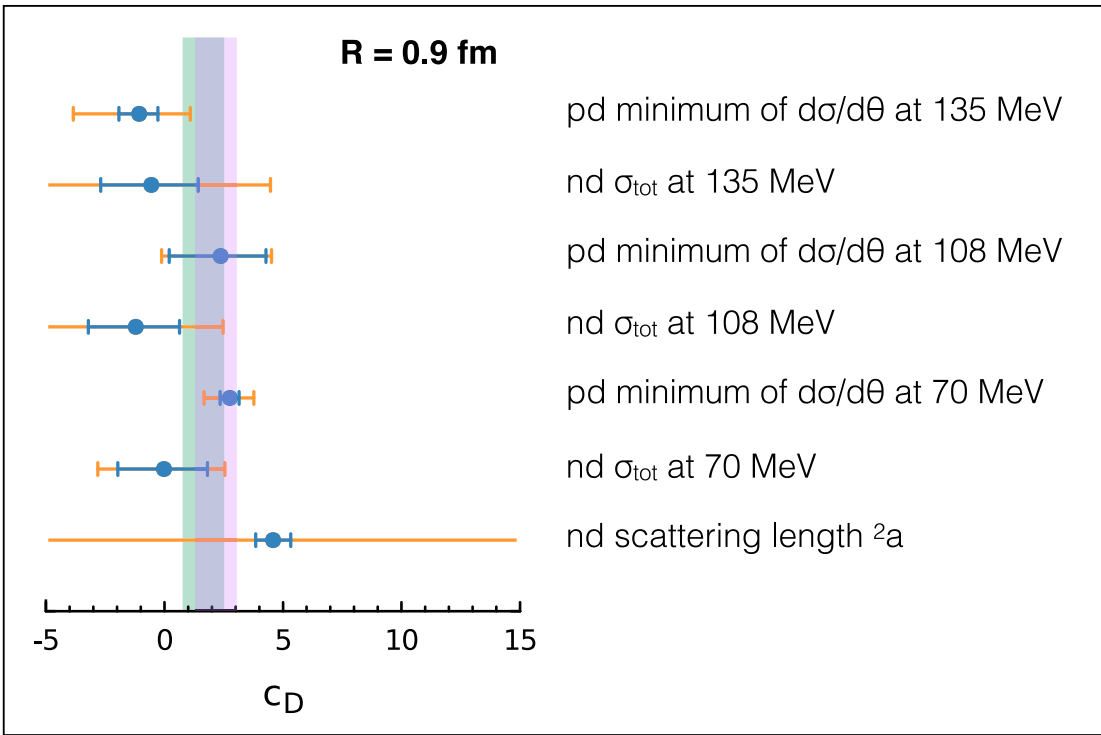


Gazit +, PRL 122, 029901(E) (2019)

+

OR

N - d scattering



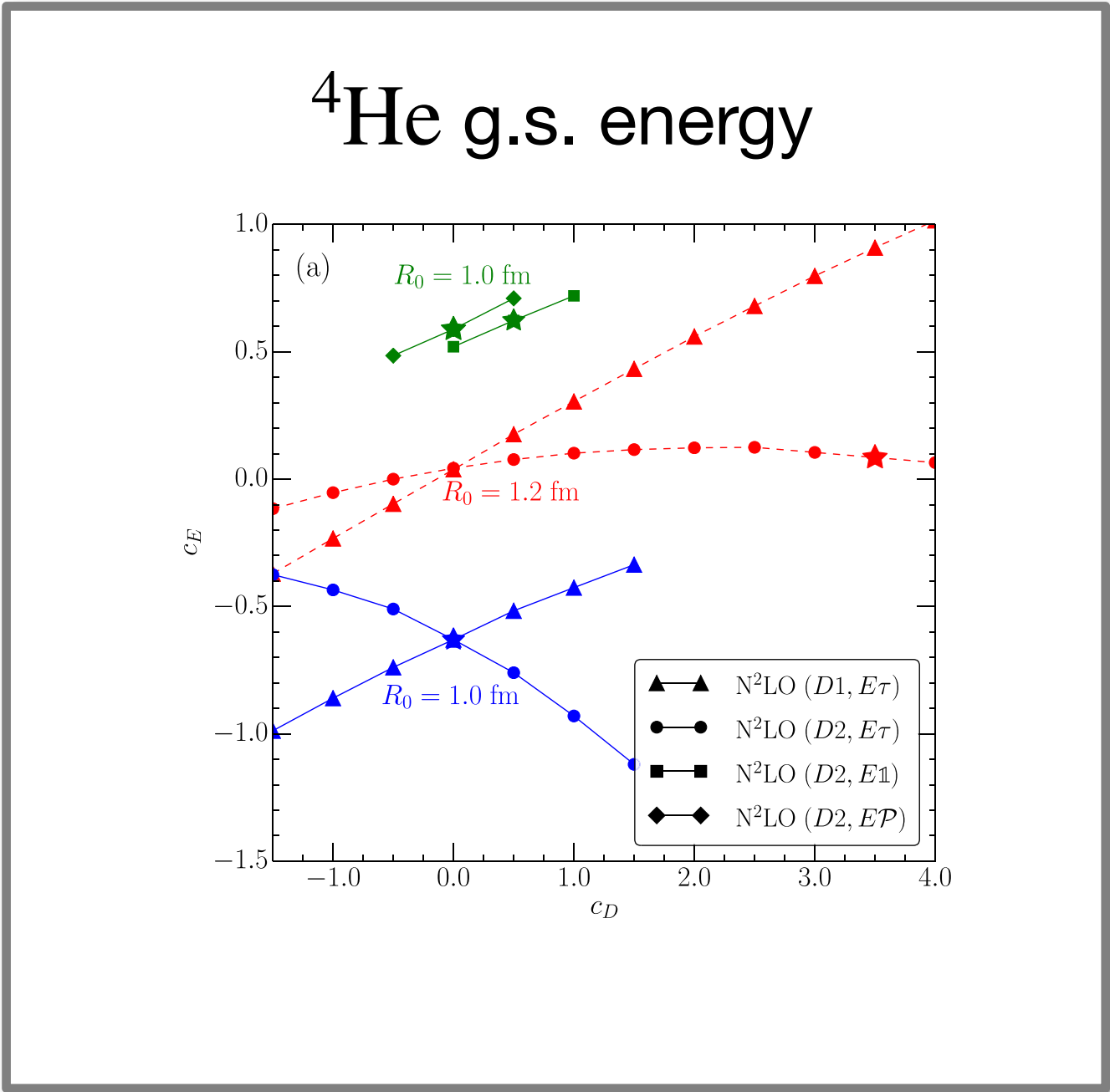
Epelbaum +, PRC 99, 024313 (2019)

Review

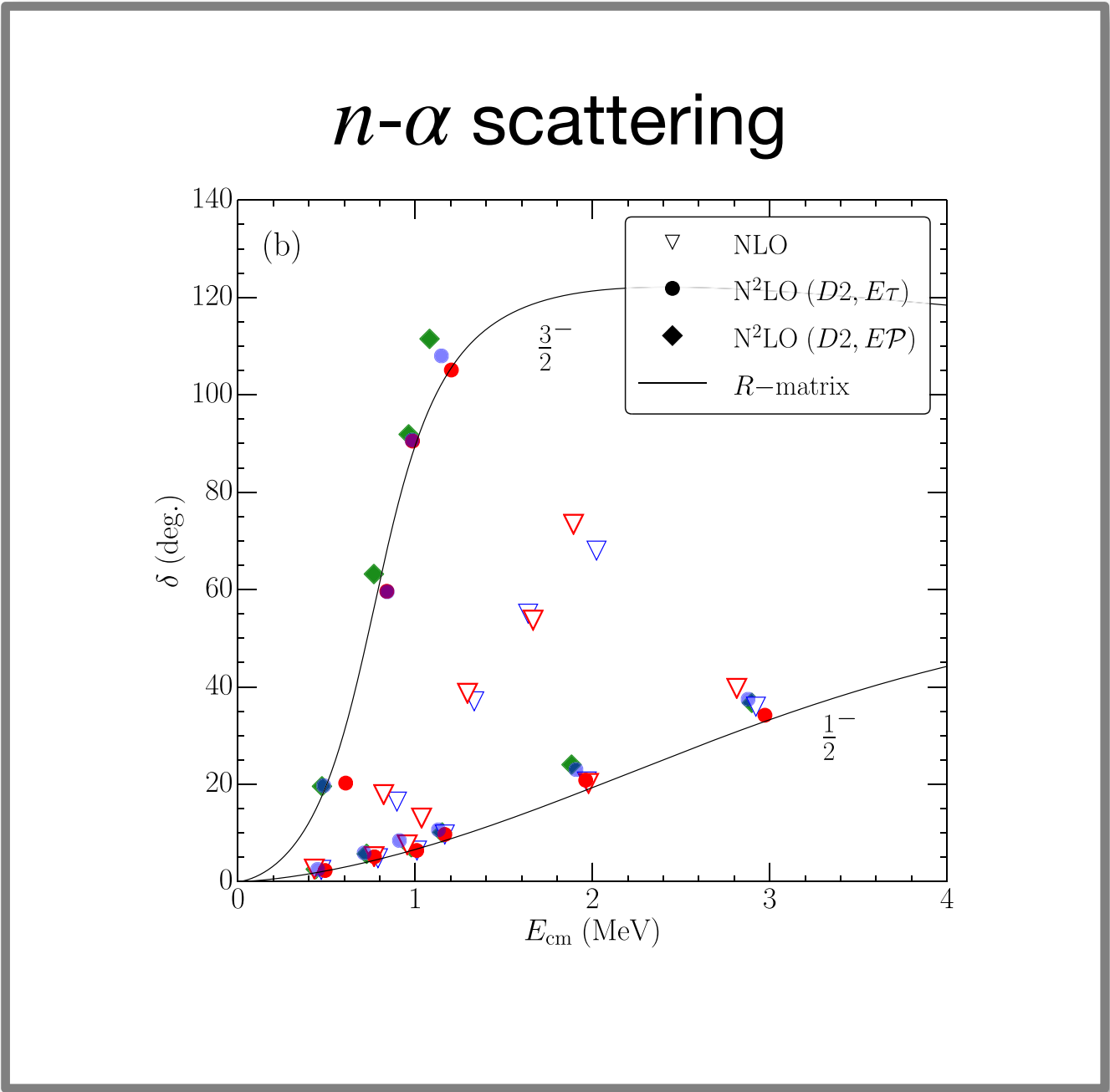
Hebeler, PR 890, 1 (2021)

Approach 2: Fixed from ^4He and/or heavier systems

Tjon, PLB **56**, 217 (1975)



+



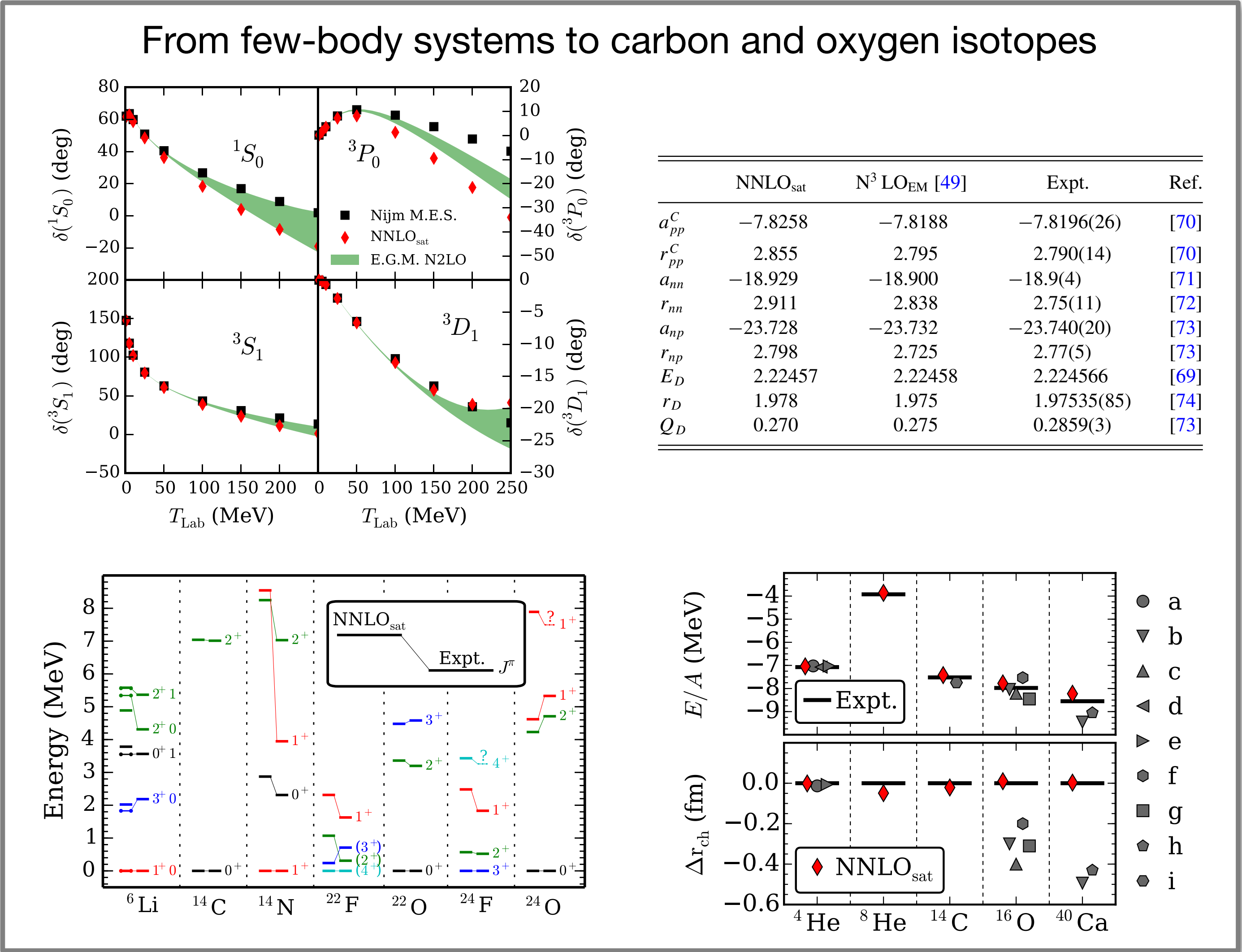
Lynn, PRL **116**, 062501 (2016)

Review
Hebeler, PR **890**, 1 (2021)

Approach 3: Totally phenomenological (NNLO_{sat})

Is it a realistic force?

Ekström, 91, 051301(R) (2015)



Chiral three-nucleon force

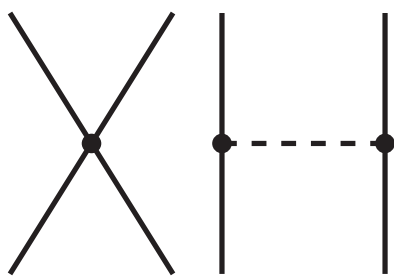
Hierarchy of chiral EFT

Entem +, PRC **96**, 024004 (2017)

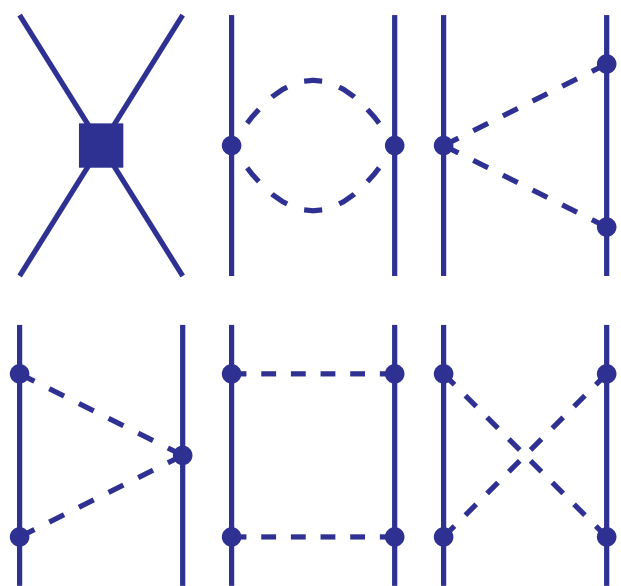
2N Force

3N Force

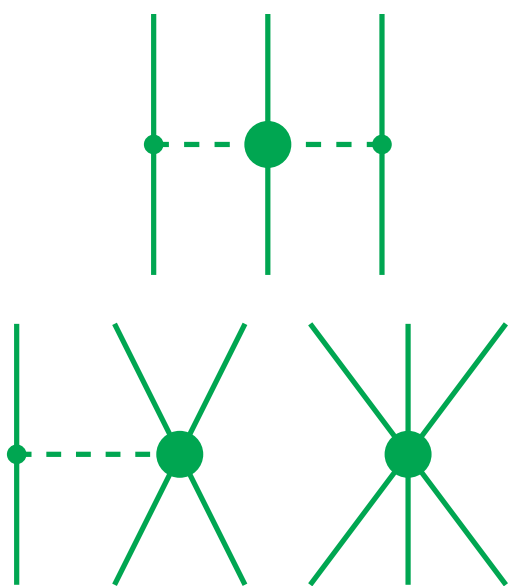
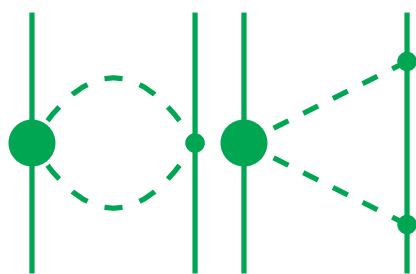
LO
 $(Q/\Lambda_\chi)^0$



NLO
 $(Q/\Lambda_\chi)^2$



NNLO
 $(Q/\Lambda_\chi)^3$



Many-nucleon forces are derived consistently in chiral EFT

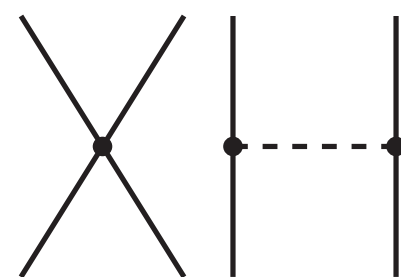
Hierarchy of chiral EFT

Entem +, PRC **96**, 024004 (2017)

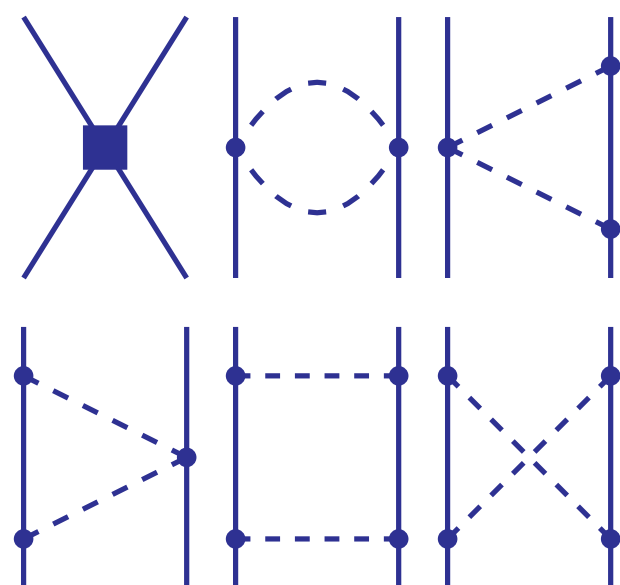
2N Force

3N Force

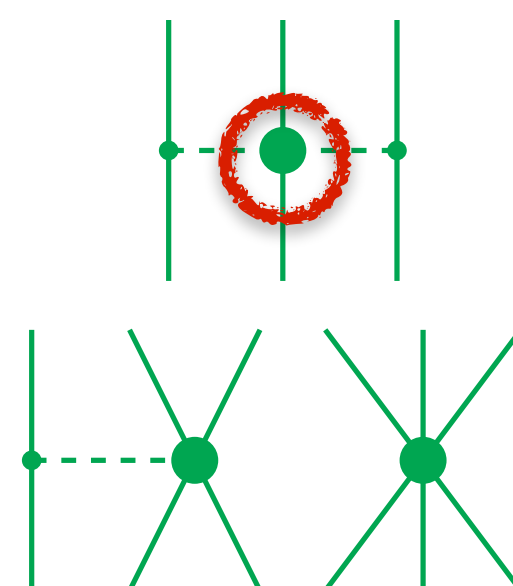
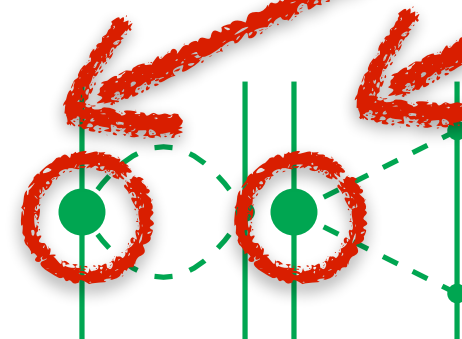
LO
 $(Q/\Lambda_\chi)^0$



NLO
 $(Q/\Lambda_\chi)^2$

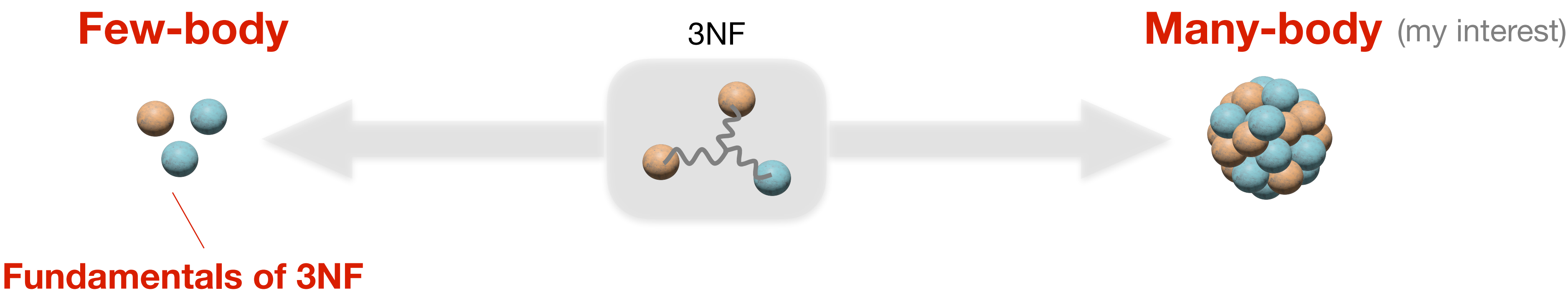


NNLO
 $(Q/\Lambda_\chi)^3$

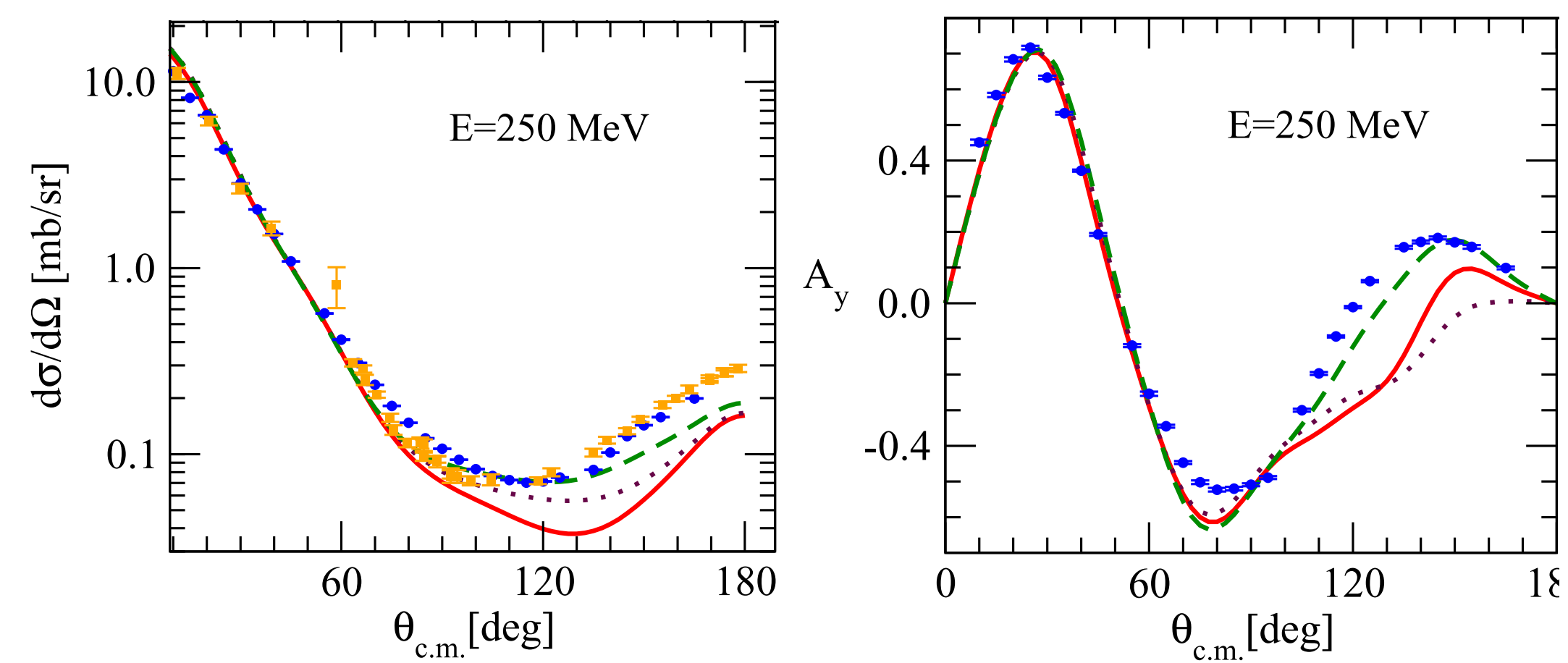


Many-nucleon forces are derived
consistently in chiral EFT

Same coupling const.



***d + p* scattering**



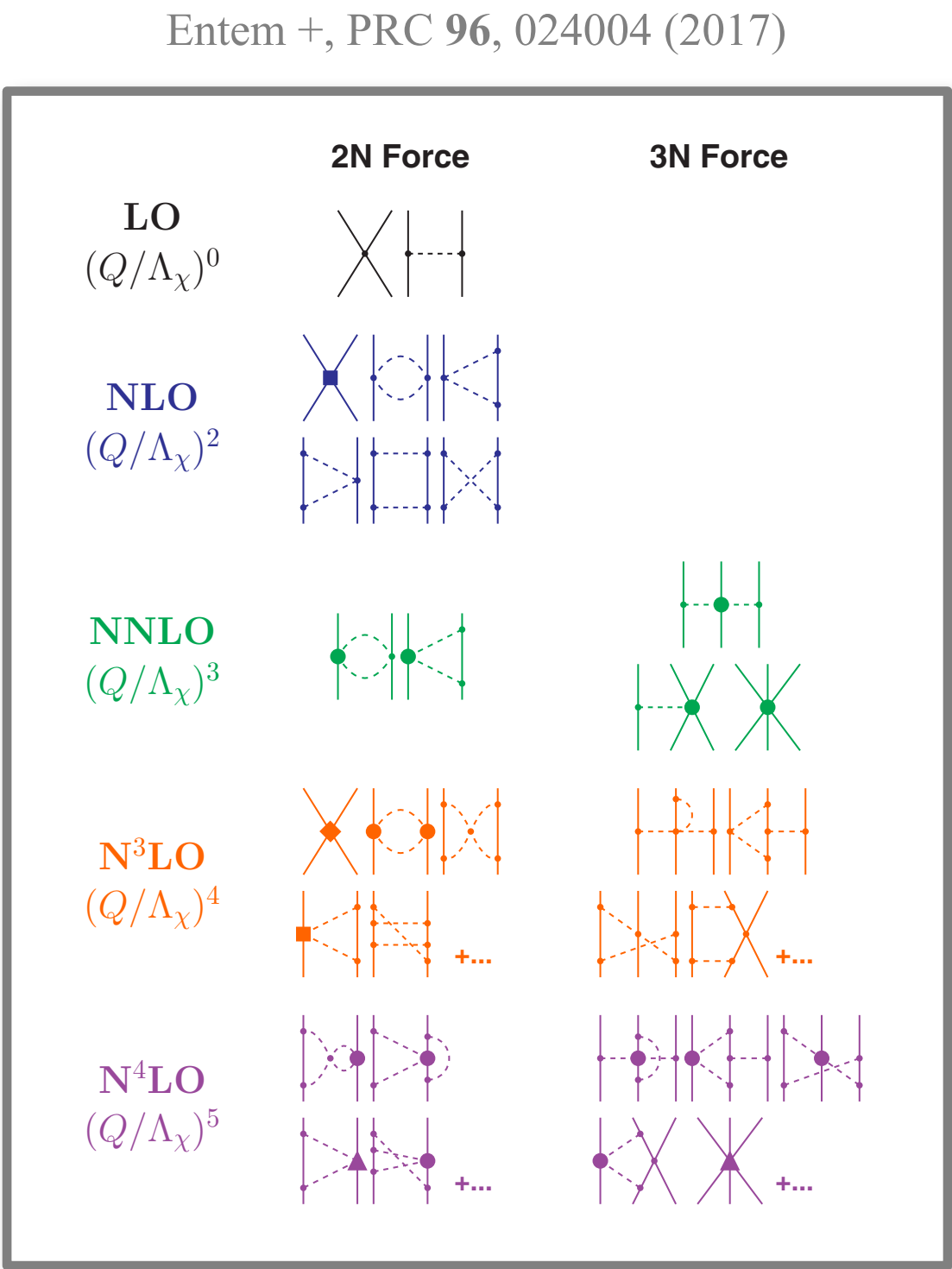
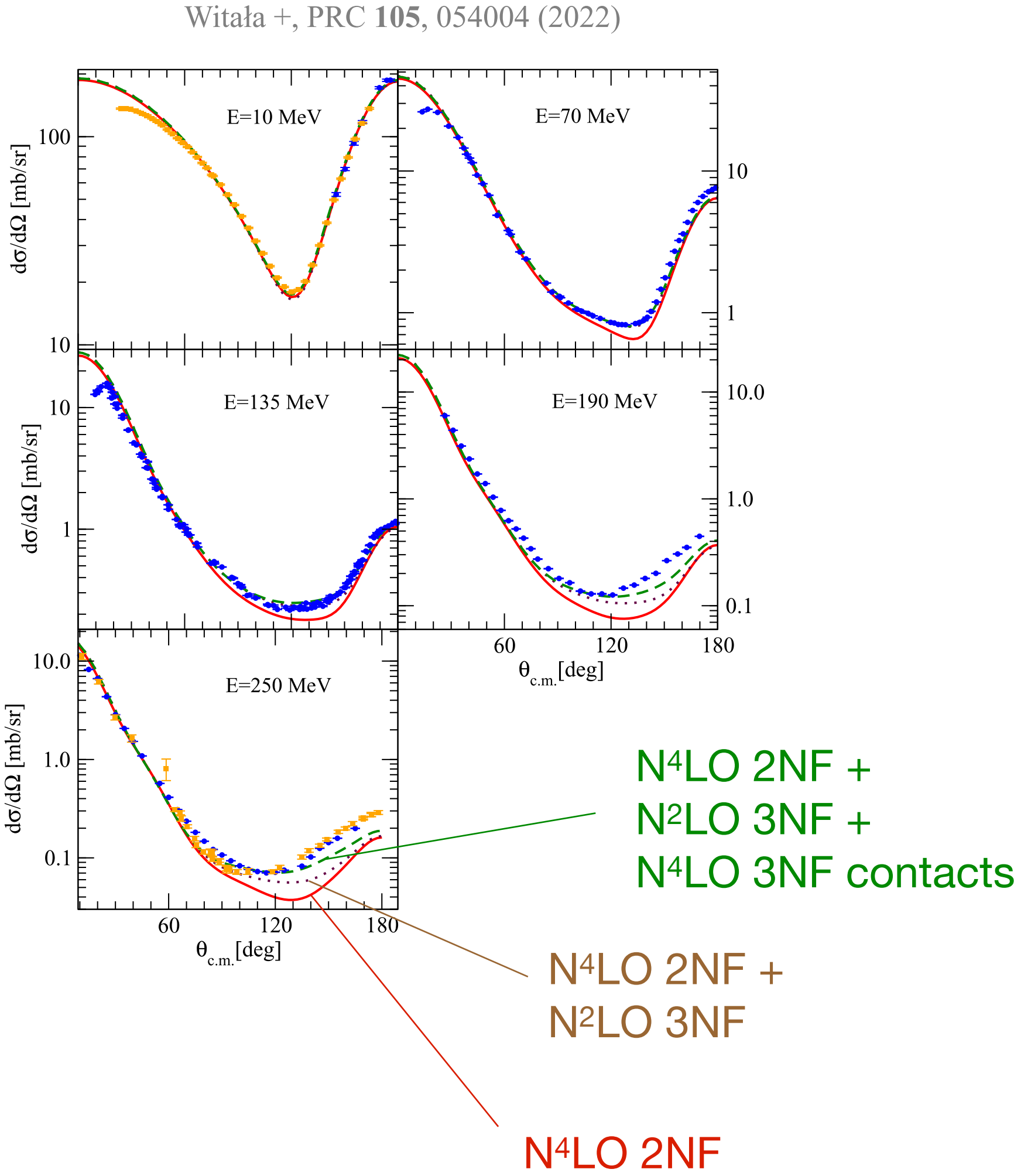
Witała +, PRC **105**, 054004 (2022)

$^3\text{H}/^3\text{He}$ structure

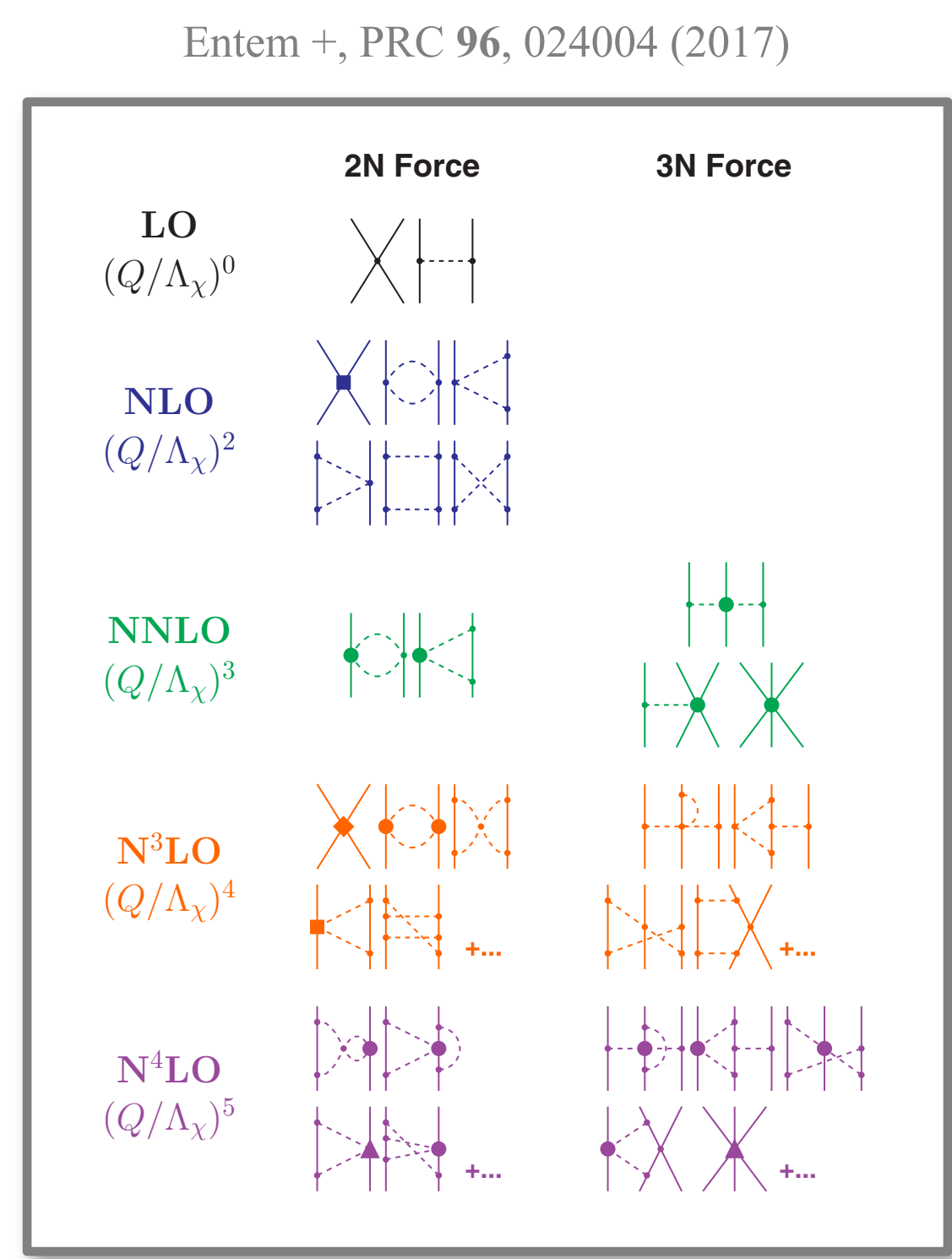
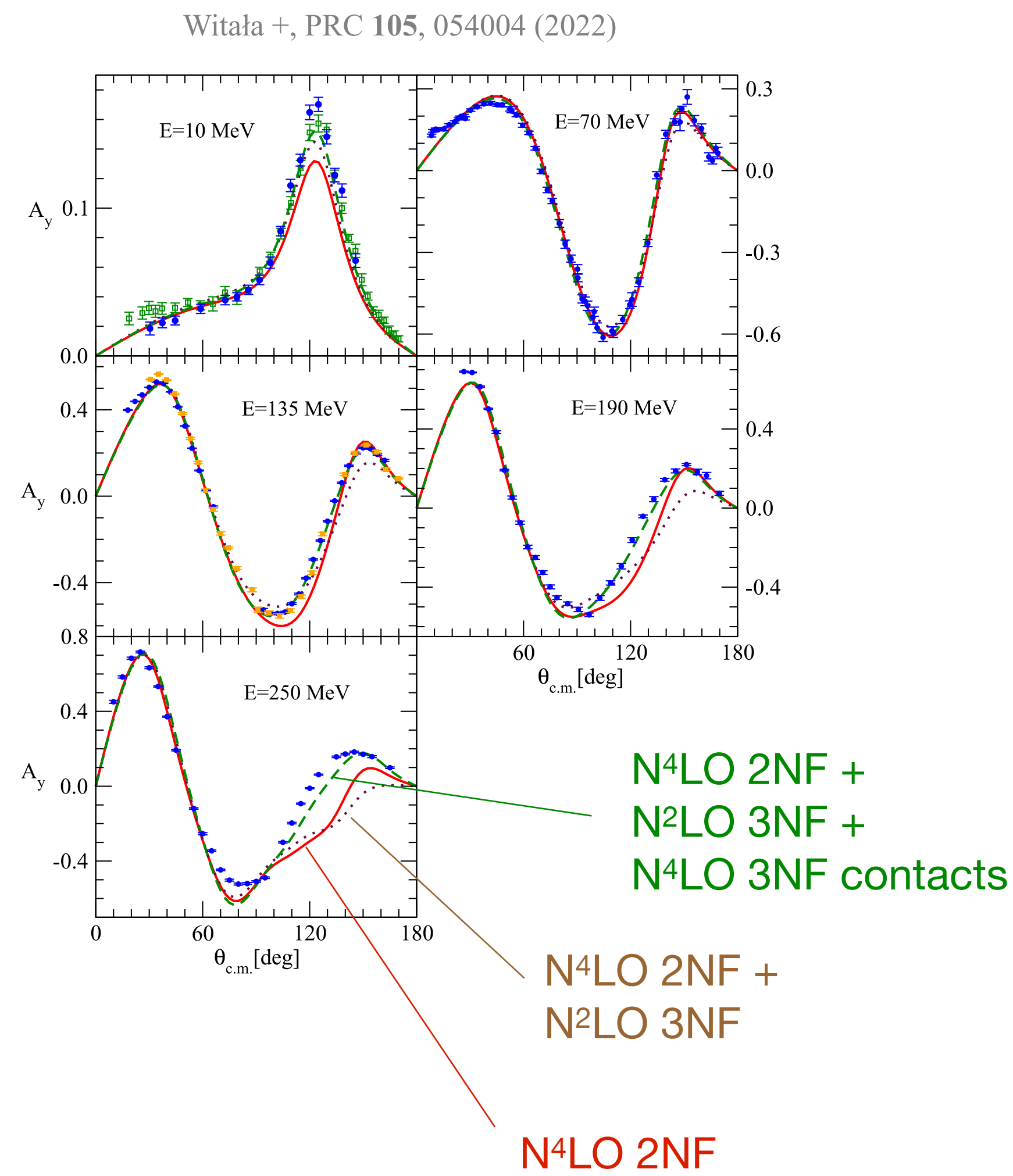
Interaction	Λ	$E(^3\text{H})$	$E(^3\text{He})$	ΔE_B
CD-Bonn + TM	4.784	−8.478	−7.735	0.743
AV18 + TM	5.156	−8.478	−7.733	0.744
AV18 + TM'	4.756	−8.448	−7.706	0.742
AV18 + Urb-IX		−8.484	−7.739	0.745
AV18 + Urb-IX (Pisa) [69]		−8.485	−7.742	0.743
AV18 + Urb-IX (Argonne) [28]		−8.47(1)		
Expt.		−8.482	−7.718	0.764

Nogga +, PRC, **65**, 054003 (2002)

Sagara discrepancy

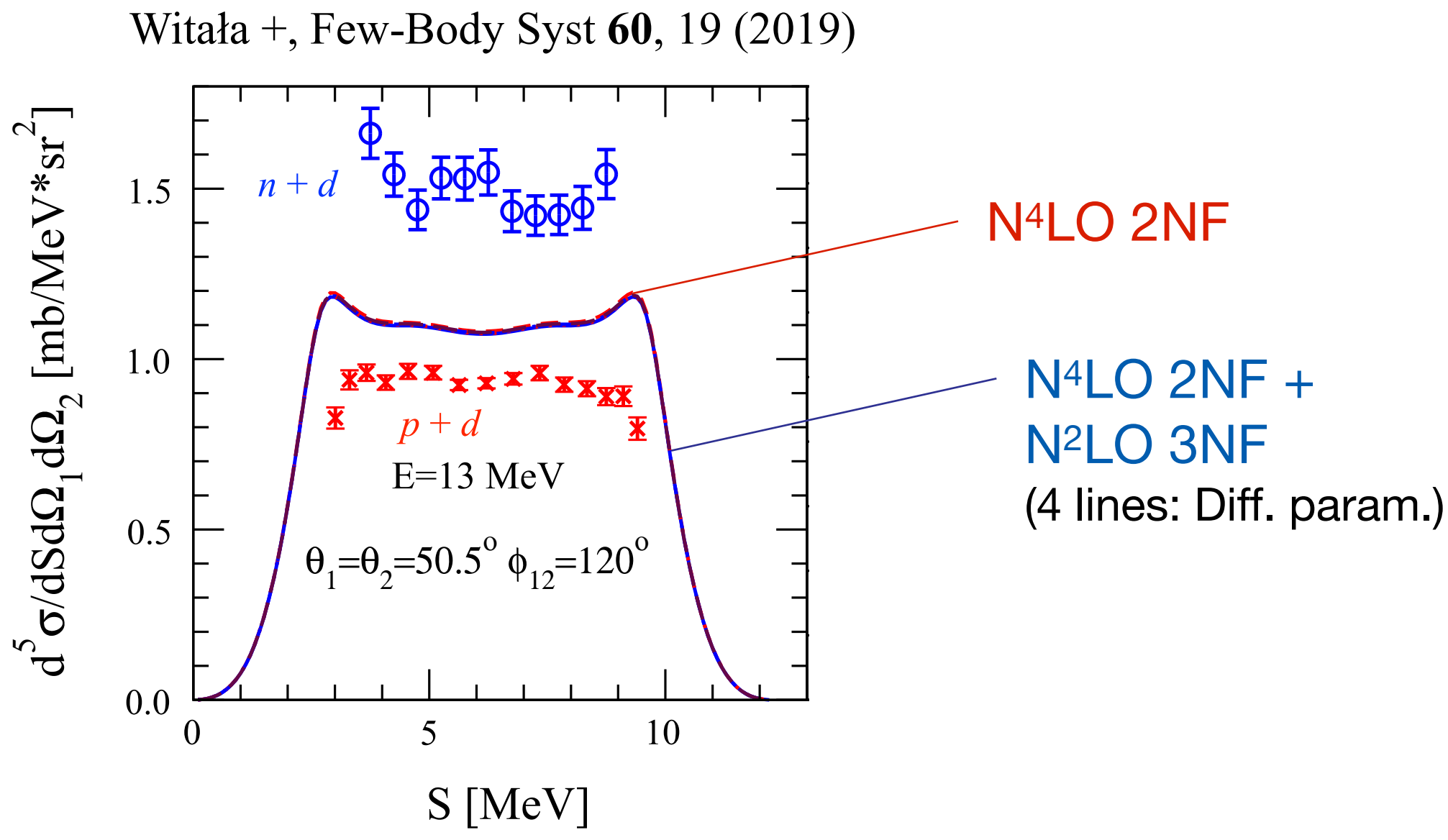


A_y puzzle

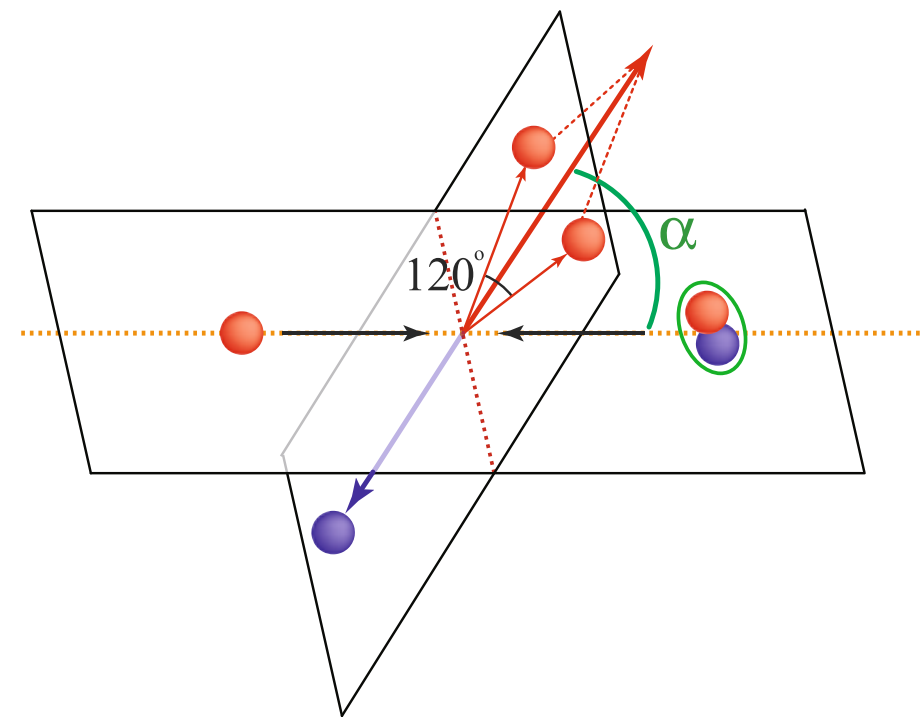


Cf. also Girlanda +, arXiv:2302.03468

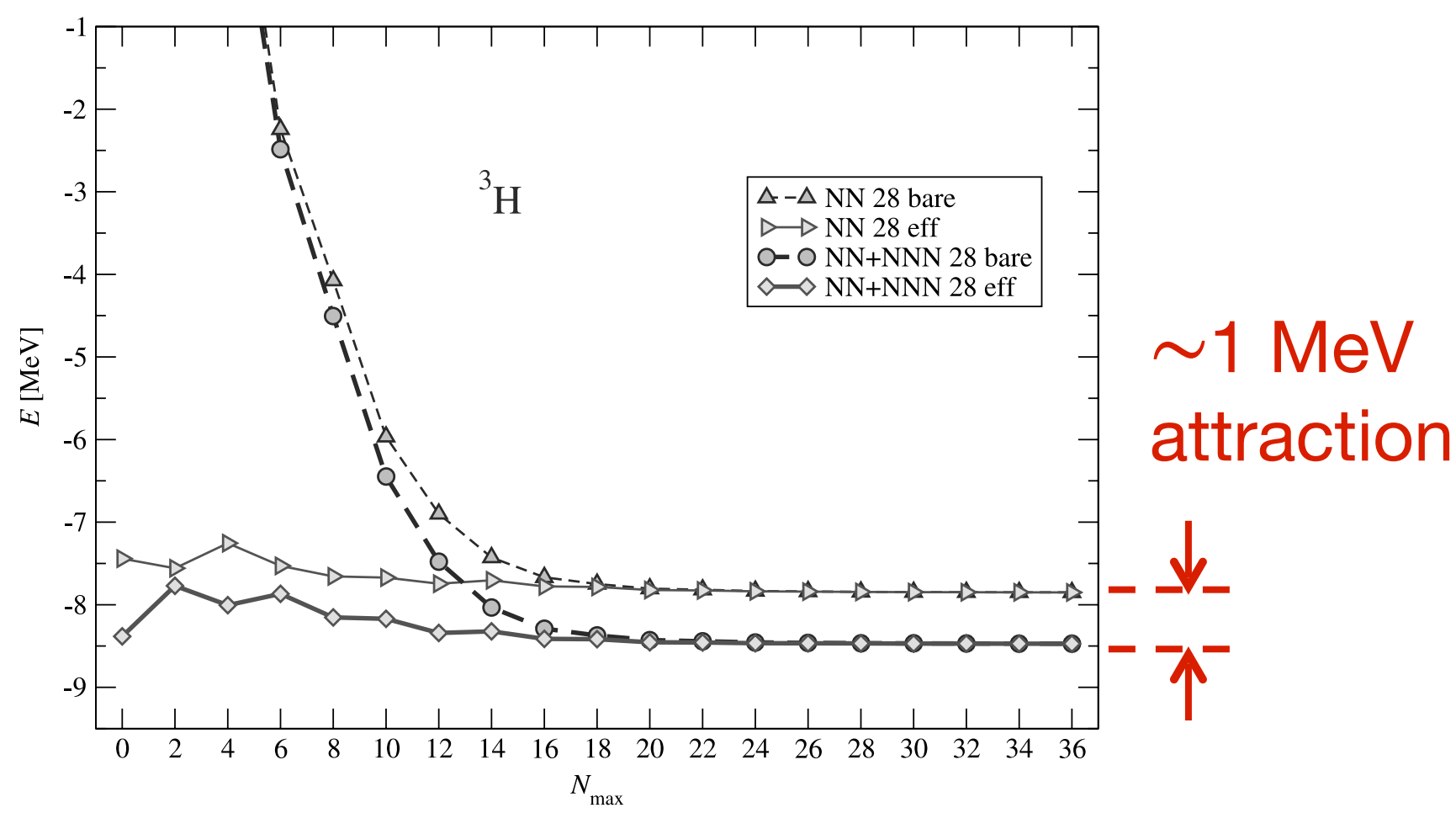
Space-star anomaly



Yabe +, Few-Body Syst **50**, 291 (2011)



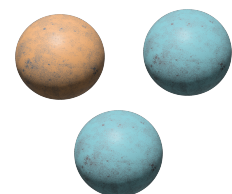
Triton-binding energy



ab initio calc.

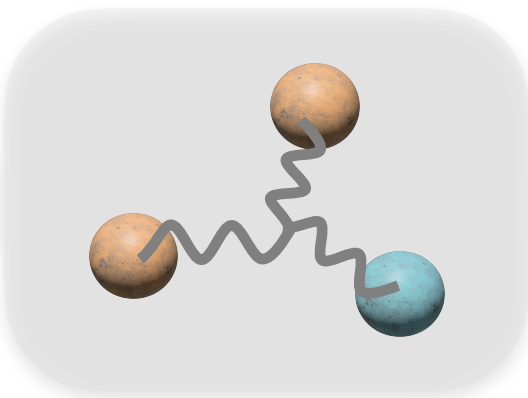
Navrátil, FBS **41**, 117 (2007)

Few-body

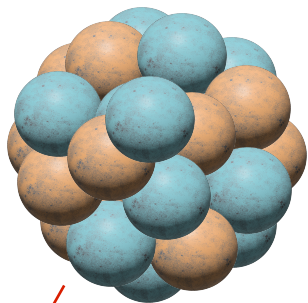


Fundamentals of 3NF

3NF

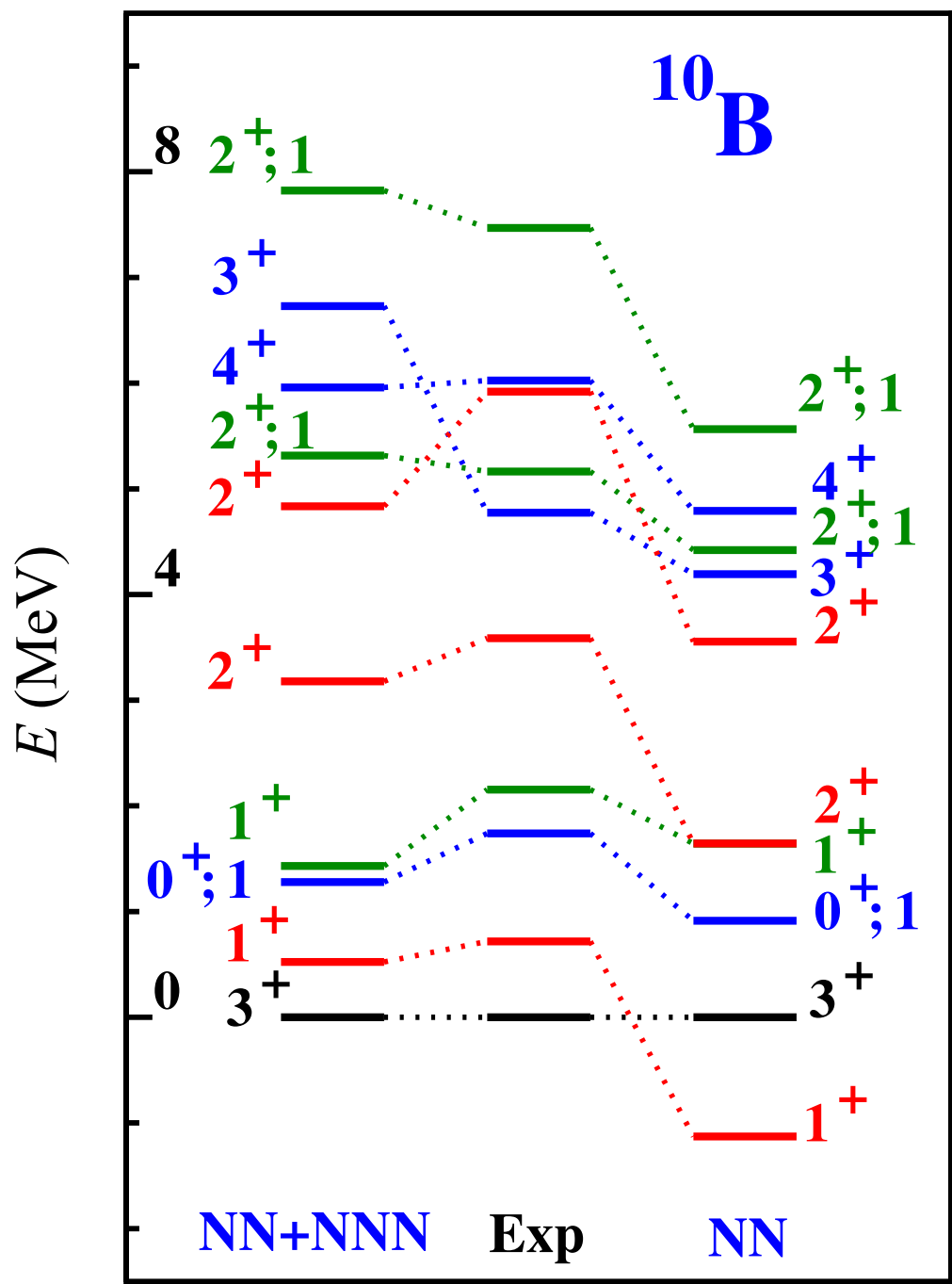


Many-body (my interest)



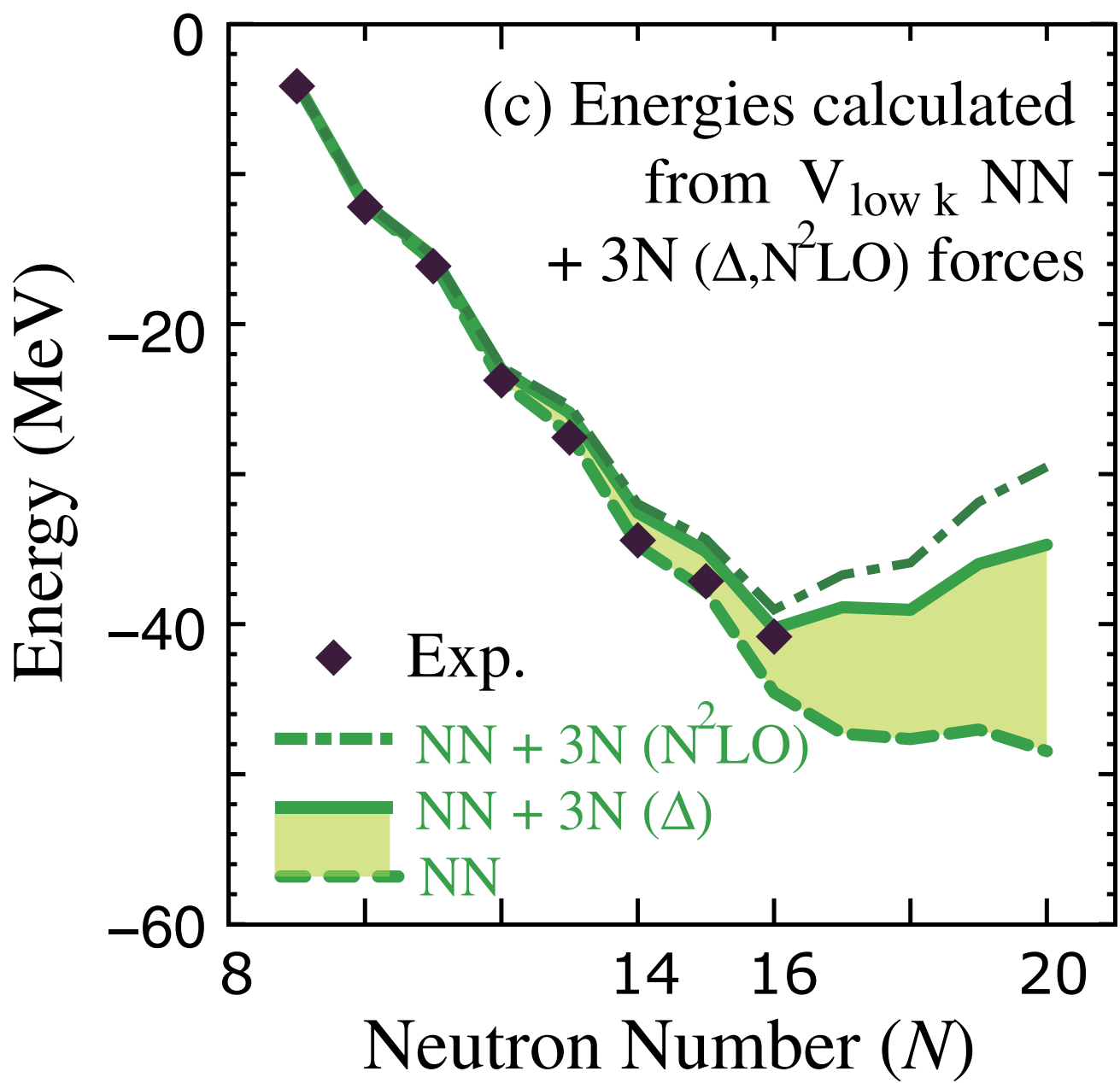
Major playground of 3NF

¹⁰B spectra



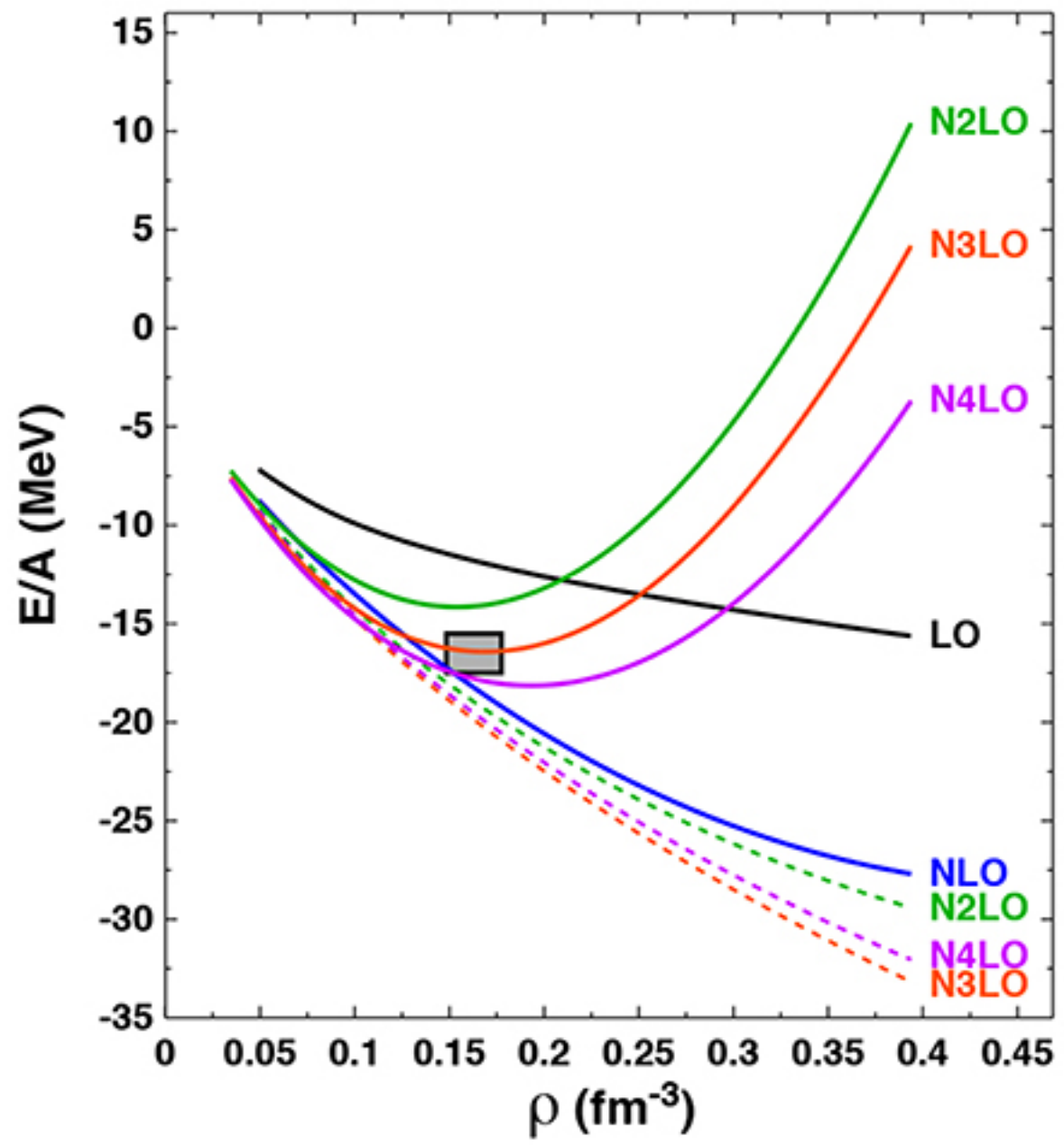
Navrátil +, PRL **99**, 042501 (2007)

Oxygen-drip line



Otsuka +, PRL **105**, 032501 (2010)

Nuclear matter saturation

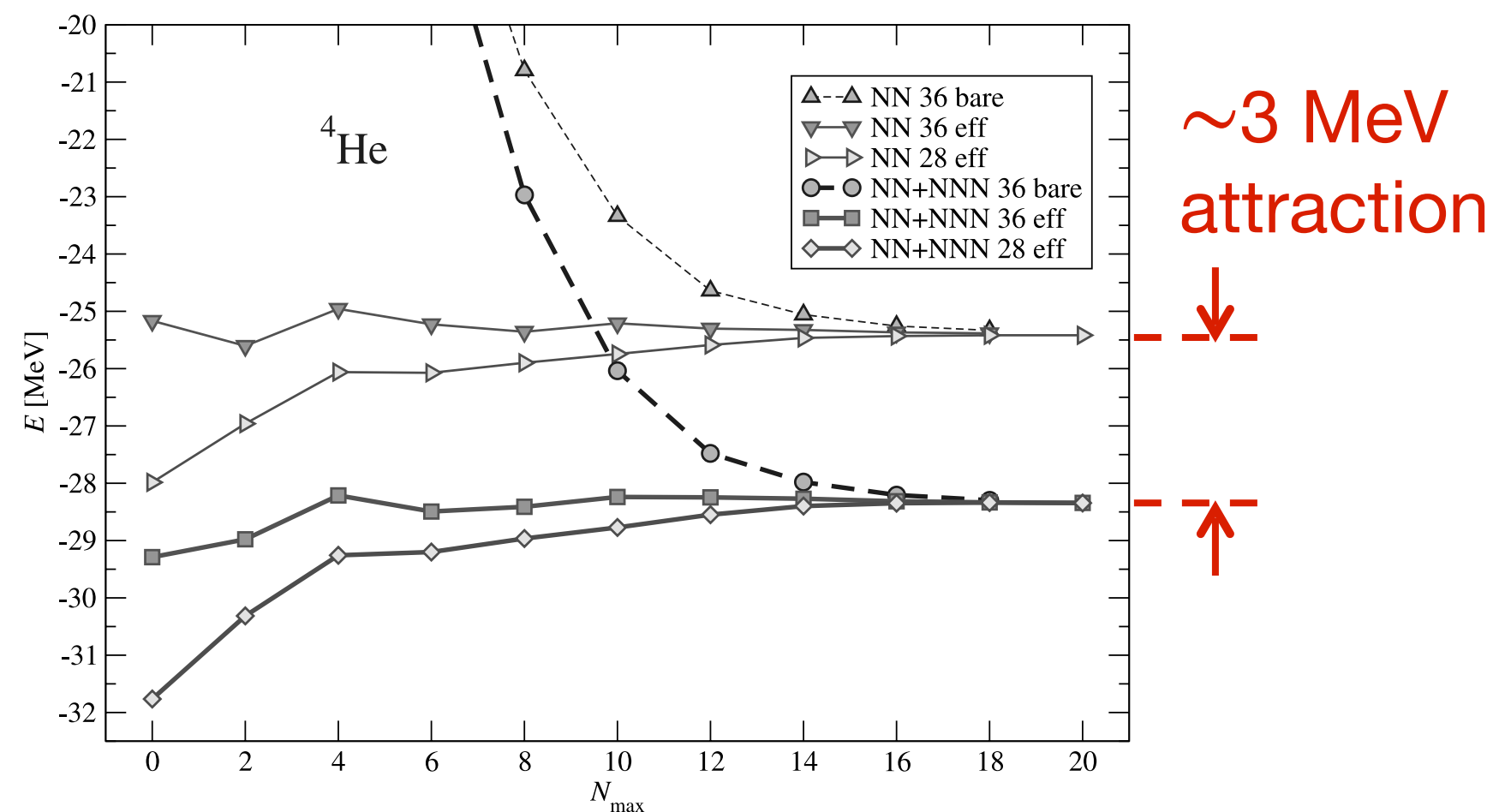
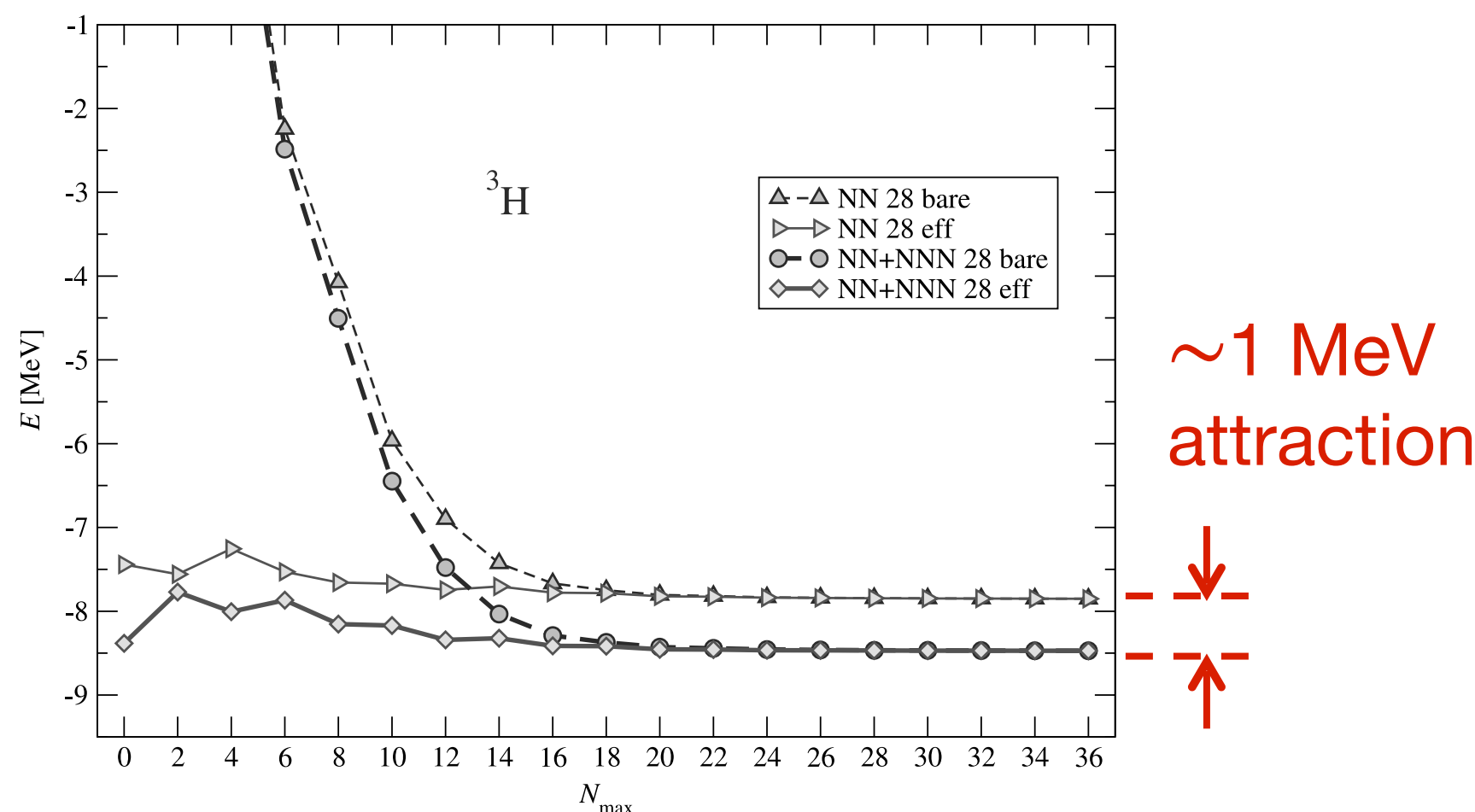


Sammarruca & Millerson, Front. Phys. **7**, 00213 (2019)

3NF contributions to ground-state energies

s-shell nuclei

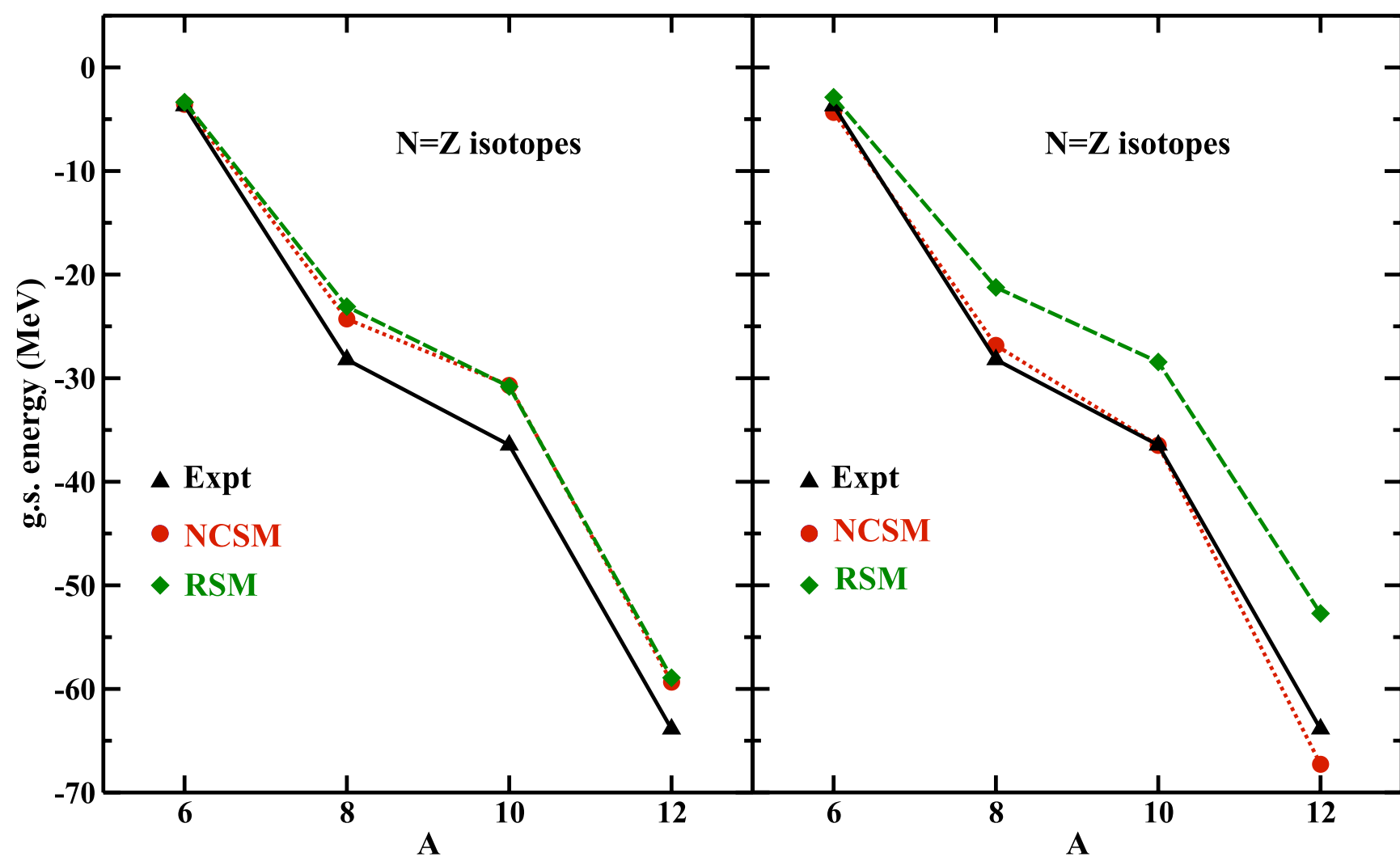
NCSM Navrátil, FBS **41**, 117 (2007)



p-shell nuclei

2NF only

2NF + 3NF



NCSM

Navrátil +, PRL **99**, 042501 (2007)

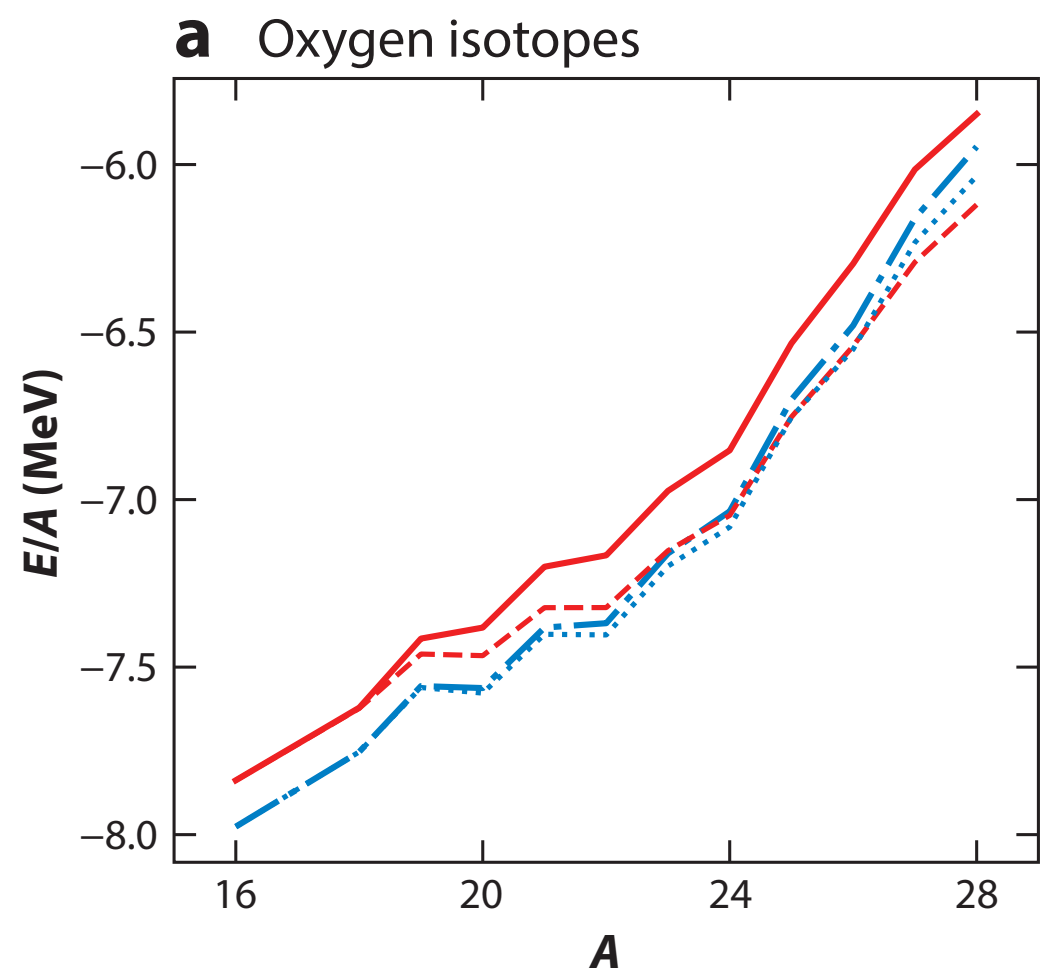
$\sim 5-10$ MeV attraction

3NF contributions to ground-state energies

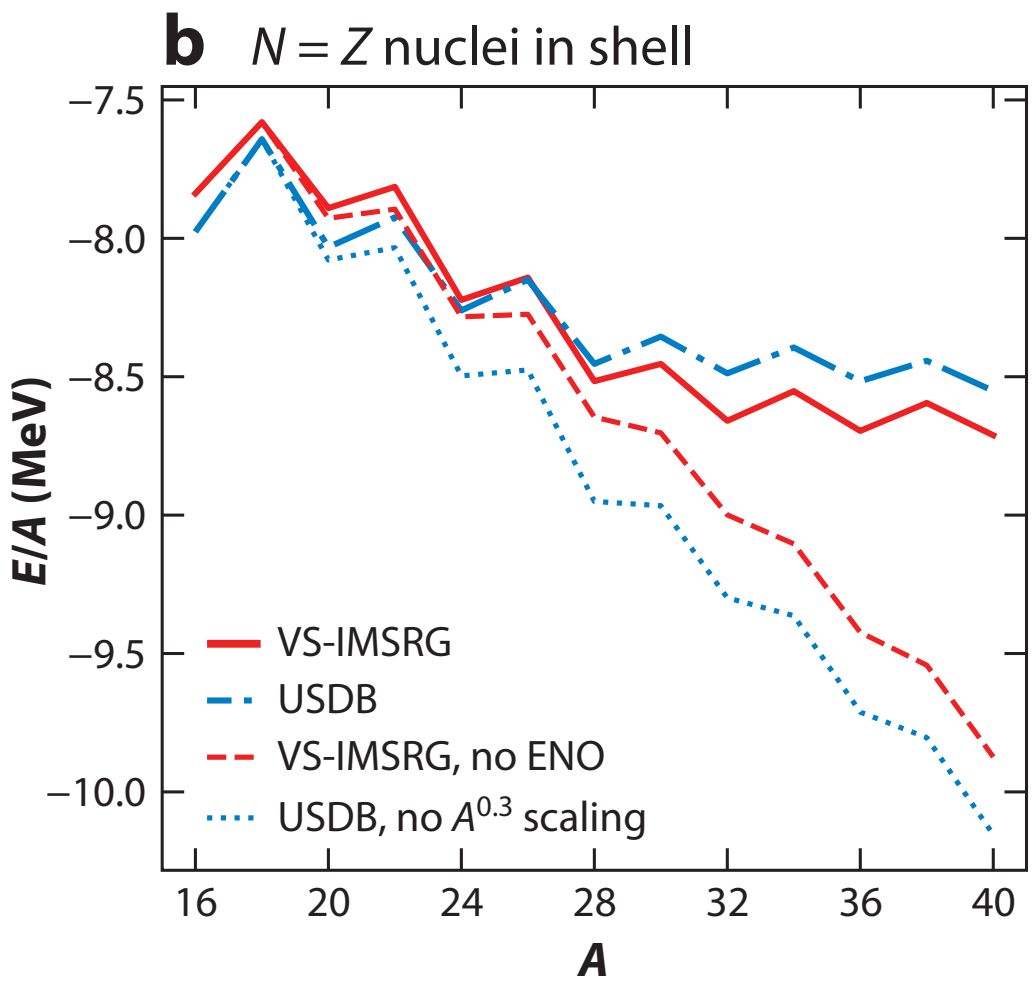
sd-shell nuclei

Valence-space in-medium SRG

Stroberg+, ARNPS **69**, 307 (2019)

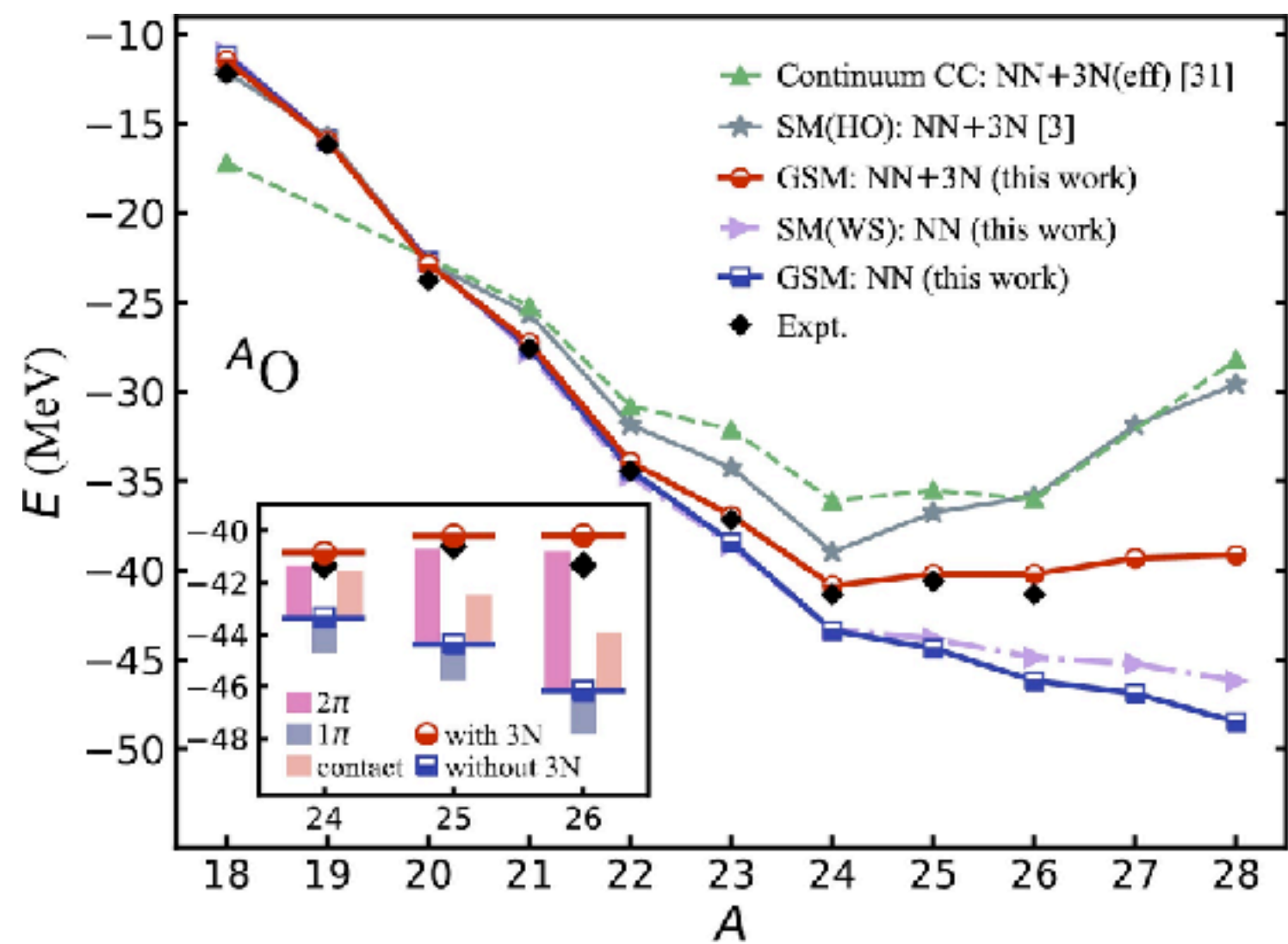


$\lesssim 0.3$ MeV/A
repulsion



$\lesssim 1.5$ MeV/A
repulsion

Oxygen (*sd*-shell + continuum)



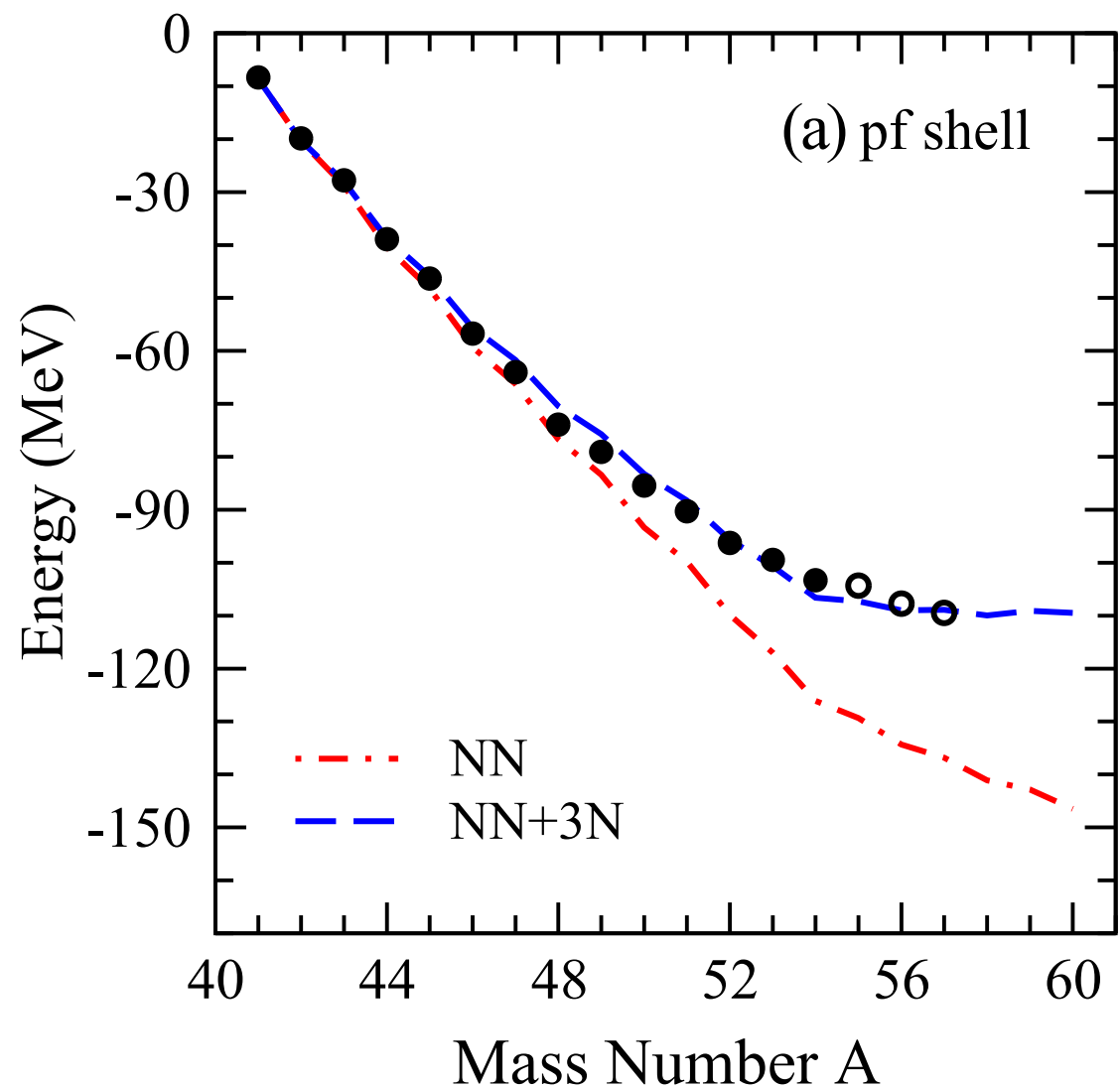
Gamow shell model

Ma +, PLB **802**, 135257 (2020)

$\lesssim 10$ MeV repulsion

3NF contributions to ground-state energies

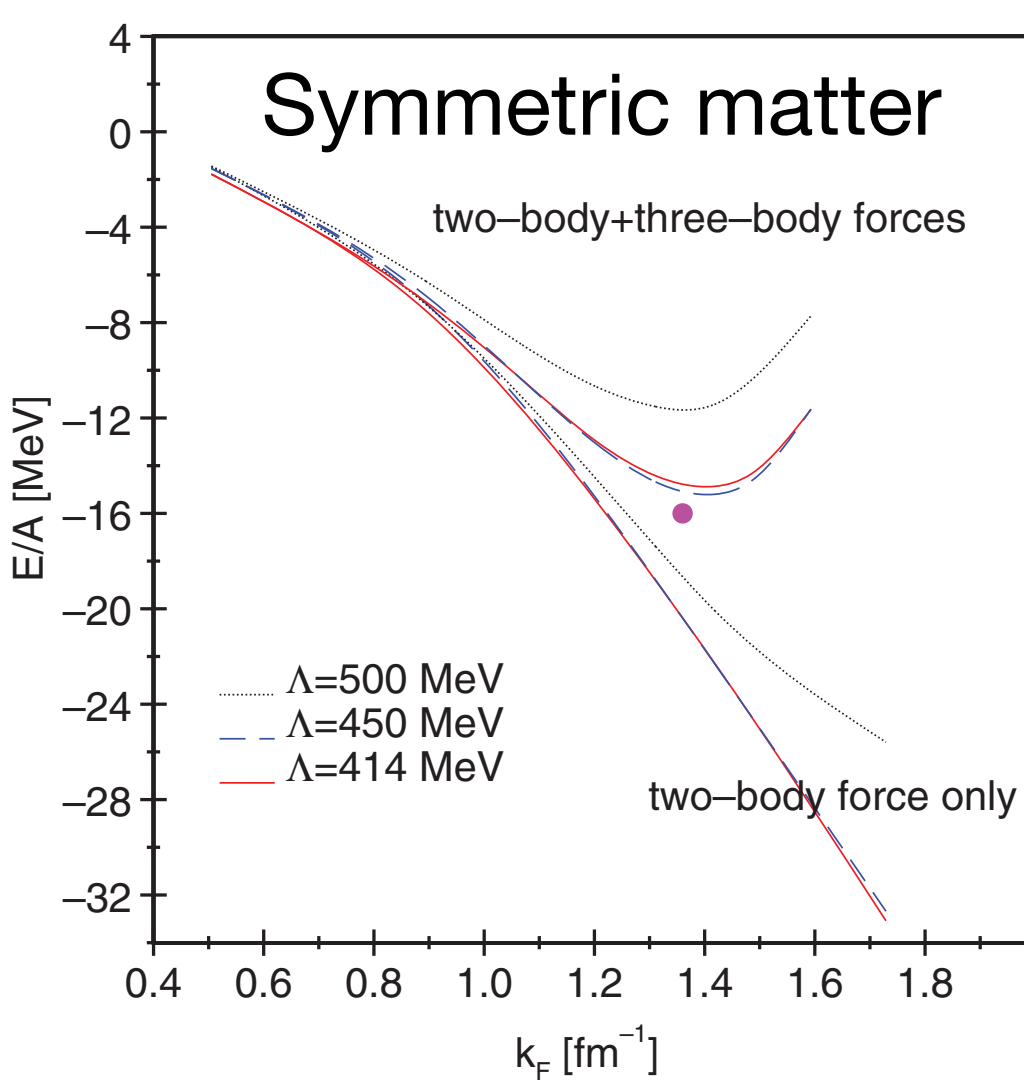
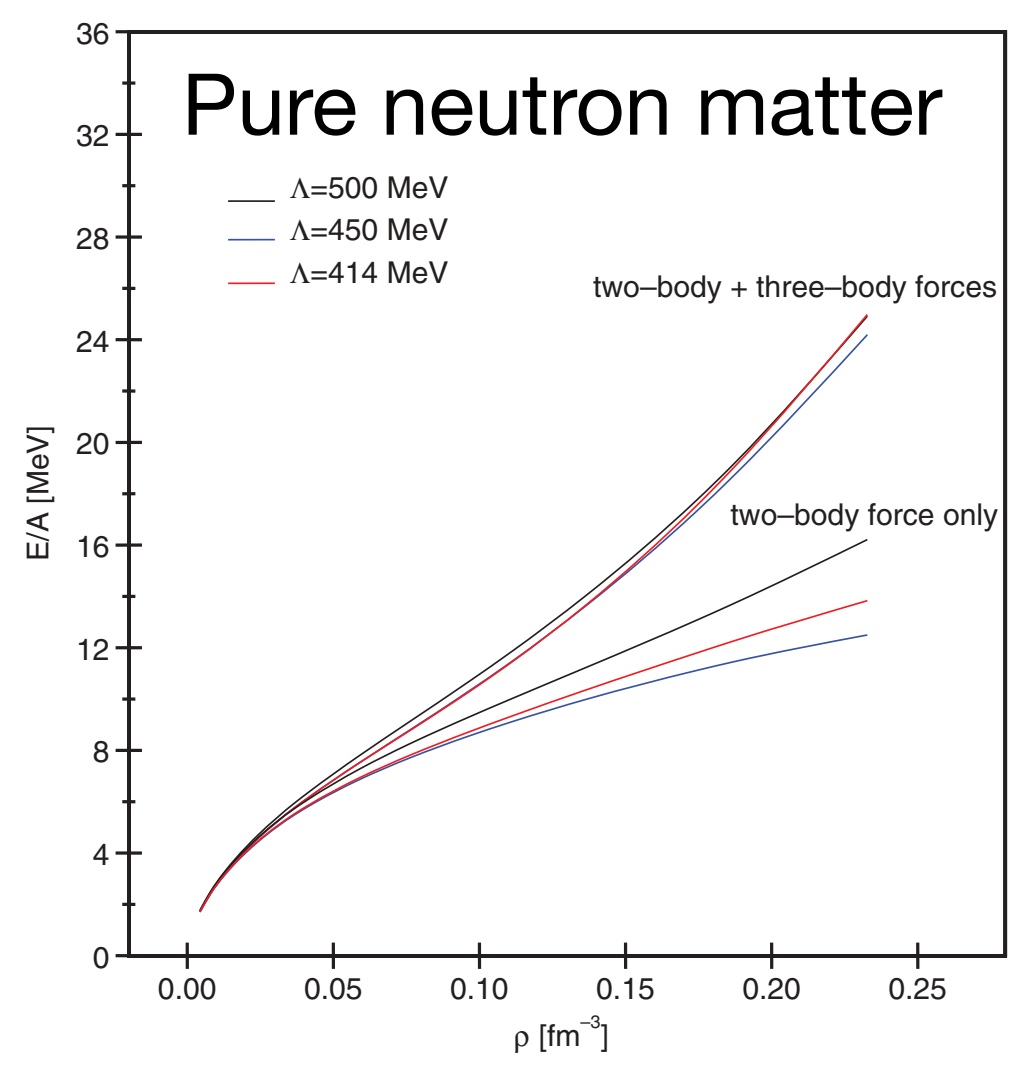
pf-shell nuclei



RSM Holt +, PRC **90**, 024312 (2014)

$\lesssim 10\text{--}20$ MeV/A repulsion

Nuclear matter



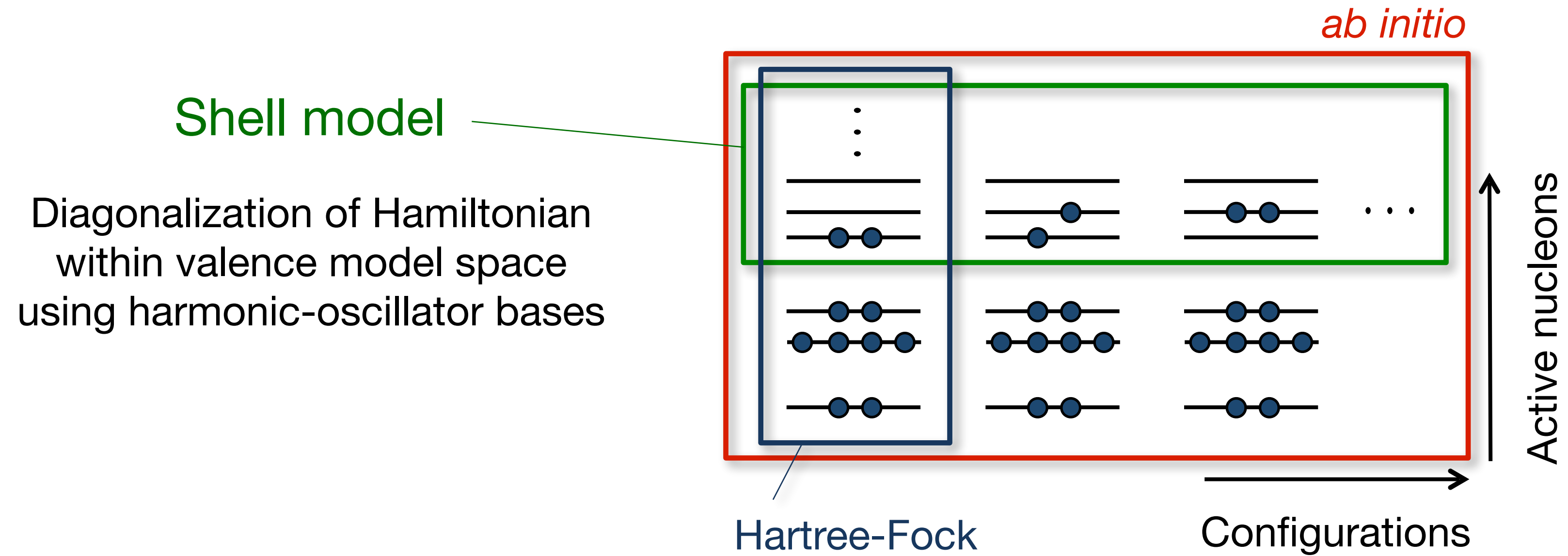
Perturbative calc.

Coraggio +, PRC **87**, 014322 (2013)
Coraggio +, PRC **89**, 044321 (2014)

$\sim 10\text{--}20$ MeV/A repulsion

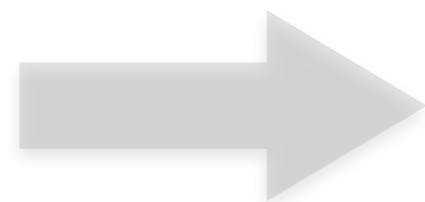
Implementation of chiral forces to shell model

Valence-space diagonalization



Realistic shell model (RSM)

= Shell model (valence space) with a realistic force



Applicable to heavier systems

Realistic Hamiltonian

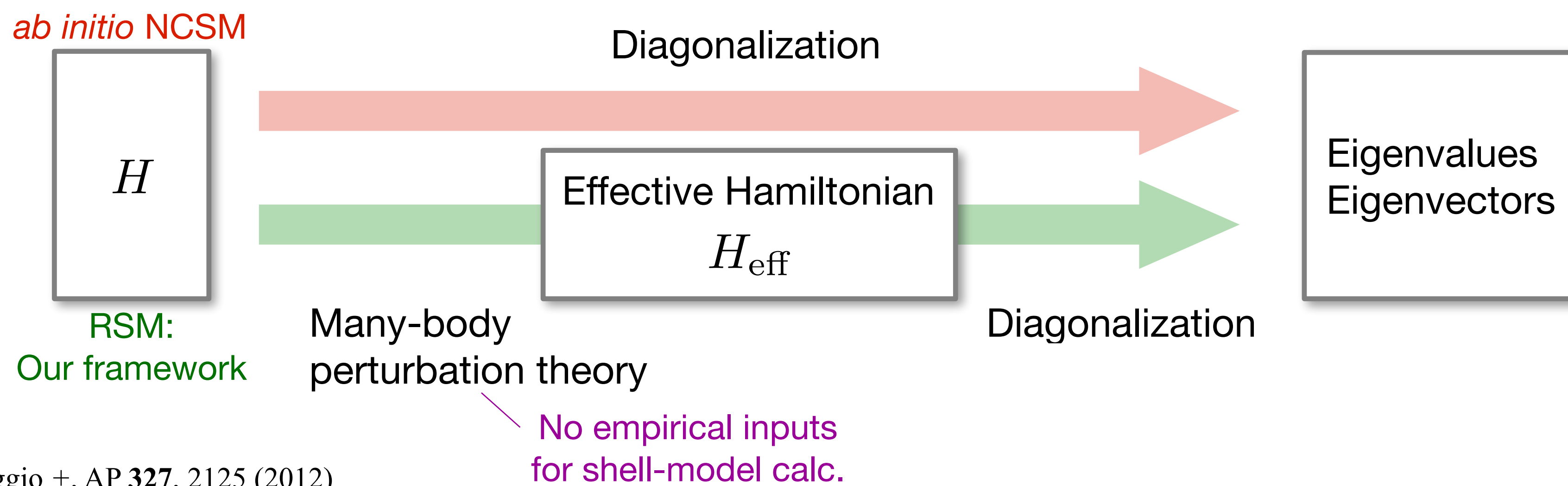
$$H = H_{1B} + H_{2B} + H_{3B}$$

Single-particle
energy

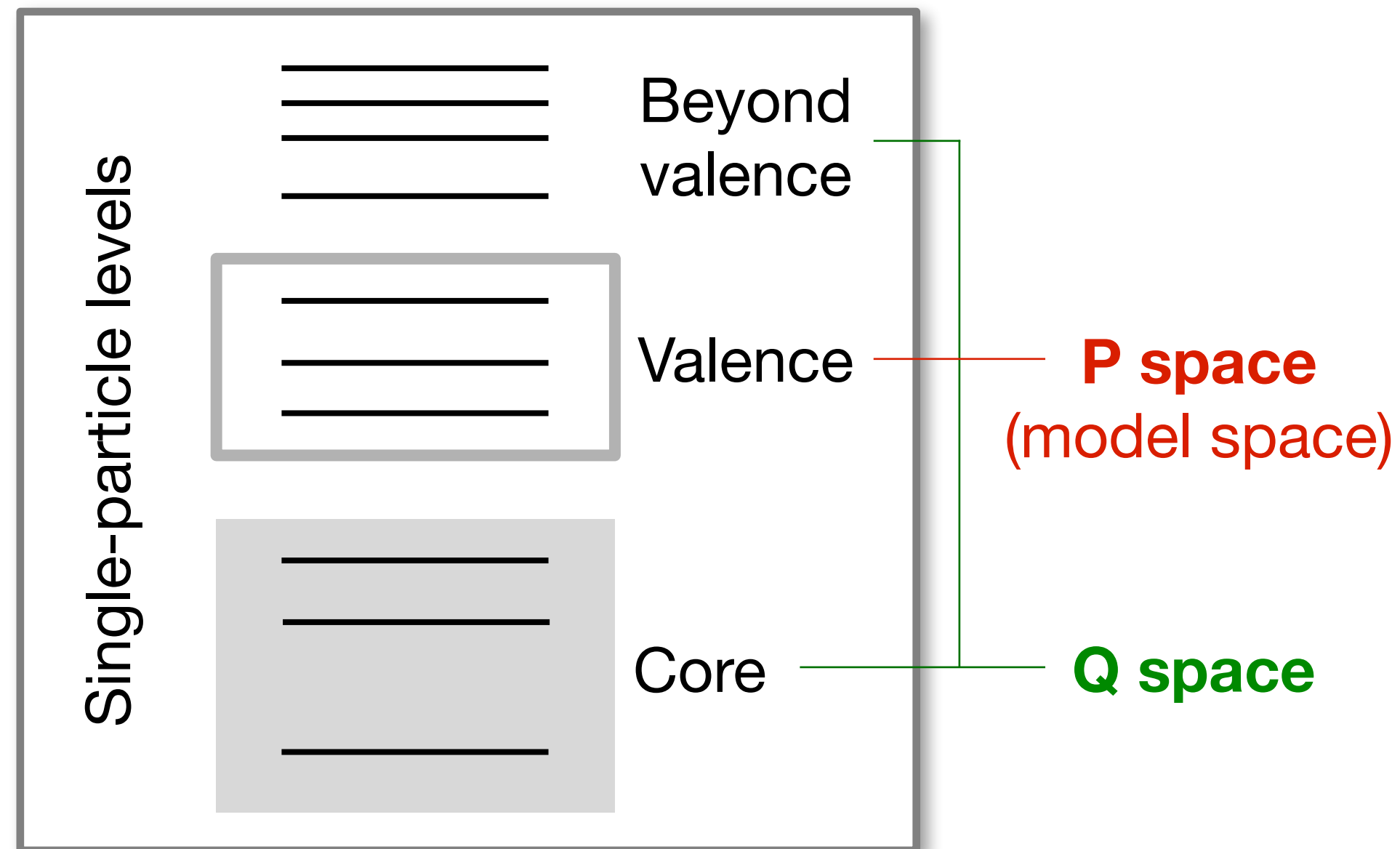
Chiral 2NF
at N³LO
+ Coulomb

Chiral 3NF
at N²LO

Shell-model framework



Many-body perturbation theory (1)



$$H = H_0 + H_1$$

$$H_{\text{eff}} = P\mathcal{H}P = PH_0P + H_1^{\text{eff}}$$

$$\mathcal{H} = X^{-1}HX$$

$$X = e^{\omega} \quad (\text{Similarity transformation})$$

Coraggio +, AP 327, 2125 (2012)

Feshbach projection operator

$$P = \sum_i |\Phi_i\rangle \langle \Phi_i|, \quad Q = 1 - P$$

$$P^2 = P, \quad Q^2 = Q, \quad PQ = QP = 0$$

	P	Q
P	H_{eff}	0
Q	0	$Q\mathcal{H}Q$

Many-body perturbation theory (2)

Coraggio +, AP 327, 2125 (2012)

Suzuki & Lee, PTP 64, 2091 (1980)

$$\omega = Q\omega P$$

$$P\omega P = Q\omega Q = P\omega Q = 0$$

$$\omega = \begin{pmatrix} P\omega P & P\omega Q \\ Q\omega P & Q\omega Q \end{pmatrix} \rightarrow \begin{pmatrix} 0 & 0 \\ Q\omega P & 0 \end{pmatrix}$$

$$P\mathcal{H}P = PHP + PHQ\omega$$

$$P\mathcal{H}Q = PHQ$$

$$Q\mathcal{H}P = QHP + QHQ\omega - \omega PHP - \omega PHQ\omega = 0$$

$$Q\mathcal{H}Q = QHQ - \omega PHQ$$

Decoupling equation

Many-body perturbation theory (3)

Coraggio +, AP 327, 2125 (2012)

$$\omega = \frac{1}{\epsilon_0 - QHQ} QH_1P - \frac{1}{\epsilon_0 - QHQ} \omega H_1^{\text{eff}}$$

ϱ box

$$\hat{Q}(\epsilon_0) = PH_1P + PH_1Q \frac{1}{\epsilon_0 - QHQ} QH_1P$$

$$H_1^{\text{eff}}(\omega) = \hat{Q}(\epsilon_0) - PH_1Q \frac{1}{\epsilon_0 - QHQ} \omega H_1^{\text{eff}}(\omega)$$

$$= \left[1 + PH_1Q \frac{1}{\epsilon_0 - QHQ} \omega \right]^{-1} \hat{Q}(\epsilon_0)$$

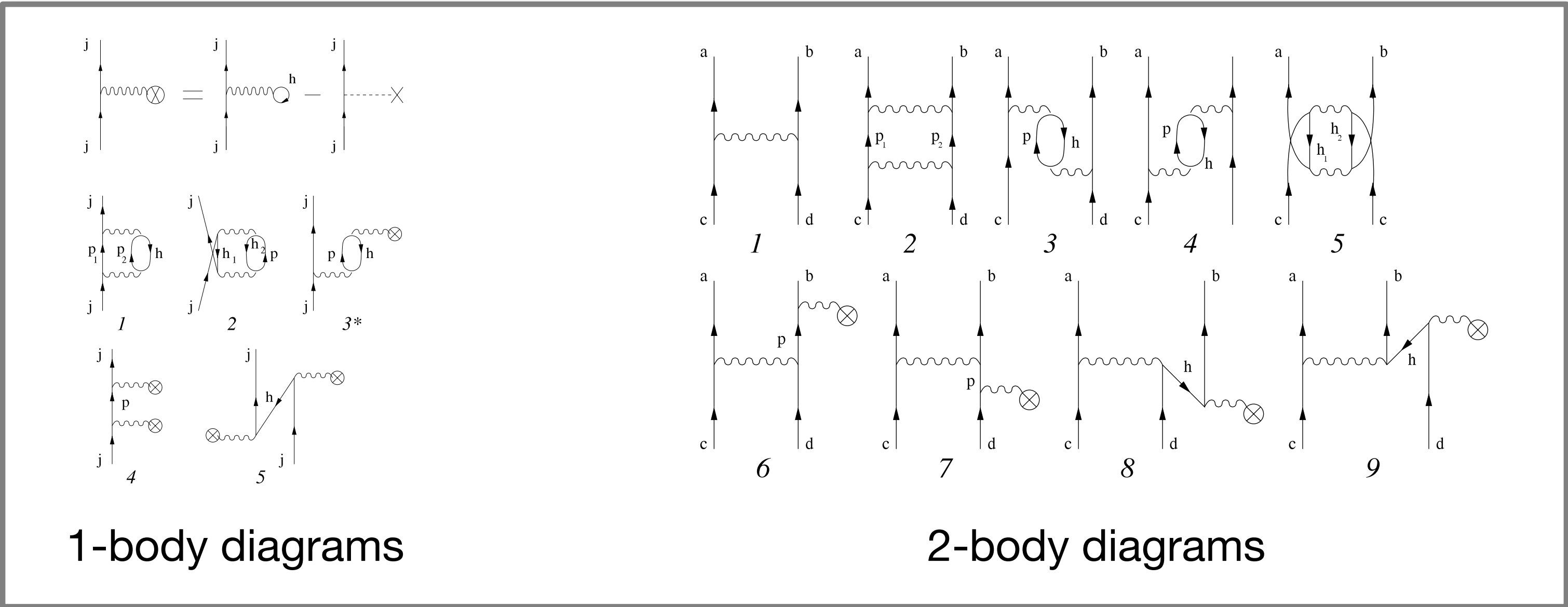
Many-body perturbation theory (4)

Coraggio +, AP 327, 2125 (2012)

Q box $\hat{Q}(\epsilon_0) = PH_1P + PH_1Q \frac{1}{\epsilon_0 - QH_0Q}QH_1P$

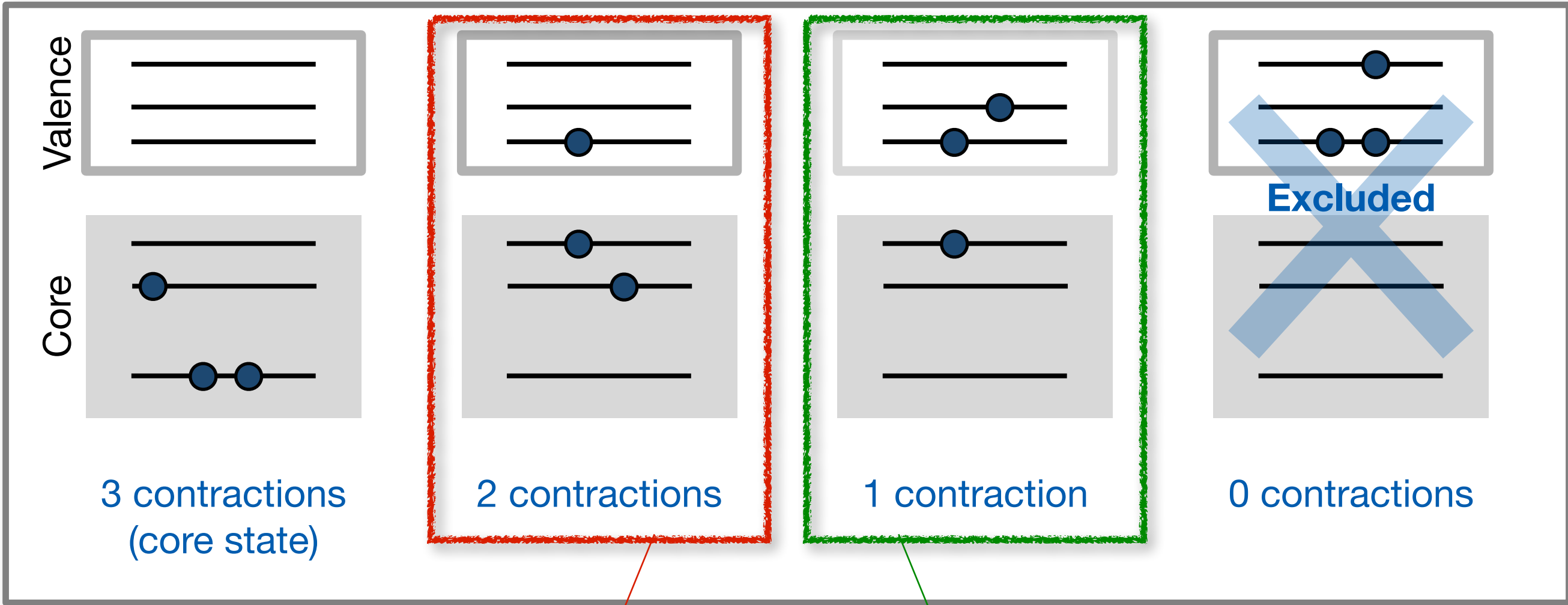
→ Perturbative expansion $\frac{1}{\epsilon_0 - QH_0Q} = \sum_{n=0}^{\infty} \frac{(QH_1Q)^n}{(\epsilon_0 - QH_0Q)^{n+1}}$

Diagrammatic expansion (examples)



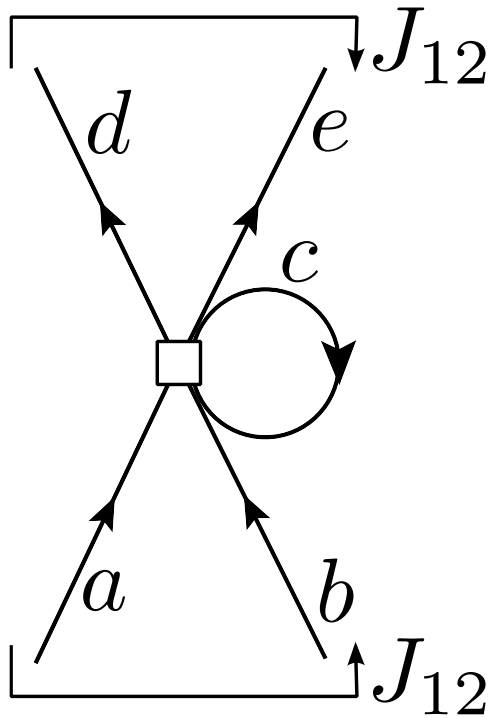
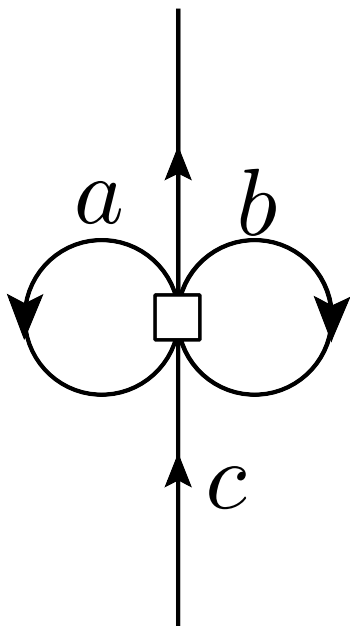
Normal-ordering approximation

→ 3NF as a modification of SPE and 2BMEs



Normal-ordered 1-body term

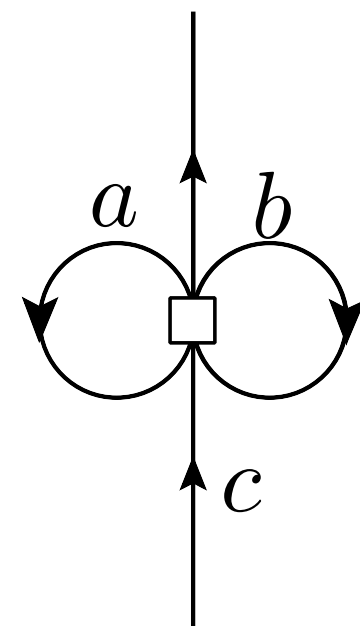
Normal-ordered 2-body term



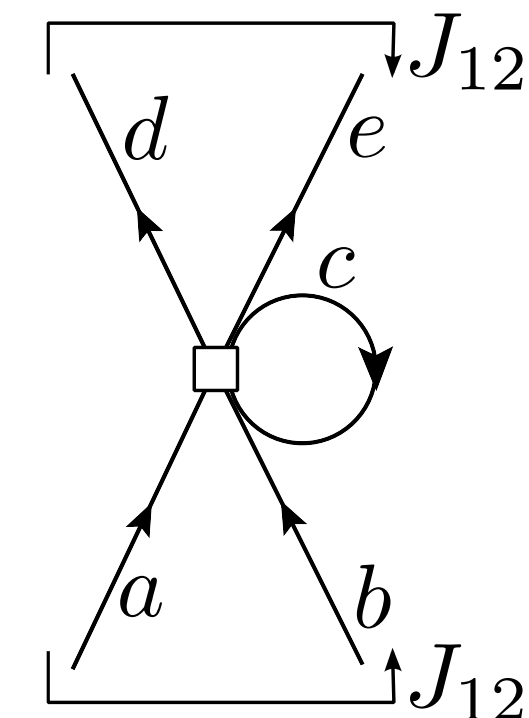
→ **3NF as a modification of SPE and 2BMEs**

$$\overline{abc}fed \equiv \overline{a_a^\dagger a_b^\dagger} a_c^\dagger a_f a_e a_d$$

Normal-ordered 1-body term



Normal-ordered 2-body term



Results

Results
to be given in Seminar tomorrow